L01 — study notes

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Class 1 — Review of basic ingredients · Map of the notes

One claim. A likelihood picks the estimator; the sampling rule decides how it behaves. The same $\theta^s(1-\theta)^{n-s}$ gives one MLE and one posterior, but different biases and p-values depending on whether n or s was fixed. **Estimators are judged by their sampling distribution, through MSE = variance + bias²**, worked out exactly, by Monte Carlo, or by the bootstrap. The bootstrap fails exactly where the statistic lives on the edge of the support. **Models are judged by what they predict and what they assume.** One Poisson rate fails the coal data's $E = V$ test and two rates fit. R^2 cannot choose a regression. An AR(1) has an equilibrium distribution only when $|\phi| < 1$. **And a p-value is not the evidence.** $p = 0.036$ can sit beside $P(H_1 \mid x) = 0.31$, and a search over 32 specifications makes "significance" likely by construction.

Three layers per topic:

Layer	What it means here
U — intuition	the two experimenters with one coin; "the bootstrap cannot recover what the sample does not contain"; the eye reading the coal series; the replication-crisis narrative; reading regression output
C — course (examinable)	the decks' and Bolfarine's definitions and results with their numbering: likelihood, MLE, sufficiency, unbiasedness, consistency, Cramér–Rao efficiency, MSE, bootstrap, dispersion test, profile likelihood, GLRT; OLS in matrix form, OLS = MLE, $R^2/\bar{R}^2/C_p$/AIC/BIC; AR(1) moments, equilibrium law, unit root, AR(p), VAR(1) mean and vec covariance; family-wise error, Bonferroni, Bayes factor vs p-value

Layer	What it means here
G — graduate	Haldane's unbiased estimator, UMVUE and invariance, $V(s^2)$ from cumulants, the integrated change-point likelihood, the best multiple $c\,x_{(n)}$, the Jeffreys–Lindley paradox in full. **Named or proved, not examinable**

Reading order

Part I — likelihood and the properties of an estimator

1. L01-01-likelihood-sampling-rules — `mle_sampling_rules.pdf` slides 1–11; B §1.3, §2.1, §3.1–3.2 (pp. 4–45, selected), §6.3 (p. 100); G §2.2 (pp. 12–18). **Binomial and negative-binomial data share one likelihood, MLE, $J(\hat{\theta})$ and posterior, but not their bias (0.1 vs 0.256) or their p-values (0.0107 vs 0.0020).** Haldane's unbiased fix, and why the Wald interval leaves $[0, 1]$.
2. L01-02-poisson-mle-properties — Poisson deck slides 1–12; B §1.1.4, §2.1–2.2, §3.1–3.2, §3.7 (pp. 2–52, selected). $\hat{\lambda} = \bar{y}$ **passes all four tests: sufficient, unbiased, consistent, efficient.** s^2 is unbiased too, but only $1/(1 + 2\lambda) = 20\%$ efficient. The Monte Carlo table, and Chebyshev's vacuous bound.

Part II — a model that fails, two that fit, and a first prior

3. L01-03-coal-mining-change-point — Poisson deck slides 13–21; B §3.4 (pp. 46–49), §6.5 (pp. 103–114); G §7.1.1 (pp. 91–95). **Overdispersion (index 1.592, $p < 0.001$) rejects one rate. The profile likelihood puts the cut at 1891 (weight 0.251), with LR = 70.0, which is not referred to χ^2_1 because τ was searched.** Profiling vs integrating.
4. L01-04-first-bayesian-poisson — handwritten "first Bayesian experience"; G §2.2 (pp. 12–18). **Exp(1/2) prior × Poisson likelihood = Gamma$(s_n + 1, a + n)$. On 1851–1891 the posterior mean is 3.084 and the mode 3.060, against the MLE 3.098: shrinkage with weight $a/(n + a) \approx 1\%$.**

Part III — judging estimators: MSE, Monte Carlo, bootstrap

5. L01-05-comparing-estimators-mse-bootstrap — MSE deck slides 1–20; handwritten MSE note; B §1.1.5, §1.3 (pp. 3–13), Ex. 2.2.5, Ex. 3.1.3; I §5.2 (pp. 212–215). **MSE = Var + bias². For $U(0, \theta)$, $\hat{\theta}_4 < \hat{\theta}_2 < \hat{\theta}_3 < \hat{\theta}_1$ for $n \geq 3$, with ties at $n = 1, 2$. The nonparametric bootstrap puts a 63% point mass on $x_{(n)}$, so it is inconsistent for the maximum, and the parametric bootstrap fixes it.**

Part IV — regression and time series

6. L01-06-gaussian-linear-regression — `multiplelinearregression.pdf` pp. 1–22; `overfitting_a_linear_model` `regressao_abstencao_2026.pdf`; W Ch. 2–3, §6.3, §8.2; AME §1.5.1.1, Ch. 2, §3.1–3.2. **OLS is a projection, and the Gaussian MLE. R^2 cannot fall, so it cannot select. $R^2_{in} = 1.0000$ predicts 10.5 times worse. The abstention table is read line by line (HC1, joint F, an R^2 gain of 0.002).**

7. L01-07-ar-var-review — `ar-ma.pdf` pp. 1–17; `ar-var-R.txt`; T §2.1–2.4 (pp. 30–57), §8.2 (pp. 399–403); W §11.1, §11.3. **AR(1) = a regression on the lag, and OLS = the conditional MLE. For $|\phi| < 1$, y_t settles into $N(\mu, \sigma^2/(1-\phi^2))$. At $\phi = 1$ the variance grows like t. VAR(1): eigenvalues, $(I - \Phi)^{-1}\phi_0$, and the vec formula on Tsay Ex. 8.3.**

Part V — evidence

8. L01-08-p-hacking-evidence — `part1-p-hacking.pdf`; `p_hacking_slides-economics.pdf`. $1 - 0.95^{32} = 0.81$ **(an upper bound when the tests share data); Bonferroni** $0.05/32$**. At** $X = 1.8$**,** $p = 0.036$ **but** $BF_{01} = 2.23$ **and** $P(H_1 \mid x) = 0.31$**. Prior odds of 20:1 are needed to reach 0.9.**

Support

9. L01-09-formulas — the exam specification (site items i–v, class-plan goals 1–11) mapped to notes; every numbered result; every number.
10. L01-10-pitfalls — sixteen confusions with minimal counterexamples, and an original 30-item True/False self-test.
11. L01-11-python — the reference implementation, section by section, with asserts. The runnable script is `L01_python_checks.py`.

The one result to hold

$$\underbrace{L(\theta \mid x)}_{\text{picks } \hat{\theta}} \longrightarrow \underbrace{\text{sampling distribution of } \hat{\theta}}_{\text{set by the design: exact, MC or bootstrap}} \longrightarrow \underbrace{\text{MSE}(\hat{\theta}) = V(\hat{\theta}) + B(\hat{\theta})^2}_{\text{the verdict}}$$

The first arrow obeys the likelihood principle. The second does not. That one fact explains the stopping-rule paradox (L01-01), the biased-but-better $x_{(n)}$ (L01-05), the bootstrap's failure at the edge of the support (L01-05 §10), the searched LR that is not χ^2_1 (L01-03 §6), and the minimum of 32 p-values that is not a p-value (L01-08).

Numbers and cases to have memorized

Quantity	Value	Meaning
$E(s/N)$, $s = 1$, $\theta = 0.1$	**0.256** (binomial: 0.100)	same MLE, different bias
p-values, $s = 1$, $n = 10$, $H_0 : \theta = 0.5$	**0.0107** (binomial) vs **0.0020** (neg. binomial)	same data, different evidence "against" H_0
Efficiency of s^2 for Poisson	$1/(1 + 2\lambda) =$ **0.2** at $\lambda = 2$	five times the data
Coal: rates, cut, LR	**3.125 / 0.917**; $\tau =$ **1891**, $w = 0.251$; LR **70.0**	one rate rejected; the year is not pinned down (1889–1892: 0.696)
Coal posterior (1851–1891)	mean **3.084**, mode 3.060, sd 0.273; MLE **3.098**	shrinkage toward the prior mean 2
Four MSEs, $\theta = 3$, $n = 10$	**2.325, 0.136, 0.300, 0.075**	$\hat{\theta}_4 < \hat{\theta}_2 < \hat{\theta}_3 < \hat{\theta}_1$
Bootstrap point mass	$1 - (1 - 1/n)^n \to$ **0.632**	inconsistent for $x_{(n)}$
Boston overfit	$R^2_{in} =$ **1.0000**, $R^2_{out} =$ **0.052**; RMSE 47.13 vs 4.48	R^2 cannot select

Quantity	Value	Meaning
AR(1), $\phi = 0.9$	equilibrium variance $\mathbf{5.263}\sigma^2$; half-life **6.58**	stationary; $\phi = 1$: $V = t\sigma^2$
Tsay Ex. 8.3	eigenvalues 0.8, 0.5; $\mu = (1, 2)'$; $\Gamma_0 = \frac{1}{9}\begin{pmatrix} 25 & 20 \\ 20 & 40 \end{pmatrix}$	stationary despite $\Phi_{22} = 1.1$
p-hacking	$1 - 0.95^{32} = \mathbf{0.806}$; $p = \mathbf{0.036}$ vs $P(H_1 \mid x) = \mathbf{0.31}$; prior odds **20:1**	significance ≠ evidence

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Figures

Built by `figures/make_figures.py`. Run: `python Notas/L01/figures/make_figures.py`. Every figure asserts at least one plotted number.

Figure	Shows	Lives in
L01-sampling-rules-pmf	exact pmf of $\hat{\theta}$: binomial vs negative binomial, $s = 1, 4$, $n = 10$; means 0.256 / 0.461	L01-01-likelihood-sampling-rules §8
L01-poisson-pmf	Poisson pmf for $\lambda = 1, 2, 3$; $\Pr(y = 2 \mid 2) = 0.271$	L01-02-poisson-mle-properties §2
L01-poisson-mle-sampling	MC sampling distribution of $\hat{\lambda}$ at $n = 10, 50, 200$ vs $N(\lambda, \lambda/n)$	L01-02-poisson-mle-properties §8
L01-poisson-running-mean	one running mean inside $\lambda \pm 2\sqrt{\lambda/n}$: consistency pathwise	L01-02-poisson-mle-properties §8
L01-coal-series	yearly counts 1851–1962 with the three MLE lines 1.705 / 3.125 / 0.917	L01-03-coal-mining-change-point §1
L01-coal-profile-likelihood	profile log-likelihood of τ (peak 1891, LR 70.0) and $w(\tau)$ on 1885–1895	L01-03-coal-mining-change-point §5
L01-poisson-prior-posterior	Exp(1/2) prior, normalised likelihood, Gamma(128, 41.5) posterior	L01-04-first-bayesian-poisson §5
L01-mse-four-estimators	exact MSE of the four estimators vs n, $\theta = 3$, log scale	L01-05-comparing-estimators-mse-bootstrap §7
L01-bootstrap-max-pointmass	point mass $1 - (1 - 1/n)^n$; nonparametric vs parametric vs true CDF of $x^*_{(n)}$	L01-05-comparing-estimators-mse-bootstrap §10
L01-r2-vs-criteria	best-subset R^2, $\bar{R}^2$, C_p, BIC on a seeded simulation (true $d = 3$)	L01-06-gaussian-linear-regression §6
L01-overfitting-train-test	Boston: training vs test RMSE and R^2 across interaction orders	L01-06-gaussian-linear-regression §7

Figure	Shows	Lives in
L01-ar1-equilibrium	$y_t \mid y_0 = 8$ converging to $N(0, 5.26)$; conditional variance bounded iff $\lvert\phi\rvert < 1$	L01-07-ar-var-review §6
L01-ar1-paths	one shock sequence through $\phi = 0.5, 0.9, 1$	L01-07-ar-var-review §7
L01-p-vs-bayes-factor	tail areas vs density heights at $X = 1.8$; $p(x)$ vs $P(H_1 \mid x)$	L01-08-p-hacking-evidence §4

Interactive companions

- The Change-Point Dial (`companions/companion-poisson-change-point.html`). Move the **cut year** τ, the **first and last year** of the sample, the **weight window** h and the **confidence** level. Watch the two rate lines with their Wald bands, the profile curve with its marker, and the **LR read-out** (70.0 at $\tau = 1891$, not referred to χ^2_1). The read-outs also give $w(\tau)$ (0.251), the rates, their ratio, and each segment's dispersion index and p-value. Linked from L01-02-poisson-mle-properties §5 and L01-03-coal-mining-change-point §6.
- The Four Estimators (`companions/companion-four-estimators-mse.html`). Move θ, n and a scale c on $x_{(n)}$, with buttons that snap c to the MLE, the unbiased value and the min-MSE value. Watch the four MSE curves, the **ranking** (with its ties at $n = 1$ and $n = 2$), $\mathrm{MSE}(c\,x_{(n)})$ against c, and the bootstrap point mass. The defaults reproduce 2.325, 0.136, 0.300 and 0.075. Linked from L01-05-comparing-estimators-mse-bootstrap §7.

Both pages are verified by `companions/check_companions.py` at three control vectors (both ends and the default) against their own JavaScript run under `node`.

Scope and ceiling

- **Examinable (C):** what the decks and Bolfarine's assigned sections state. That covers the likelihood, MLE and observed/expected information for Bernoulli, binomial, negative binomial, Poisson and uniform; sufficiency, unbiasedness, consistency and Cramér–Rao efficiency; MSE and the four $U(0, \theta)$ estimators; Monte Carlo; the nonparametric and parametric bootstrap; the dispersion test, profile likelihood and GLRT; the Exp–Gamma update; OLS in matrix form, OLS = MLE, R^2, $\bar{R}^2$, C_p, AIC, BIC and best subset; AR(1) moments, estimation, equilibrium law and unit root; AR(p) as a regression; VAR(1) stationarity, mean and covariance; family-wise error, Bonferroni, and Bayes factor vs p-value.
- **Named only (G):** Haldane's estimator and its proof, UMVUE and Rao–Blackwell, invariance and the delta method beyond the deck, the integrated change-point likelihood (L01-03 §8), the best $c\,x_{(n)}$, exact (Clopper–Pearson) intervals, and the Bayesian sample-size criterion.
- **Belongs to later classes, not developed here:** the Gibbs switch-point model (G §7.1.1, Classes 3+4); ridge, lasso, elastic net, normal–gamma and horseshoe (the later pages of `multiplelinearregression.pdf`, Classes 3+4); `hierarchical-beta-binomial.html` (Classes 3+4); Bayes factors for composite hypotheses (Classes 3+4); Dickey–Fuller tests and ARCH/GARCH (Class 8); impulse responses and structural VARs (Class 9); cross-validation (Class 2).
- **Final exam:** no code. Expect derivations and the reading of printed output (L01-09-formulas §0).

Errata found

- **Bolfarine & Sandoval, Table 1.1 (p. 12)** prints $\text{MSE}[x_{(n)}] = \theta^2/662$ at $n = 10$ and $\theta^2/2312$ at $n = 20$. Formula (17) gives $\theta^2/66$ and $\theta^2/231$, and the ratio column should read 0.36, 0.18, 0.059, 0.017. The book's conclusion is unaffected → L01-05-comparing-estimators-mse-bootstrap §5.
- **Handwritten MSE note** (`Class1-comparing-estimators-MSE.pdf`, p. 2) states the ranking "for $n \geq 2$" with strict inequalities. It holds strictly for $n \geq 3$, since at $n = 2$, $\text{MSE}(x_{(n)}) = \text{MSE}(2\bar{x}) = \theta^2/6$. The same page writes the MSE definition as $E\{(\hat{\theta} - E(\theta))^2\}$ → L01-05-comparing-estimators-mse-bootstrap §2, §5.
- **p-hacking deck** (`part1-p-hacking.pdf`, slide 5) counts "$2^5 = 32$" specifications. Its own table of options has $3 \times 3 \times 3 \times 2 \times 3 = 162$ combinations, for which the independent-test rate is 0.9998. The 0.81 is also an upper bound, because the specifications share the same 30 students → L01-08-p-hacking-evidence §3.
- Minor: the handwritten Poisson table rounds $\Pr(y = 3 \mid 2) = 0.18045$ to 0.181 (L01-02-poisson-mle-properties §2). `bootstrap-example.html` prints the σ^2 interval upper limit first (L01-05-comparing-estimators-mse-bootstrap §12.2). The economics deck misnames the authors of Brodeur et al. (2016) (L01-08-p-hacking-evidence §6).

Course-wide position

Class	Relation to Class 1
1	**likelihood, estimators and their sampling distributions; OLS; AR(1)/AR(p)/VAR(1); p-values vs evidence**
2 — statistical learning	MSE becomes **prediction error** and its bias–variance split; L01-06's optimism $2p\sigma^2/n$ becomes **validation and cross-validation**; the iid Bernoulli becomes **logistic regression**; the Poisson rate becomes **Poisson regression** → L02 §1, §3, §4, §6
3+4 — Bayesian learning	the Beta and Gamma updates of L01-01/L01-04 become **conjugate analysis**; the integrated change point becomes **Gibbs sampling** (G 7.3); the Bayes factor of L01-08 becomes **model probabilities**; the prior weight $a/(n + a)$ becomes **shrinkage** (ridge, lasso, horseshoe, hierarchical models)
5+6 — causal inference	L01-06's regression machinery and the abstention example's **IDHM sign flip** become **omitted-variable bias**, IV, DiD and RDD; "association, not a causal effect" becomes the identification question
7 — more econometric models	HC1 standard errors become **WLS/GLS** and Student-t errors; the likelihood toolkit becomes **Tobit and probit**; shrinkage becomes **hierarchical models**

Class	Relation to Class 1
8 — univariate time series	AR(1) becomes **ARMA** with forecasts; the unit root becomes **Dickey–Fuller tests** and long memory; Ljung–Box on squared returns becomes **ARCH/GARCH and stochastic volatility**
9 — multivariate time series	VAR(1) becomes **VAR(p), impulse responses and structural VARs**; the vec covariance and the eigenvalue condition carry over; shrinkage becomes the **Bayesian VAR**

Rules packs: rules/04 (AR, unit root) · rules/05 (VAR) · scope rules/09.

Bibliography actually used in these notes

Book / source	Where
The Class 1 decks and site pages (`Aula/class-01/materials/`) — the spine	everywhere
Bolfarine & Sandoval (2010) §1.1.4–1.1.5, §1.3 (Table 1.1), §2.1–2.2, §3.1–3.2, §3.4, §3.7, §4.4, §6.3, §6.5	the C layer of notes 01–05
Greenberg 1e §2.2 (pp. 12–18), §7.1.1 (pp. 91–95)	notes 01, 03, 04
Wooldridge 6e Ch. 2–3, §4.5 (4.41), §5.1–5.2, §6.3 (6.21)–(6.22), §8.2 (8.4), §11.1, §11.3	notes 06–07
Tsay AFTS 3e §2.1–2.4, §8.2 (Example 8.3)	note 07
ISLP (2023) §5.2 (eq. 5.8), Exercise 5.2	note 05
Izbicki & Mendonça (2020) §1.5.1.1, Ch. 2, §3.1–3.2	note 06
Jarrett (1979) (coal series, via `.agent/notes/coal-mining-data.md`); **Lindley & Phillips (1976)**; **Haldane (1945)**; **Brodeur et al. (2016, 2020)** as cited by the decks	notes 01, 03, 08

Related project material: Aula/COURSE.md (the site's Class 1 list) · Map/class-plan.md (goals, readings, practice) · Map/exercises-index.md (exercise IDs) · rules/04 · rules/05.

Practice (number and printed page only; statements in the books): B 1.4 (p. 13), 1.6 (p. 13), 1.9 (p. 14), 1.12 (p. 14), 2.2 (p. 32), 2.4 (p. 32), 2.10 (p. 33), 2.11 (p. 33), 3.13 (p. 54), 3.17 (p. 55), 4.9 (p. 72), 5.11 (p. 89), 6.11 (p. 116); G 2.2 (p. 19), 3.2 (p. 39), 3.7–3.8 (p. 40), 7.3 (p. 106); I 3.7 (p. 129), 3.9 (pp. 129–130), 5.2 (pp. 224–225), 5.9 (pp. 227–228); T 2.2 (p. 104), 8.1–8.2 (p. 462); W 11.C8 (p. 370), 12.1 (p. 396).

Sampling rules · Same likelihood, different sampling distributions

> **Claim.** Bernoulli trials, a binomial count and a negative-binomial waiting time give **the same likelihood** for θ, up to a constant. So the **MLE, the observed information, the plug-in standard error, the Wald interval and the Bayesian posterior are all identical.** What differs is which of (s, n) is random. That changes the **exact** distribution of $\hat{\theta}$, and with it **bias, exact variance, coverage and p-values.** With $s = 1$ success in $n = 10$ trials, $\hat{\theta} = 0.1$ under both designs. Under inverse sampling, though, its mean is **0.256**, and the one-sided p-value against $\theta = 0.5$ is **0.0020 instead of 0.0107**. Prev: L01-00-map · Next: L01-02-poisson-mle-properties · Map: L01-00-map

Notation.

Symbol	Meaning	Range
θ	success probability	$(0, 1)$
y_i	outcome of trial i (1 = success)	$\{0, 1\}$
n	number of trials	$1, 2, \ldots$ (fixed for Bernoulli/binomial, random N for negative binomial)
s	number of successes, $s = \sum_i y_i$	$0, \ldots, n$ (random S for binomial, fixed for negative binomial)
$L(\theta)$, $\ell(\theta)$	likelihood, log-likelihood	—
$J(\theta) = -\ell''(\theta)$	observed information	> 0
$I(\theta) = E[J(\theta)]$	expected (Fisher) information of the whole sample	> 0
a, b	Beta prior parameters	> 0

---

1 · U — one coin, two experimenters

Two researchers watch the same ten tosses, $y = (0, 0, 0, 0, 0, 0, 0, 0, 0, 1)$. The first had decided in advance to toss **ten times** and count heads. The second had decided to toss **until the first head**, and it happened to arrive on toss ten. The data are identical, the likelihoods are proportional, and the MLE is $1/10$ for both. **But they ran different experiments.** If either of them repeated the experiment, the set of outcomes that *could* have happened would differ: counts $0, 1, \ldots, 10$ for the first, waiting times $1, 2, 3, \ldots$ for the second.

> **Every frequentist property is a statement about outcomes that did not happen.** Bias, standard error, coverage and p-values average over the experiment's sample space. The likelihood only looks at the outcome that did happen. The rest of the deck sorts the quantities into those two kinds.

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2 · C — three sampling rules, one likelihood (slide 1)

Rule	Random quantity	Probability of the data
Bernoulli, n fixed	$Y_1, \ldots, Y_n$ iid $\text{Ber}(\theta)$	$\prod_i \theta^{y_i}(1-\theta)^{1-y_i}$
Binomial, n fixed	$S \sim \text{Bin}(n, \theta)$	$\binom{n}{s}\theta^s(1-\theta)^{n-s}$

Rule	Random quantity	Probability of the data
Negative binomial, s fixed	N = trials until the s-th success	$\binom{n-1}{s-1}\theta^s(1-\theta)^{n-s}$

Collect the Bernoulli product by adding exponents:

$$L(\theta; y) = \theta^{\sum_i y_i}(1-\theta)^{\sum_i (1-y_i)} = \theta^s(1-\theta)^{n-s}. \tag{1}$$

The other two differ from (1) only by a factor that does not contain θ:

$$L(\theta; s) = \binom{n}{s}\theta^s(1-\theta)^{n-s} \;\propto\; \theta^s(1-\theta)^{n-s}, \tag{2}$$

$$L(\theta; n) = \binom{n-1}{s-1}\theta^s(1-\theta)^{n-s} \;\propto\; \theta^s(1-\theta)^{n-s}. \tag{3}$$

The negative-binomial coefficient counts the arrangements in which the s-th success falls **exactly** on trial n: the last trial is a success, and $s-1$ successes fall among the first $n-1$ trials.

> **Likelihood principle (slide 1).** (1), (2) and (3) are in the same equivalence class, so **any likelihood-based or Bayesian inference about θ is the same.** *But* the random quantity is different (S against N), **so the sampling distribution of anything computed from L is different.**

The likelihood here is Bolfarine's (1.3.1) (p. 5) and Def. 3.1.1 (p. 35): the joint probability of the observed sample, read as a function of θ.

---

3 · C — the MLE and the observed information are the same function of the data (slide 2)

Take logs of (1). The constants in (2)–(3) only add $\log\binom{\cdot}{\cdot}$, which vanishes on differentiation:

$$\ell(\theta) = s\log\theta + (n-s)\log(1-\theta). \tag{4}$$

Differentiate (4) (the score):

$$\ell'(\theta) = \frac{s}{\theta} - \frac{n-s}{1-\theta}. \tag{5}$$

Set (5) to zero, which is Bolfarine's likelihood equation (3.1.3), p. 36:

$$\frac{s}{\hat\theta} = \frac{n-s}{1-\hat\theta}$$

$$s(1-\hat\theta) = (n-s)\hat\theta$$

$$s - s\hat{\theta} = n\hat{\theta} - s\hat{\theta}$$

$$\boxed{\hat{\theta} = \frac{s}{n}} \tag{6}$$

Second-order check (B (3.1.4)). Differentiate (5) once more and change sign:

$$J(\theta) = -\ell''(\theta) = \frac{s}{\theta^2} + \frac{n-s}{(1-\theta)^2} > 0. \tag{7}$$

So (6) is a maximum. Evaluate (7) at (6), substituting $\hat{\theta} = s/n$ and $1 - \hat{\theta} = (n-s)/n$:

$$J(\hat{\theta}) = \frac{s}{(s/n)^2} + \frac{n-s}{\big((n-s)/n\big)^2} = \frac{n^2}{s} + \frac{n^2}{n-s} = \frac{n^2(n-s) + n^2 s}{s(n-s)} = \frac{n^3}{s(n-s)}.$$

Divide the numerator and the denominator by n^2:

$$\boxed{J(\hat{\theta}) = \frac{n^3}{s(n-s)} = \frac{n}{\hat{\theta}(1-\hat{\theta})}} \tag{8}$$

At $s = 1$, $n = 10$: $J(\hat{\theta}) = 1000/(1 \cdot 9) = 111.1 = 10/(0.1 \times 0.9)$.

> **What is the same and what is not.** (4)–(8) contain only the numbers (s, n), so they are identical in all three experiments. **Which of** (s, n) **is random** is the only difference (slide 2):

	fixed	random	estimator
Bernoulli / binomial	n	S	$\hat{\theta} = S/n$
Negative binomial	s	N	$\hat{\theta} = s/N$

4 · C — binomial: $\hat{\theta} = S/n$ (slide 3)

Exact distribution. $S = \sum_i Y_i \sim \text{Bin}(n, \theta)$, and $\hat{\theta}$ takes the value k/n exactly when $S = k$:

$$P\Big(\hat{\theta} = \frac{k}{n}\Big) = \binom{n}{k}\theta^k(1-\theta)^{n-k}, \qquad k = 0, 1, \ldots, n. \tag{9}$$

Since $E(S) = n\theta$ and $\text{Var}(S) = n\theta(1-\theta)$ (B §1.1.3, p. 2),

$$E(\hat{\theta}) = \frac{n\theta}{n} = \theta \quad \text{(unbiased)}, \qquad \text{Var}(\hat{\theta}) = \frac{n\theta(1-\theta)}{n^2} = \frac{\theta(1-\theta)}{n}. \tag{10}$$

Expected (Fisher) information. Take the expectation of (7) with S random, substituting $E(S) = n\theta$ and $E(n - S) = n(1-\theta)$:

$$I(\theta) = E\Big[\frac{S}{\theta^2} + \frac{n-S}{(1-\theta)^2}\Big] = \frac{n\theta}{\theta^2} + \frac{n(1-\theta)}{(1-\theta)^2} = \frac{n}{\theta} + \frac{n}{1-\theta} = \frac{n}{\theta(1-\theta)}. \tag{11}$$

The variance (10) equals $1/I(\theta)$ from (11), so S/n attains the Cramér–Rao bound (B Thm 2.1.1, p. 18). **Asymptotics** (B (3.2.3), p. 44; here the CLT and the MLE theory coincide):

$$\hat{\theta} \overset{a}{\sim} N\Big(\theta,\ \frac{\theta(1-\theta)}{n}\Big). \tag{12}$$

Bernoulli or binomial makes no difference. S is sufficient, and the order of the y_i's carries no information about θ.

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5 · C — negative binomial: $\hat{\theta} = s/N$ (slide 4)

Exact distribution. From (3) with n replaced by the random $N = m$:

$$P(N = m) = \binom{m-1}{s-1}\theta^s(1-\theta)^{m-s}, \qquad m = s, s+1, \ldots \tag{13}$$

and $\hat{\theta} = s/m$ exactly when $N = m$, so $P(\hat{\theta} = s/m)$ is (13). **Its support is** $\{1,\ s/(s+1),\ s/(s+2), \ldots\}$**.** It never reaches 0 and piles up near 0.

Fisher information. Now s is fixed and N is random. Use $E(N) = s/\theta$, hence $E(N-s) = s/\theta - s = s(1-\theta)/\theta$. Substitute into (7), with n replaced by N:

$$I(\theta) = E\Big[\frac{s}{\theta^2} + \frac{N-s}{(1-\theta)^2}\Big] = \frac{s}{\theta^2} + \frac{s(1-\theta)/\theta}{(1-\theta)^2} = \frac{s}{\theta^2} + \frac{s}{\theta(1-\theta)}.$$

Put both terms over $\theta^2(1-\theta)$:

$$I(\theta) = \frac{s(1-\theta) + s\theta}{\theta^2(1-\theta)} = \frac{s}{\theta^2(1-\theta)}. \tag{14}$$

Asymptotics ($s \to \infty$):

$$\hat{\theta} \overset{a}{\sim} N\Big(\theta,\ \frac{\theta^2(1-\theta)}{s}\Big). \tag{15}$$

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6 · C — the same answer via the delta method (slide 5)

N is the sum of s iid Geometric(θ) waiting times, each with mean $1/\theta$ and variance $(1-\theta)/\theta^2$. By the CLT,

$$N \overset{a}{\sim} N\Big(\frac{s}{\theta},\ \frac{s(1-\theta)}{\theta^2}\Big). \tag{16}$$

Take $g(x) = s/x$, so $g'(x) = -s/x^2$. Evaluate at the mean $x = s/\theta$:

$$g'\Big(\frac{s}{\theta}\Big) = -\frac{s}{(s/\theta)^2} = -\frac{s\theta^2}{s^2} = -\frac{\theta^2}{s}. \tag{17}$$

The delta method (B (3.2.4), p. 44) gives $\mathrm{Var}(g(N)) \approx g'(\mu)^2\,\mathrm{Var}(N)$. Substitute (17) and the variance in (16):

$$\mathrm{Var}(\hat\theta) \approx \Big(-\frac{\theta^2}{s}\Big)^2 \cdot \frac{s(1-\theta)}{\theta^2} = \frac{\theta^4}{s^2}\cdot\frac{s(1-\theta)}{\theta^2} = \frac{\theta^2(1-\theta)}{s},$$

which is (15) again, and equals $1/I(\theta)$ from (14).

Plug-in check (slide 5 remark). Put $\hat\theta = s/n$ into (15), writing $\theta^2 = (s/n)^2$:

$$\frac{\hat\theta^2(1-\hat\theta)}{s} = \frac{(s/n)^2(1-\hat\theta)}{s} = \frac{s}{n}\cdot\frac{1-\hat\theta}{n} = \frac{\hat\theta(1-\hat\theta)}{n}. \tag{18}$$

(18) is the plug-in variance from (12), and both equal $J(\hat\theta)^{-1}$ from (8).

The plug-in standard errors coincide, so the Wald intervals are identical. The exact distributions (9) and (13) are not. The asymptotic standard error depends only on the likelihood's curvature, and the curvature does not depend on the sampling rule.

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7 · C — bias: binomial against negative binomial (slide 6)

- **Binomial:** $E(S/n) = \theta$ exactly, by (10).
- **Negative binomial:** $1/x$ is strictly convex, so **Jensen** gives

$$E\Big(\frac{s}{N}\Big) > \frac{s}{E(N)} = \frac{s}{s/\theta} = \theta. \tag{19}$$

So $\hat\theta = s/N$ is **biased upward**.

Closed form for $s = 1$ (geometric). Put $s = 1$ in (13): $P(N = m) = \theta(1-\theta)^{m-1}$. Then

$$E\Big(\frac{1}{N}\Big) = \sum_{m\ge1}\frac{1}{m}\,\theta(1-\theta)^{m-1} = \frac{\theta}{1-\theta}\sum_{m\ge1}\frac{(1-\theta)^m}{m}.$$

Use the series $\sum_{m\ge1} x^m/m = -\log(1-x)$ for $|x| < 1$ with $x = 1-\theta$, so $1-x = \theta$:

$$\boxed{E\Big(\frac{1}{N}\Big) = \frac{\theta}{1-\theta}(-\log\theta) = \frac{-\theta\log\theta}{1-\theta}} \tag{20}$$

Worked number at $\theta = 0.1$, with $-\log 0.1 = 2.302585$:

$$E(\hat\theta) = \frac{0.1\times 2.302585}{0.9} = \frac{0.2302585}{0.9} = 0.2558 \;\approx\; \mathbf{0.256}.$$

That is **more than twice θ.**

Haldane (1945). For $s \geq 2$, the estimator $\tilde{\theta} = (s-1)/(N-1)$ is exactly unbiased. *Proof (two lines, not on the slide).* Since $\frac{s-1}{m-1}\binom{m-1}{s-1} = \binom{m-2}{s-2}$,

$$E\Big[\frac{s-1}{N-1}\Big] = \sum_{m \geq s} \binom{m-2}{s-2} \theta^s (1-\theta)^{m-s} = \theta \sum_{m' \geq s-1} \binom{m'-1}{s-2} \theta^{s-1}(1-\theta)^{m'-(s-1)} = \theta \cdot 1,$$

with $m' = m-1$. The last sum is (13) for $s-1$ successes summed over its whole support.

> **Moral (slide 6): unbiasedness is a property of the sampling rule, not of the likelihood.** s/N and S/n are the same function of the data, the same MLE, and only one of them is unbiased.

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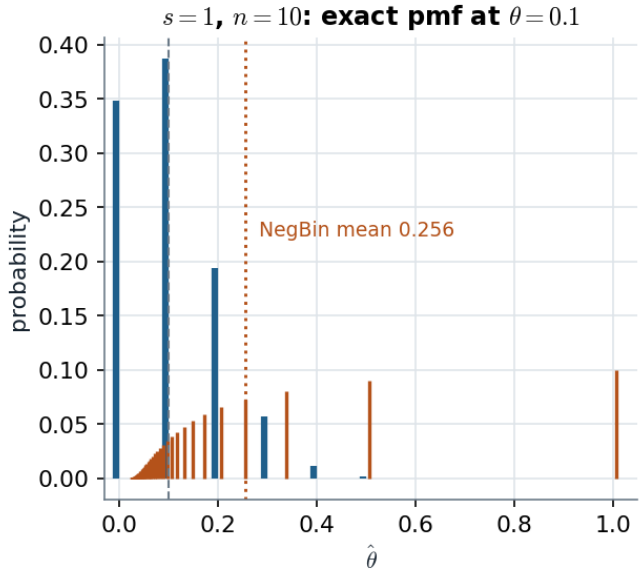
8 · Worked — the deck's two data sets (slides 7–9)

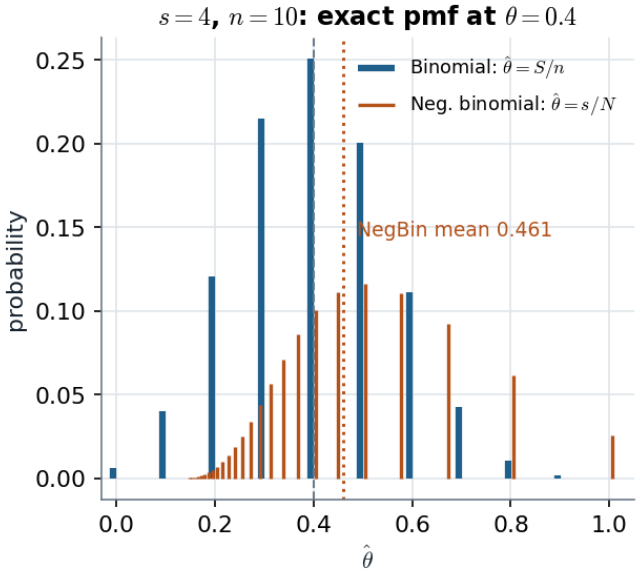
Everything is evaluated at $\theta = \hat{\theta}$. The asymptotic se comes from (18): $\sqrt{0.1 \times 0.9/10} = \sqrt{0.009} = 0.0949$, and $\sqrt{0.4 \times 0.6/10} = \sqrt{0.024} = 0.1549$. The exact moments of s/N are the sums $\sum_m (s/m) P(N=m)$ and $\sum_m (s/m)^2 P(N=m)$ over (13), computed in L01-11-python.

	$s=1, n=10$ Bin	NegBin	$s=4, n=10$ Bin	NegBin
$\hat{\theta}$	0.100	0.100	0.400	0.400
Asymptotic se	0.095	0.095	0.155	0.155
Exact $E(\hat{\theta})$	0.100	**0.256**	0.400	**0.461**
Exact sd$(\hat{\theta})$	0.095	**0.281**	0.155	**0.178**
Wald 95% CI	$(-0.086,\ 0.286)$	same	$(0.096,\ 0.704)$	same

The Wald interval, arithmetic in view:

$0.1 \pm 1.96 \times 0.0949 = 0.1 \pm 0.1859 = (-0.086,\ 0.286)$, $\quad$ $0.4 \pm 1.96 \times 0.1549 = 0.4 \pm 0.3036 = (0.096,\ 0.704)$.





The exact pmf (9) against (13) at $\theta = \hat{\theta}$, $n = 10$. The dotted line is the negative-binomial mean (0.256 and 0.461). Source: `mle_sampling_rules.pdf` *slides 7–8.*

With $s = 1$ the binomial $\hat{\theta}$ lives on the grid $\{0, 0.1, \ldots, 1\}$ and is centred at 0.1. The inverse-sampling $\hat{\theta}$ cannot be 0. It has 10% probability of being exactly 1 (a success on the first trial), and the rest is a comb accumulating near 0. That gives a mean of 0.256 and an sd three times the asymptotic one. With $s = 4$ the two shapes are closer, but the negative binomial is still visibly right-skewed (mean 0.461).

> **Read the table bottom-up.** The asymptotic se is **blind to the sampling rule**; the exact distribution is not. And with $s = 1$ the Wald interval **leaves** $[0, 1]$, so the asymptotics are useless at this sample size whichever rule produced the data (slide 9).

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9 · C — the classic consequence: p-values (slide 10)

Test $H_0 : \theta = 0.5$ against $H_1 : \theta < 0.5$ (Lindley & Phillips, 1976). The p-value is Bolfarine's *nível descritivo* (Def. 6.3.1, p. 100): "o menor nível de significância α para o qual a hipótese nula H_0 seria rejeitada". It is the null probability of a result at least as extreme as the one observed, and **"extreme" is defined by the sample space**:

- **Binomial** ($n = 10$ fixed): extreme means **few successes**, $p = P(S \leq s \mid n, \theta = 0.5)$.
- **Negative binomial** (s fixed): extreme means **many trials**, $p = P(N \geq n \mid s, \theta = 0.5)$.

$s = 1$**,** $n = 10$**.** Binomial:

$$p = P(S \leq 1) = \frac{\binom{10}{0} + \binom{10}{1}}{2^{10}} = \frac{1 + 10}{1024} = \frac{11}{1024} = \mathbf{0.0107}.$$

Negative binomial: $N \geq 10$ means the first 9 trials were all failures:

$$p = P(N \geq 10) = 0.5^9 = \frac{1}{512} = \mathbf{0.0020}.$$

$s = 4$**,** $n = 10$**.** Binomial:

$$p = P(S \leq 4) = \frac{1 + 10 + 45 + 120 + 210}{1024} = \frac{386}{1024} = \mathbf{0.377}.$$

Negative binomial: $N \geq 10$ means **at most 3** successes in the first 9 trials:

$$p = P(\mathrm{Bin}(9, 0.5) \leq 3) = \frac{1 + 9 + 36 + 84}{512} = \frac{130}{512} = \mathbf{0.254}.$$

	$s = 1, n = 10$	$s = 4, n = 10$
Binomial p-value	0.0107	0.377
Neg. binomial p-value	0.0020	0.254

> **Same data, same likelihood, different evidence "against"** H_0, because the reference set of unobserved outcomes depends on the experimenter's **intentions** (slide 10). At the 1% level the first experimenter does not reject and the second does, on identical tosses.

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10 · C — the Bayesian contrast (slide 11; Greenberg §2.2)

With prior $\theta \sim \text{Beta}(a, b)$, density $\propto \theta^{a-1}(1-\theta)^{b-1}$, Bayes' theorem in proportional form (Greenberg (2.3), p. 14) gives, under **any** of (1)–(3):

$$\pi(\theta \mid \text{data}) \;\propto\; \underbrace{c \cdot \theta^{s}(1-\theta)^{n-s}}_{\text{(1),(2) or (3)}} \cdot \theta^{a-1}(1-\theta)^{b-1} \;\propto\; \theta^{a+s-1}(1-\theta)^{b+n-s-1},$$

where $c \in \{1, \binom{n}{s}, \binom{n-1}{s-1}\}$ **cancels in the normalisation**. Hence

$$\theta \mid \text{data} \;\sim\; \text{Beta}(a+s,\; b+n-s), \qquad E(\theta \mid \text{data}) = \frac{a+s}{a+b+n}. \tag{21}$$

This is Greenberg's beta–Bernoulli result (p. 16) and his (2.5) (p. 17). It splits (21) as a weighted average of the prior mean and the MLE:

$$\frac{a+s}{a+b+n} = \frac{a+b}{a+b+n} \cdot \frac{a}{a+b} + \frac{n}{a+b+n} \cdot \frac{s}{n}. \tag{22}$$

Worked (uniform prior, $a = b = 1$). $s = 1$: $\text{Beta}(1+1,\, 1+9) = \text{Beta}(2, 10)$, mean $2/12 = \mathbf{0.167}$. $s = 4$: $\text{Beta}(5, 7)$, mean $5/12 = \mathbf{0.417}$. By (22), with prior mean $1/2$ and weight $2/12$: $\frac{2}{12} \cdot 0.5 + \frac{10}{12} \cdot 0.1 = 0.0833 + 0.0833 = 0.167$. ✓

> **Take-home (slide 11).** The stopping rule is **irrelevant to the posterior**: it enters only through the constants $\binom{n}{s}$ or $\binom{n-1}{s-1}$, which cancel. **Likelihood-based conclusions obey the likelihood principle; the frequentist properties of $\hat{\theta}$ (bias, exact variance, p-values, coverage) do not.**

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11 · Worked — the site's Bernoulli code (`bernoulli-iid-logistic-R.txt`, Part one)

Data $y = (1, 1, 1, 1, 0, 0, 0, 1, 1, 1)$: $n = 10$, $s = 7$.

MLE and Wald intervals. $\hat{\theta} = 7/10 = 0.7$, and by (18) $\text{se} = \sqrt{0.7 \times 0.3/10} = \sqrt{0.021} = 0.1449$:

Level	z	$0.7 \pm z \times 0.1449$	Interval
90%	1.645	0.7 ± 0.2384	(0.462, 0.938)
95%	1.960	0.7 ± 0.2840	(0.416, 0.984)
99%	2.576	0.7 ± 0.3733	(0.327, **1.073**)

The 99% interval again leaves $[0, 1]$.

Bayes with Beta(3,3). The prior mean is 0.5 and the prior sd is $\sqrt{9/(36 \cdot 7)} = 0.189$. By (21), $\text{Beta}(3+7,\, 3+3) = \text{Beta}(10, 6)$, with

$$E(\theta \mid y) = \frac{10}{16} = 0.625, \qquad \text{sd} = \sqrt{\frac{10 \cdot 6}{16^2 \cdot 17}} = \sqrt{0.013787} = 0.117,$$

and 95% credible interval (0.384, 0.837). The posterior predictive probability that the next trial is a success is $\Pr(y_{11} = 1 \mid y) = E(\theta \mid y) = 0.625$. Compared with the MLE 0.7, the estimate is **shrunk toward the prior mean 0.5**, with weight 6/16 on it by (22).

Bridge to Class 2. Part two of the same file gives each trial a regressor x_i and lets $\Pr(y_i = 1 \mid x_i) = 1/(1 + e^{-(\alpha+\beta x_i)})$. The y_i are then **independent but not identically distributed**, (1) no longer applies, and the MLE ($\hat{\alpha} = 2.877$, $\hat{\beta} = 17.995$ in the file's comments) has no closed form. That is the logistic regression of L02 §4.

---

12 · U — how many trials? (`optimal-samplesize.html`)

The Wald margin of error is $E = z_{1-\alpha/2}\sqrt{\theta(1-\theta)/n}$. Solve for n:

$$n \;\geq\; \frac{z_{1-\alpha/2}^2\,\theta(1-\theta)}{E^2}. \tag{23}$$

$\theta(1-\theta) \leq 1/4$, with equality at $\theta = 1/2$ (the parabola's vertex), so the worst case is $n \geq z^2/(4E^2)$. With the page's rounding $z \approx 2$ this becomes $n \geq 1/E^2$: $n \geq 400$ **for** $E = 0.05$**, and** $n \geq 10{,}000$ **for** $E = 0.01$**.** With $z = 1.96$ exactly, (23) gives $1.96^2 \times 0.25/0.05^2 = 384.1$, so $n = 385$. The page's Bayesian alternative (the average-length criterion of M'Lan, Joseph & Wolfson, 2008, with a Beta prior) is **named only**. It belongs with the priors of Classes 3+4.

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13 · G — what the deck does not settle

- **Which is "right"?** Nothing in this deck. The frequentist answer depends on the design by construction. The likelihood principle (Birnbaum, 1962) says it should not. The deck states the conflict and lets the numbers make it.
- **Optional stopping.** "Toss until the p-value drops below 0.05" is a sampling rule with its own sample space. The likelihood and the posterior ignore it, the nominal p-value does not. The p-hacking version of this point is L01-08-p-hacking-evidence.
- **Exact intervals** (Clopper–Pearson for the binomial) repair the $s = 1$ coverage problem. They are named only; they are not in the deck.

Links

- Next: the same MLE checklist (sufficiency, unbiasedness, consistency, efficiency) for the Poisson, L01-02-poisson-mle-properties. Bias as one part of MSE: L01-05-comparing-estimators-mse-bootstrap. p-values against Bayes factors: L01-08-p-hacking-evidence. All results on one page: L01-09-formulas. Traps: L01-10-pitfalls. Code with asserts: L01-11-python.
- Class 2: logistic regression drops the iid assumption, L02 §4.
- Rules: rules/07 (conjugate Beta–Bernoulli); scope rules/09.
- Reading: B §1.3 (pp. 4–6), §2.1 (pp. 15–19), §3.1 (pp. 35–37, Ex. 3.1.2), §3.2.2 (pp. 44–45), §6.3 Def. 6.3.1 (p. 100); G §2.2 (pp. 12–18). Class plan: Class 1.
- Practice (statements in the books; see exercises-index): **B 1.9** (p. 14), **B 2.2** (p. 32), **B 2.10** (p. 33), **B 2.11** (p. 33, the geometric case of §5–§7), **B 3.17** (p. 55).

Poisson MLE · $\hat{\lambda} = \bar{y}$ is sufficient-based, unbiased, consistent and efficient

Claim. For iid Poisson counts the likelihood depends on the data only through $s_n = \sum y_i$, and its maximiser is the **sample mean**. The sample mean passes all four classical tests: **sufficiency** (B Ex. 2.2.3), **unbiasedness**, **consistency** ($V = \lambda/n \to 0$) and **efficiency** (V equals the Cramér–Rao bound λ/n). The sample variance s^2 is also unbiased for λ, but its variance is $\lambda/n + 2\lambda^2/(n-1)$, so it is only $1/(1+2\lambda) = \mathbf{20}\%$ efficient at $\lambda = 2$. In practice it **needs five times the data**. Prev: L01-01-likelihood-sampling-rules · Next: L01-03-coal-mining-change-point · Map: L01-00-map

Notation.

Symbol	Meaning	Range
y_i	count in period i (coal-mining disasters in year i)	$\{0, 1, 2, \dots\}$
n	number of periods	$1, 2, \dots$
λ	occurrence rate, $E(y_i) = V(y_i) = \lambda$	$(0, \infty)$
$s_n = \sum_{i=1}^{n} y_i$	total count	$\{0, 1, \dots\}$
$\bar{y}_n = s_n/n$	sample mean	$[0, \infty)$
$s^2 = \frac{1}{n-1}\sum_i (y_i - \bar{y})^2$	sample variance	$[0, \infty)$
$\ell(\lambda)$, $I(\lambda)$	log-likelihood; expected Fisher information of the whole sample	—
N	Monte Carlo replications	1000

---

1 · C — the Poisson model (slide 2; B §1.1.4, pp. 2–3)

$$y_1, \dots, y_n \mid \lambda \overset{iid}{\sim} \text{Poisson}(\lambda), \qquad \Pr(y_i \mid \lambda) = \frac{\lambda^{y_i} e^{-\lambda}}{y_i!}, \quad \lambda > 0. \tag{1}$$

Mean and variance, derived. Drop the $k = 0$ term, cancel k against $k!$, and use $\sum_{j \geq 0} \lambda^j / j! = e^{\lambda}$:

$$E(y) = \sum_{k \geq 1} k \frac{\lambda^k e^{-\lambda}}{k!} = \lambda e^{-\lambda} \sum_{k \geq 1} \frac{\lambda^{k-1}}{(k-1)!} = \lambda e^{-\lambda} e^{\lambda} = \lambda. \tag{2}$$

The factorial moment works the same way:

$$E[y(y-1)] = \lambda^2 e^{-\lambda} \sum_{k \geq 2} \frac{\lambda^{k-2}}{(k-2)!} = \lambda^2.$$

So $E(y^2) = \lambda^2 + \lambda$. Substitute into $V(y) = E(y^2) - [E(y)]^2$:

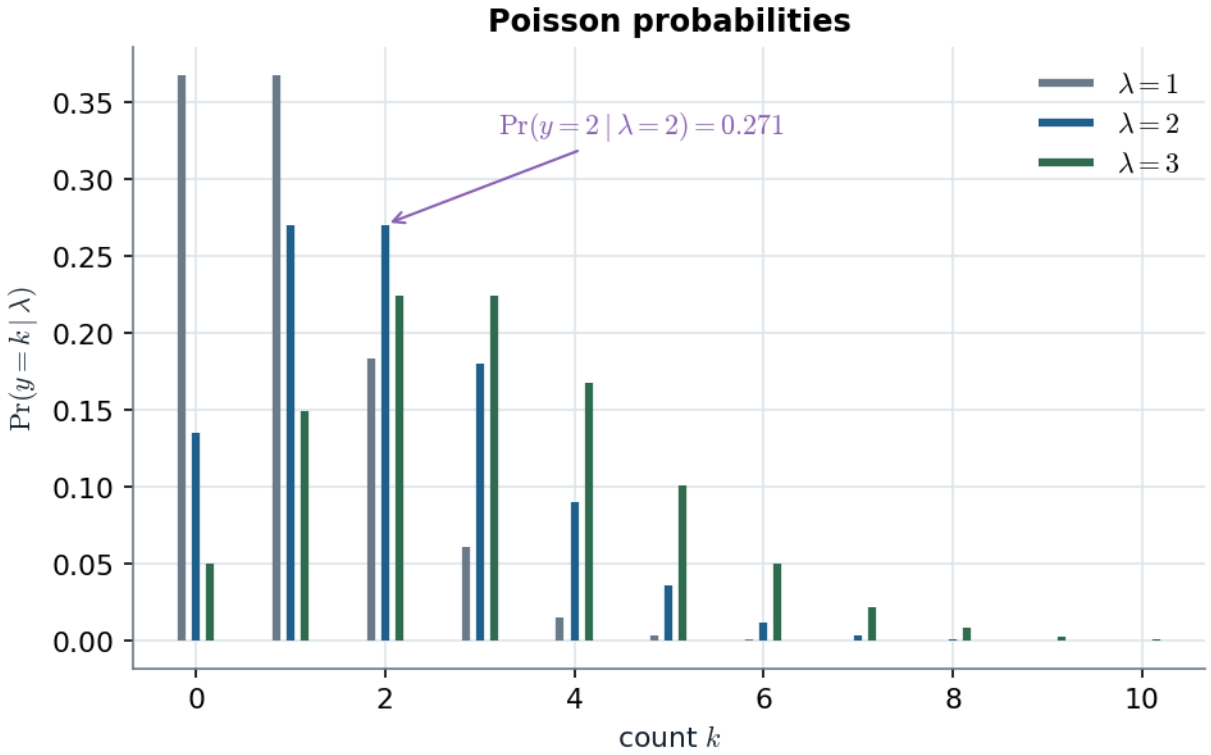
$$V(y) = (\lambda^2 + \lambda) - \lambda^2 = \lambda. \tag{3}$$

Equidispersion, $E = V = \lambda$, is the model's testable fingerprint. It is what the dispersion test of L01-03-coal-mining-change-point §3 checks, and what fails on the full coal series.

2 · Worked — the probabilities on slide 3 (class1-poisson.pdf p. 1; class1-poisson.html)

k	$\lambda = 1$: $e^{-1}/k!$	value	$\lambda = 2$: $2^k e^{-2}/k!$	value
0	0.36788/1	0.368	1×0.13534	0.135
1	0.36788/1	0.368	2×0.13534	0.271
2	0.36788/2	0.184	$(4/2) \times 0.13534$	0.271
3	0.36788/6	0.061	$(8/6) \times 0.13534$	0.180
Sum		**0.981**		**0.857**

So $\Pr(y = 2 \mid \lambda = 2) = 0.271$ and $\Pr(y \le 3 \mid \lambda = 2) = 0.857$, as on the html page. (The handwritten table rounds the last entry to 0.181; the exact value is 0.18045.)



Poisson pmf (1) for $\lambda = 1, 2, 3$; $\Pr(y = 2 \mid \lambda = 2) = 0.271$ is marked. Source: slide 3; `class1-poisson.html` §1.2.

For integer λ the two bars at $k = \lambda - 1$ and $k = \lambda$ are always equal, because $\Pr(k)/\Pr(k-1) = \lambda/k$. As λ grows the mass moves right **and spreads**, which is $V = \lambda$ seen directly.

---

3 · C — likelihood and sufficiency (slide 4; class1-poisson.pdf p. 2; B §2.2)

Multiply (1) over i and collect exponents:

$$p(y_1, \dots, y_n \mid \lambda) = \prod_{i=1}^{n} \frac{\lambda^{y_i} e^{-\lambda}}{y_i!} = \frac{\lambda^{s_n} e^{-n\lambda}}{\prod_{i=1}^{n} y_i!}. \tag{4}$$

As a function of λ:

$$L(\lambda \mid y_1, \dots, y_n) = c\, \lambda^{s_n} e^{-n\lambda}, \qquad c^{-1} = \prod_{i=1}^{n} y_i! \ \text{(constant)}. \tag{5}$$

The likelihood depends on the data only through (s_n, n), so s_n is a sufficient statistic for λ (slide 4). (5) has exactly the form of the Neyman factorisation criterion, B Thm 2.2.1 (2.2.2), p. 22: $h(y) = c$ and $g_\lambda(T) = \lambda^T e^{-n\lambda}$ with $T = s_n$. This is B Ex. 2.2.4.

By the definition (B Def. 2.2.1, p. 20: the conditional distribution of the sample given T must not depend on λ). This is B Ex. 2.2.3, p. 21. A sum of independent Poissons is Poisson, so $T = s_n \sim \text{Poisson}(n\lambda)$. For $\sum_i y_i = t$, divide (4) by $\Pr(T = t)$:

$$P(Y = y \mid T = t) = \frac{\lambda^t e^{-n\lambda} / \prod_i y_i!}{(n\lambda)^t e^{-n\lambda}/t!} = \frac{t!}{\prod_i y_i!} \cdot \frac{\lambda^t}{n^t \lambda^t} = \frac{t!}{y_1! \cdots y_n!} \Big(\frac{1}{n}\Big)^t. \tag{6}$$

(6) is free of λ. It is the multinomial$(t;\, 1/n, \dots, 1/n)$: given the total, the counts are spread over the n years like balls dropped uniformly into n boxes, **which tells nothing about the rate.**

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4 · C — the MLE is the sample mean (slide 5; class1-poisson.pdf p. 3)

Take logs of (5):

$$\ell(\lambda) \equiv \log L(\lambda \mid y) = \log c + s_n \log \lambda - n\lambda. \tag{7}$$

Differentiate (the score):

$$\ell'(\lambda) = \frac{s_n}{\lambda} - n. \tag{8}$$

Set (8) to zero (B (3.1.3), p. 36):

$$\frac{s_n}{\hat\lambda} = n \quad \Longrightarrow \quad \boxed{\hat\lambda_{MLE} = \frac{s_n}{n} = \bar y_n} \tag{9}$$

Is it a maximum? (B (3.1.4)) Differentiate (8) again:

$$\ell''(\lambda) = -\frac{s_n}{\lambda^2} < 0 \quad (s_n > 0), \tag{10}$$

so (9) is a maximum, and indeed the global one, since ℓ is strictly concave.

Edge case (not on the slides). If $s_n = 0$, (7) becomes $\log c - n\lambda$, which increases as $\lambda \downarrow 0$. On the open space $\lambda > 0$ the MLE **does not exist**; it would sit on the boundary. This happens with probability $e^{-n\lambda}$, negligible for the coal data but not for small $n\lambda$.

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5 · C — unbiasedness and consistency (slides 5–6; class1-poisson.pdf pp. 3–4; B §3.7)

Unbiased. Take the expectation of (9) and use (2):

$$E(\hat{\lambda} \mid \lambda) = \frac{1}{n}\sum_{i=1}^{n} E(y_i \mid \lambda) = \frac{1}{n} \cdot n\lambda = \lambda. \tag{11}$$

Variance. By independence and (3):

$$V(\hat{\lambda} \mid \lambda) = \frac{1}{n^2}\sum_{i=1}^{n} V(y_i \mid \lambda) = \frac{n\lambda}{n^2} = \frac{\lambda}{n} \xrightarrow[n\to\infty]{} 0. \tag{12}$$

Unbiased with vanishing variance means MSE $= V + \text{bias}^2 = \lambda/n + 0 \to 0$. So $\hat{\lambda} \to \lambda$ in probability, i.e. $\hat{\lambda}$ **is consistent**.

In Bolfarine's form (Def. 3.7.1, p. 52, $\lim_n P(|\hat{\theta} - \theta| > \epsilon) = 0$), **Chebyshev** (Ex. 3.7.1) with (11)–(12) gives

$$P(|\hat{\lambda} - \lambda| > \epsilon) \le \frac{V(\hat{\lambda})}{\epsilon^2} = \frac{\lambda}{n\epsilon^2} \xrightarrow[n\to\infty]{} 0. \tag{13}$$

CLT (slide 6; B (3.2.3), p. 44):

$$\hat{\lambda}_{MLE} \overset{a}{\sim} N(\lambda, \lambda/n), \qquad \text{SE}(\hat{\lambda}_{MLE}) = \sqrt{\hat{\lambda}_{MLE}/n}. \tag{14}$$

> **Interactive companion.** The Change-Point Dial (`companions/companion-poisson-change-point.html`): each segment's rate is drawn with its Wald band $\hat{\lambda}_j \pm z\sqrt{\hat{\lambda}_j/n_j}$ from (14). Move the **confidence** slider to change z. Move the **cut year** or the **sample window** to change n_j, and the bands widen or narrow like the $\sqrt{1/n}$ in (12).

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6 · C — efficiency: the Cramér–Rao bound (slide 7; class1-poisson.pdf pp. 4–5; B §2.1)

An unbiased estimator is **efficient** if its variance equals the Cramér–Rao lower bound (B Def. 2.1.1, p. 15: $e(\hat{\theta}) = LI(\theta)/\text{Var}[\hat{\theta}]$):

$$\text{CRLB} = \frac{1}{I(\lambda)}, \qquad I(\lambda) = n\, E_{y|\lambda}\Big[\Big(\frac{\partial \log p(y \mid \lambda)}{\partial \lambda}\Big)^2\Big]. \tag{15}$$

Here $I(\lambda)$ is the expected Fisher information of the sample: n times the information of one observation, B (2.1.6), p. 17. The bound itself is B Thm 2.1.1, p. 18, $\text{Var}[\hat{\theta}] \ge 1/(nI_F(\theta))$.

For one Poisson observation, $\log p(y \mid \lambda) = y\log\lambda - \lambda - \log y!$, so

$$\frac{\partial \log p(y \mid \lambda)}{\partial \lambda} = \frac{y}{\lambda} - 1. \tag{16}$$

Substitute (16) into (15) and expand the square:

$$I(\lambda) = n\,E\Big[\Big(\frac{y}{\lambda} - 1\Big)^2\Big] = n\,E\Big[\frac{y^2}{\lambda^2} - \frac{2y}{\lambda} + 1\Big] = n\Big[\frac{1}{\lambda^2}E(y^2) - \frac{2}{\lambda}E(y) + 1\Big].$$

Substitute $E(y^2) = V(y) + [E(y)]^2 = \lambda + \lambda^2 = \lambda(\lambda+1)$ (from (2)–(3)) and $E(y) = \lambda$:

$$I(\lambda) = n\Big[\frac{\lambda(\lambda+1)}{\lambda^2} - \frac{2\lambda}{\lambda} + 1\Big] = n\Big[\frac{\lambda+1}{\lambda} - 1\Big] = \frac{n}{\lambda}. \tag{17}$$

Compare with (12):

$$\boxed{I(\lambda) = \frac{n}{\lambda} = \frac{1}{V(\hat{\lambda}_{MLE} \mid \lambda)}} \quad \Longrightarrow \quad e(\hat{\lambda}_{MLE}) = \frac{\{I(\lambda)\}^{-1}}{V(\hat{\lambda}_{MLE} \mid \lambda)} = \frac{\lambda/n}{\lambda/n} = 1. \tag{18}$$

$\hat{\lambda}_{MLE}$ **is efficient.** Bolfarine's route (Ex. 2.1.2, p. 19) uses the second derivative instead. Differentiate (16): $\partial^2 \log p/\partial\lambda^2 = -y/\lambda^2$, so $E[-y/\lambda^2] = -\lambda/\lambda^2 = -1/\lambda$, and $LI(\lambda) = \lambda/n$. Same bound.

> **The bound checks efficiency; it does not build estimators.** In Bolfarine's words (p. 19), "a desigualdade da informação … não é um método de construção de estimadores". The MLE is the construction, and (18) is the certificate.

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7 · C — a competitor: the sample variance (slide 8)

By (3), $V(y_i) = \lambda$ too, so s^2 is **also unbiased for** λ, since $E(s^2) = \sigma^2 = \lambda$ for any iid sample. Its variance (slide 8) is

$$V(s^2 \mid \lambda) = \frac{\lambda}{n} + \frac{2\lambda^2}{n-1} \;>\; \frac{\lambda}{n} = \text{CRLB}. \tag{19}$$

Where (19) comes from (G, not on the slide). For any iid sample, $V(s^2) = \mu_4/n - \sigma^4(n-3)/[n(n-1)]$, with μ_4 the fourth central moment. Every cumulant of a Poisson equals λ, and $\mu_4 = \kappa_4 + 3\kappa_2^2$, so $\mu_4 = \lambda + 3\lambda^2$. Substitute this and $\sigma^4 = \lambda^2$:

$$V(s^2) = \frac{\lambda + 3\lambda^2}{n} - \frac{\lambda^2(n-3)}{n(n-1)} = \frac{\lambda}{n} + \lambda^2\,\frac{3(n-1) - (n-3)}{n(n-1)} = \frac{\lambda}{n} + \lambda^2\frac{2n}{n(n-1)} = \frac{\lambda}{n} + \frac{2\lambda^2}{n-1}.$$

Efficiency of s^2. Divide the CRLB by (19):

$$e(s^2) = \frac{\lambda/n}{\lambda/n + 2\lambda^2/(n-1)}. \tag{20}$$

Multiply the numerator and the denominator by n/λ:

$$e(s^2) = \frac{1}{1 + 2\lambda\,n/(n-1)} \;\xrightarrow[n\to\infty]{}\; \boxed{\frac{1}{1+2\lambda}} \tag{21}$$

At $\lambda = 2$, (21) gives $1/(1+4) = \mathbf{0.2}$. At $n = 10$, (20) gives $0.2/(0.2 + 8/9) = 0.2/1.0889 = 0.184$.

Reading (21). The bigger the rate, the worse s^2 does. It spends the data estimating spread, which carries information about λ only through the model's $V = E$ restriction, and that information is much noisier than the mean's.

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8 · Worked — the Monte Carlo exercise (slides 8–12)

Design (slide 8). $\lambda = 2$, $N = 1000$ replications, $n \in \{10, 50, 200\}$. For each n: draw $y_1^{(j)}, \dots, y_n^{(j)}$ iid Poisson(2), $j = 1, \dots, N$, and record $\hat{\lambda}_{MLE}^{(j)} = \bar{y}^{(j)}$ and $s^{2(j)}$. (The deck's R uses `matrix(rpois(N*n, lambda), N, n)`, `rowMeans`, `apply(y, 1, var)`; the Python port is in L01-11-python.)

Slide 10, the deck's table. The "theoretical" rows come from (12), (19) and (20); the "MC" rows are the deck's simulation:

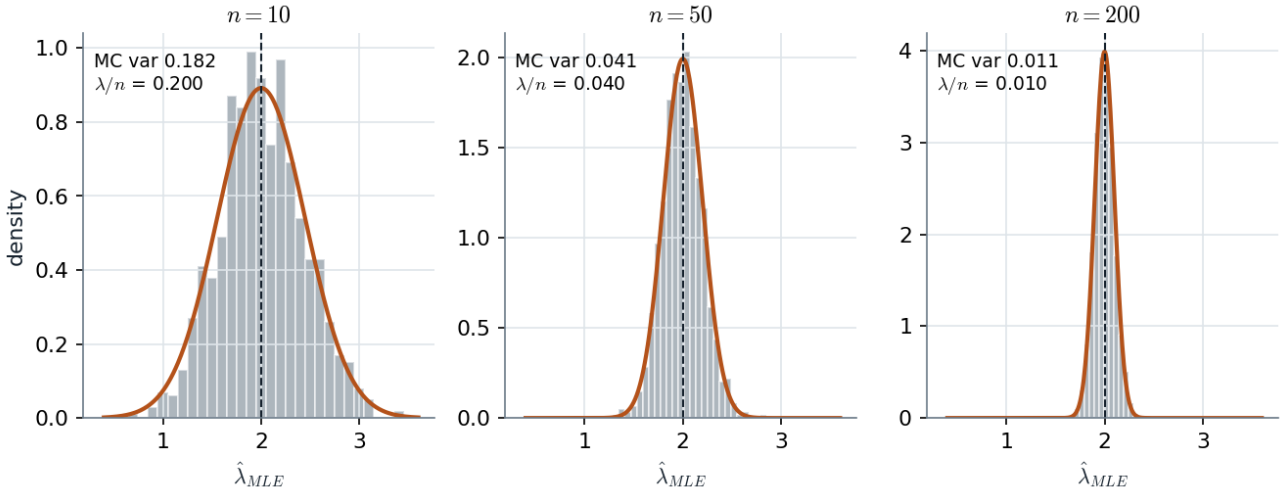
	$\bar{y}$, $n = 10$	50	200	s^2, $n = 10$	50	200
MC mean	2.003	1.994	1.998	2.036	1.990	1.994
MC variance	0.193	0.038	0.011	1.245	0.187	0.051
Theoretical var.	**0.200**	**0.040**	**0.010**	**1.089**	**0.203**	**0.050**
Efficiency (MC)	1	1	1	0.161	0.214	0.197
Efficiency (theory)	1	1	1	**0.184**	**0.197**	**0.199**

Theoretical rows, arithmetic. For $\bar{y}$, (12) gives $2/10 = 0.200$, $2/50 = 0.040$ and $2/200 = 0.010$. For s^2, (19) gives:

$$n = 10: \ 0.2 + \tfrac{8}{9} = 0.2 + 0.8889 = 1.089; \qquad n = 50: \ 0.04 + \tfrac{8}{49} = 0.04 + 0.1633 = 0.203; \qquad n = 200: \ 0.01 + \tfrac{8}{199} = 0.0$$

Efficiency (theory) then follows from (20): $0.2/1.089 = 0.184$, $0.04/0.203 = 0.197$, $0.01/0.050 = 0.199$.

MC efficiency is the CRLB over the MC variance of s^2: $0.2/1.245 = 0.161$, $0.04/0.187 = 0.214$, $0.01/0.051 = 0.196$. That is 0.197 on the slide, where the unrounded variance was used.



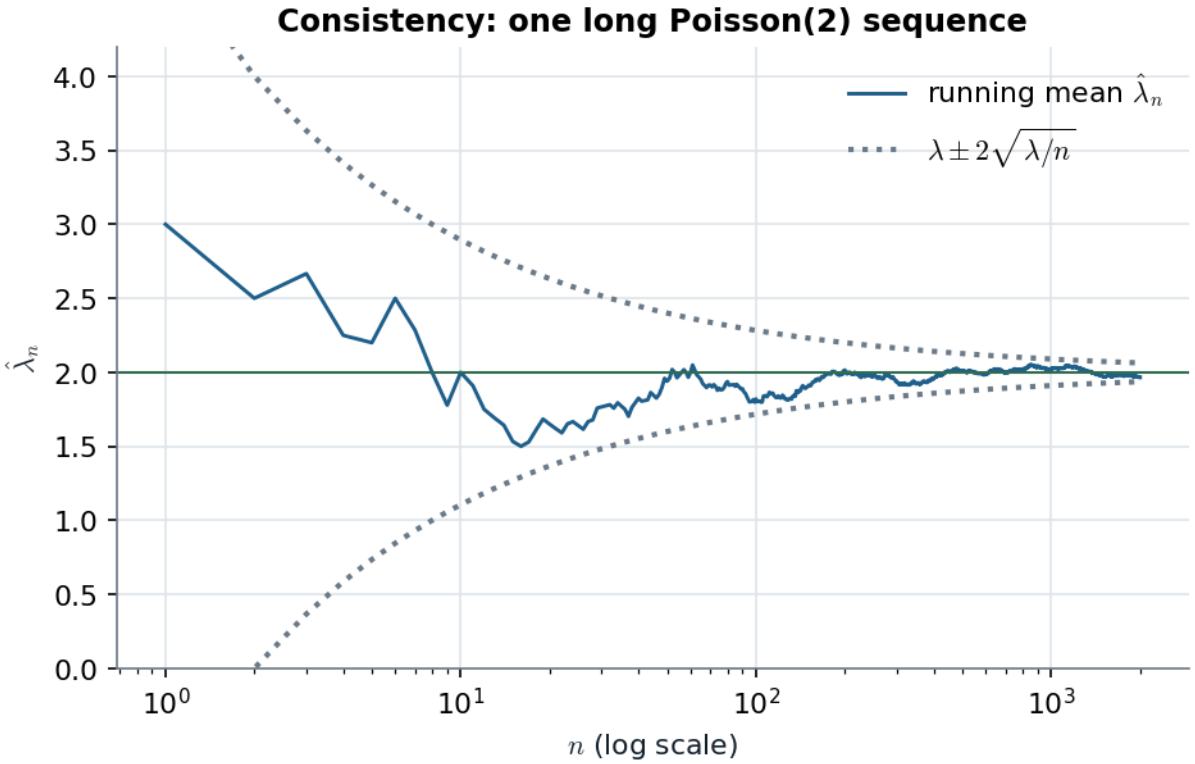
Sampling distribution of $\hat{\lambda}_{MLE}$ (1000 replications, seeded) against the CLT density (14). Source: slide 9; theory (12), (14).

The support is the grid $\{0, 1/n, 2/n, \dots\}$, so the $n = 10$ histogram is lumpy. It is already centred at 2 with the predicted spread. By $n = 200$ the normal curve and the histogram are indistinguishable, and the spread has shrunk by $\sqrt{200/10} = 4.5$. (This figure's own seed gives MC variances 0.182, 0.041 and 0.011; the deck's run gave 0.193, 0.038 and 0.011. Both scatter around λ/n, as Monte Carlo should.)

Slide 12, one long sequence and the coverage of a ±0.25 window. The deck's MC reports $\Pr(|\hat{\lambda} - \lambda| < 0.25) \approx 0.420,\ 0.799,\ 0.984$ for $n = 10, 50, 200$. They can be checked exactly, because $n\hat{\lambda} = s_n \sim \text{Poisson}(n\lambda)$, and approximately with (14):

n	deck MC	exact Poisson($2n$)	normal (14): $2\Phi(0.25/\sqrt{2/n}) - 1$	Chebyshev (13) lower bound $1 - \frac{2}{n(0.25)^2}$
10	0.420	0.424	$2\Phi(0.559) - 1 = 0.424$	$1 - 3.2 = -2.2$ (empty)
50	0.799	0.789	$2\Phi(1.250) - 1 = 0.789$	$1 - 0.64 = 0.36$
200	0.984	0.987	$2\Phi(2.500) - 1 = 0.988$	$1 - 0.16 = 0.84$

Chebyshev proves consistency but is far too crude to quantify it. At $n = 10$ its bound is vacuous, while the true probability is 0.42. Use (14) for numbers, and (13) only for the limit.



Running mean $\hat{\lambda}_n$ along one sequence of 2000 Poisson(2) draws, with $\lambda \pm 2\sqrt{\lambda/n}$ from (12). Source: slide 12.

One path, not a distribution. Early on the running mean jumps with every draw. Later it stays inside the funnel, whose width falls like $1/\sqrt{n}$. This is consistency seen pathwise.

> **Slide 10's verdict.** Both estimators are unbiased; the MLE attains the CRLB λ/n; s^2 **has efficiency** $\approx 1/(1+2\lambda) = 0.2$**, so it needs about 5 times more data to match** $\bar{y}$**.**

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9 · G — beyond the slides (named, not examinable here)

- **Invariance.** The MLE of $g(\lambda)$ is $g(\hat{\lambda})$: e.g. $e^{-\bar{y}}$ for $\Pr(y=0) = e^{-\lambda}$, with $\sqrt{n}(e^{-\bar{y}} - e^{-\lambda}) \xrightarrow{a} N(0, \lambda e^{-2\lambda})$ by (3.2.4) (B Ex. 3.2.5, p. 45).
- **UMVUE.** $\bar{y}$ is the unique minimum-variance unbiased estimator of λ (Lehmann–Scheffé, B §2.5, pp. 29–32). Rao–Blackwellising $1\{y_1 = 0\}$ gives $((n-1)/n)^{s_n}$ for $e^{-\lambda}$, an unbiased competitor to $e^{-\bar{y}}$.
- **Poisson regression.** Let λ depend on covariates through $\log \lambda_i = x_i'\beta$. The score (8) becomes $\sum_i x_i(y_i - \lambda_i)$, and the coefficients read as rate ratios: L02 §6.

Links

- Prev: the same checklist for the Bernoulli, L01-01-likelihood-sampling-rules. Next: the model fails on the full coal series, L01-03-coal-mining-change-point. Then the first prior on λ, L01-04-first-bayesian-poisson; MSE when unbiasedness is not on offer, L01-05-comparing-estimators-mse-bootstrap. Formulas: L01-09-formulas; pitfalls: L01-10-pitfalls; code with asserts: L01-11-python.
- Class 2: the Poisson GLM and the mean–variance check, L02 §6.
- Rules: rules/08 (Monte Carlo integration); scope rules/09.
- Companion: The Change-Point Dial (`companions/companion-poisson-change-point.html`) (CI bands from (14)).
- Reading: B §1.1.4 (pp. 2–3), §2.1 (pp. 15–19, Ex. 2.1.2), §2.2 (pp. 20–23, Ex. 2.2.3–2.2.4), §3.1 (pp. 35–42), §3.2 (pp. 42–45, Ex. 3.2.5), §3.7 (p. 52).
- Practice (statements in the books; see exercises-index): **B 2.4** (p. 32, sufficiency by the definition, Poisson), **B 2.2** (p. 32, CRLB and efficiency), **B 3.13** (p. 54, invariance and the delta method), **B 3.17** (p. 55, consistency), **B 1.4** (p. 13, two unbiased-or-not variance estimators compared by MSE).

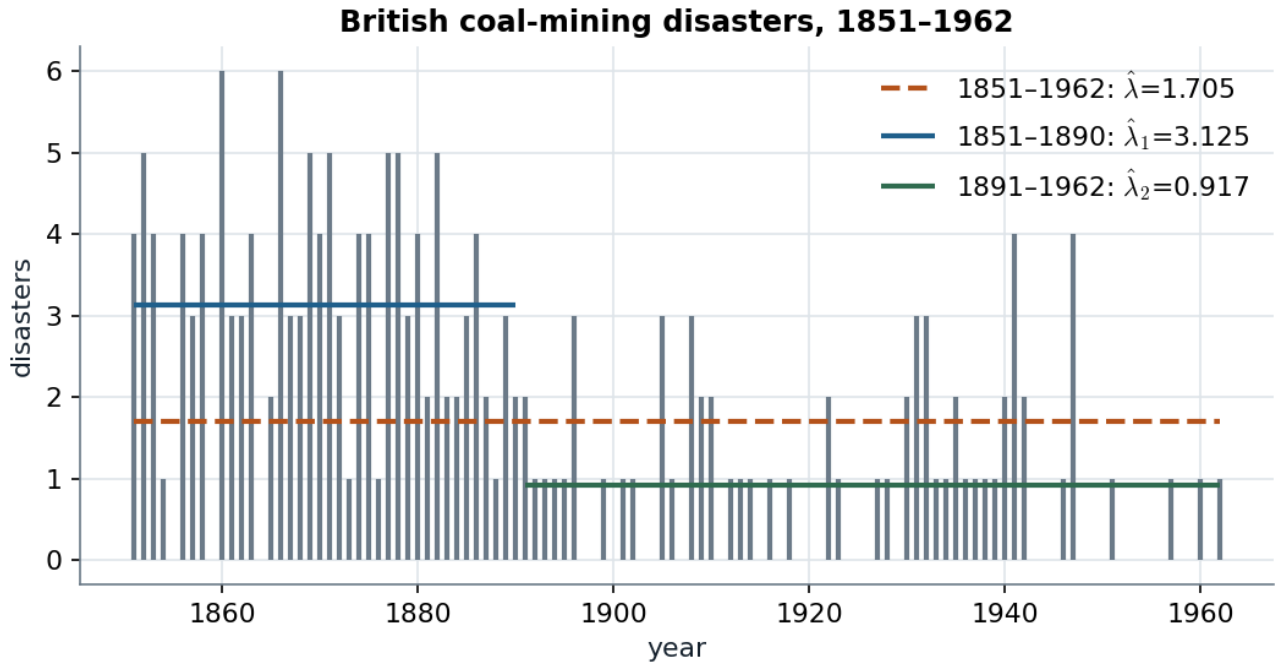
Coal-mining disasters · One Poisson rate does not fit; two do, and the break is around 1890

Claim. The 112 yearly counts have mean 1.705 and variance 2.714. That is **overdispersed** ($s^2/\bar{y} = 1.59$, dispersion test $p < 0.001$), so a single λ is rejected by the model's own $E = V$ fingerprint. Splitting at 1890 gives two series that each **pass** the test: rates **3.125** and **0.917**, with Wald intervals far apart. Profiling the two rates out of a two-regime likelihood puts the cut at $\tau = 1891$ (weight 0.251; 1889–1892 together carry 0.696). The likelihood-ratio statistic against one rate is **70.0**. The cut was **searched for**, so the statistic is not referred to χ^2_1, but no reference distribution rescues the single-rate model. Prev: L01-02-poisson-mle-properties · Next: L01-04-first-bayesian-poisson · Map: L01-00-map

Notation.

Symbol	Meaning	Range / value
y_t	disasters in year t	$\{0, \dots, 6\}$ observed
n, s	years, total disasters (full sample)	112, 191
τ	cut year = **last year of the high-rate regime**	$1851, \dots, 1961$
n_1, s_1 / n_2, s_2	years and disasters up to τ / after τ	$n_1 + n_2 = 112$, $s_1 + s_2 = 191$
λ_1, λ_2	rates before and after the cut	$(0, \infty)$
$\ell_p(\tau)$	profile log-likelihood of τ	—
$w(\tau)$	normalised profile likelihood over a window W of cut years	$[0, 1]$
D	dispersion statistic $(n-1)s^2/\bar{y}$	≥ 0

1 · U — the series (slide 13)



Yearly counts with the three MLEs, $\bar{y}$ over each span. Source: slide 13; data Jarrett (1979), `.agent/notes/coal-mining-data.md`.

The eye does the change-point analysis before the algebra does. Before about 1890, years with three or more disasters are the norm. After it, most years have zero or one. The dashed full-period line at 1.705 sits **between** the two clouds and describes neither. Everything below puts numbers on that impression.

---

2 · Worked — one Poisson per period (slide 14)

Each column applies the MLE $\hat{\lambda} = s_n/n$ and the Wald interval $\hat{\lambda} \pm 1.96\sqrt{\hat{\lambda}/n}$ from L01-02-poisson-mle-properties (9) and (14):

	1851–1962	1851–1890	1891–1962
n	112	40	72
s_n	191	125	66
$\hat{\lambda}_{MLE} = \bar{y}$	$191/112 =$ **1.705**	$125/40 =$ **3.125**	$66/72 =$ **0.917**
SE $= \sqrt{\hat{\lambda}/n}$	$\sqrt{0.015227} = 0.123$	$\sqrt{0.078125} = 0.280$	$\sqrt{0.012731} = 0.113$
95% CI	$1.705 \pm 0.242 =$ (1.464, 1.947)	$3.125 \pm 0.548 =$ (2.577, 3.673)	$0.917 \pm 0.221 =$ (0.696, 1.138)
s^2	2.714	2.574	1.063
$s^2/\bar{y}$	1.592	0.824	1.160
Dispersion test p-value	< 0.001	0.774	0.168

> **The rate falls from about 3.1 to about 0.9 disasters a year, and the two intervals are nowhere near each other** (2.577 against 1.138).

---

3 · C — the dispersion test (slide 14)

Under the Poisson model, $V(y) = E(y) = \lambda$ (L01-02-poisson-mle-properties (2)–(3)), so $s^2/\bar{y}$ should be near 1. The statistic is

$$D = \frac{(n-1)s^2}{\bar{y}} \overset{a}{\sim} \chi^2_{n-1} \quad \text{under the Poisson model.} \tag{1}$$

The logic is the Gaussian fact $(n-1)s^2/\sigma^2 \sim \chi^2_{n-1}$ (B Thm 5.1, p. 73), with $\sigma^2 = \lambda$ replaced by its estimate $\bar{y}$. For counts it is an approximation. The deck reports the **upper-tail** p-value $P(\chi^2_{n-1} \geq D)$, a test against **over**dispersion.

Arithmetic.

$$\text{1851–1962: } D = \frac{111 \times 2.7142}{1.7054} = 176.7, \quad P(\chi^2_{111} \geq 176.7) = 0.00007 < 0.001;$$

$$\text{1851–1890: } D = \frac{39 \times 2.5737}{3.125} = 32.1, \quad P(\chi^2_{39} \geq 32.1) = 0.774;$$

$$\text{1891–1962: } D = \frac{71 \times 1.0634}{0.9167} = 82.4, \quad P(\chi^2_{71} \geq 82.4) = 0.168.$$

Overdispersion is a symptom of a change in rate (slide 21). The full series fails, and each part is consistent with the Poisson.

Why a mixture inflates the variance (law of total variance). Let year t's rate be $\lambda_1 = 3.125$ with weight $\pi = 40/112$ and $\lambda_2 = 0.917$ otherwise. Then

$$V(y) = E[V(y \mid \lambda)] + V[E(y \mid \lambda)] = E(\lambda) + \pi(1-\pi)(\lambda_1 - \lambda_2)^2. \tag{2}$$

Substitute the numbers:

$$V(y) = 1.705 + \frac{40}{112} \cdot \frac{72}{112} (3.125 - 0.917)^2 = 1.705 + 0.3571 \times 0.6429 \times 4.876 = 1.705 + 1.120 = 2.825.$$

The excess over the mean, 1.12, is the between-regime term. The sample confirms the mechanism. Its own split $(n-1)s^2 = \sum_j (n_j - 1)s_j^2 + \sum_j n_j(\bar{y}_j - \bar{y})^2$ is $301.3 = 175.9 + 125.4$, and the **between** part alone, $125.4/111 = 1.13$, is what pushes s^2 from about 1.6 up to 2.714.

Observed against fitted frequencies (slide 15). Under Poisson(1.705) the share of zero years is $e^{-1.705} = 0.182$; the data have $33/112 = 0.295$. The share of years with four or more is 0.094 under the model, against $20/112 = 0.179$ in the data. **Too few zeros and too few large counts is the signature of a mixture of two rates.** The two separate fits have no such gap.

Relative likelihoods (slide 16). $L(\lambda)/L(\hat{\lambda}) = (\lambda/\hat{\lambda})^{s_n} e^{-n(\lambda - \hat{\lambda})}$ from L01-02-poisson-mle-properties (5). Evaluate each part's relative likelihood at the full-period $\hat{\lambda} = 1.705$. For 1851–1890:

$$\log \frac{L_1(1.705)}{L_1(3.125)} = 125 \log \frac{1.705}{3.125} - 40(1.705 - 3.125) = 125(-0.6057) + 56.79 = -18.92,$$

a ratio of 6×10^{-9}. For 1891–1962 the same computation gives -15.81 (1.4×10^{-7}). **The full-period likelihood sits in between, supported by neither** (slide 16).

---

4 · C — the two-regime model and its profile likelihood (slide 17; B §3.4)

$$y_t \mid \lambda_1, \lambda_2, \tau \;\sim\; \begin{cases} \text{Poisson}(\lambda_1), & t = 1851, \dots, \tau, \\ \text{Poisson}(\lambda_2), & t = \tau + 1, \dots, 1962. \end{cases} \tag{3}$$

The log-likelihood is the sum of two Poisson log-likelihoods (L01-02-poisson-mle-properties (7)), one per segment:

$$\ell(\lambda_1, \lambda_2, \tau) = s_1 \log \lambda_1 - n_1 \lambda_1 + s_2 \log \lambda_2 - n_2 \lambda_2 - \sum_t \log y_t! \tag{4}$$

where (n_j, s_j) depend on τ. **Step 1: fix τ and maximise over the rates.** (4) separates, so each rate solves its own likelihood equation:

$$\frac{\partial \ell}{\partial \lambda_j} = \frac{s_j}{\lambda_j} - n_j = 0 \quad \Longrightarrow \quad \hat{\lambda}_j(\tau) = \frac{s_j}{n_j}, \qquad j = 1, 2. \tag{5}$$

Step 2: substitute (5) back into (4). This is Bolfarine's *verossimilhança perfilada* $g(\theta_2; x) = l(\hat{\theta}_1(\theta_2), \theta_2; x)$ (p. 47), with $\theta_2 = \tau$ and $\theta_1 = (\lambda_1, \lambda_2)$:

$$\ell_p(\tau) = s_1 \log \hat{\lambda}_1 - n_1 \Big(\frac{s_1}{n_1}\Big) + s_2 \log \hat{\lambda}_2 - n_2 \Big(\frac{s_2}{n_2}\Big) + \text{const} = s_1 \log \hat{\lambda}_1 - s_1 + s_2 \log \hat{\lambda}_2 - s_2 + \text{const}. \tag{6}$$

Since $s_1 + s_2 = 191$ for every τ, the $-s_j$ terms add to the constant -191. Only $s_1 \log \hat{\lambda}_1 + s_2 \log \hat{\lambda}_2$ moves with τ. The slide writes (6) as $s_1 \log \hat{\lambda}_1 - n_1 \hat{\lambda}_1 + s_2 \log \hat{\lambda}_2 - n_2 \hat{\lambda}_2$ (+ const.), which is the same expression. A segment with $s_j = 0$ uses $0 \log 0 = 0$.

Step 3: normalise over a window W of candidate cuts. The slide uses $W = \{1885, \dots, 1895\}$, "5 years on each side of 1890":

$$w(\tau) = \frac{e^{\ell_p(\tau)}}{\sum_{\tau' \in W} e^{\ell_p(\tau')}}. \tag{7}$$

> **What $w(\tau)$ is (slide 17).** It is the posterior of τ under a **uniform prior on** τ when (λ_1, λ_2) are **profiled out** (plugged in at their MLEs) rather than **integrated out**. §8 shows the integrated version.

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5 · Worked — the cut points around 1890 (slide 18)

One row in full, $\tau = 1891$. $n_1 = 41$, $s_1 = 127$ (the first 41 years); $n_2 = 71$, $s_2 = 191 - 127 = 64$. By (5), $\hat{\lambda}_1 = 127/41 = 3.0976$ and $\hat{\lambda}_2 = 64/71 = 0.9014$, with ratio 3.436. By (6), dropping the constant:

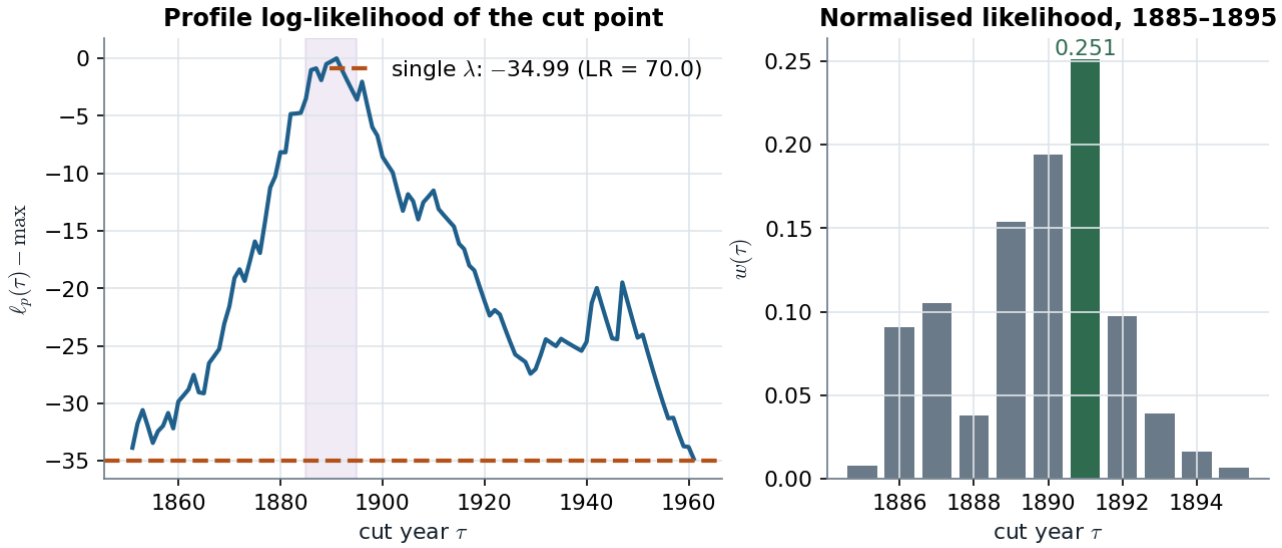
$$\ell_p(1891) = 127\log 3.0976 - 127 + 64\log 0.9014 - 64 = 127(1.13062) - 127 + 64(-0.10380) - 64$$

$$= 143.588 - 127 - 6.643 - 64 = -54.055.$$

This is the maximum over all τ, so $\ell_p(1891) - \max = 0.00$. Every other row is computed the same way. All are reproduced to three decimals in L01-11-python:

τ	n_1/n_2	$\hat{\lambda}_1$	$\hat{\lambda}_2$	$\hat{\lambda}_1/\hat{\lambda}_2$	$\ell_p(\tau) - \max$	$w(\tau)$
1885	35/77	3.229	1.013	3.187	−3.50	0.008
1886	36/76	3.250	0.974	3.338	−1.02	0.091
1887	37/75	3.216	0.960	3.350	−0.87	0.105
1888	38/74	3.158	0.959	3.291	−1.89	0.038
1889	39/73	3.154	0.932	3.386	−0.49	0.154
1890	40/72	3.125	0.917	3.409	−0.26	0.194
1891	**41/71**	**3.098**	**0.901**	**3.436**	**0.00**	**0.251**
1892	42/70	3.048	0.900	3.386	−0.94	0.098
1893	43/69	3.000	0.899	3.339	−1.86	0.039
1894	44/68	2.955	0.897	3.294	−2.74	0.016
1895	45/67	2.911	0.896	3.251	−3.59	0.007

$w(\tau)$, arithmetic. (7) needs only the differences $d_\tau = \ell_p(\tau) - \max$: $w(\tau) = e^{d_\tau} / \sum_{\tau'} e^{d_{\tau'}}$. The sum is $e^{-3.50} + e^{-1.02} + \cdots + e^{-3.59} = 0.030 + 0.362 + 0.420 + 0.150 + 0.613 + 0.772 + 1 + 0.389 + 0.156 + 0.065 + 0.028 = 3.985$ (terms from the unrounded gaps; exponentiating the rounded gaps shown gives 3.983), so $w(1891) = 1/3.985 = 0.251$ and $w(1890) = 0.772/3.985 = 0.194$.



Profile log-likelihood (6) over all admissible cuts (left) and $w(\tau)$ from (7) over the slide's window (right). Source: slides 18–19.

The profile is high and **bumpy** around 1885–1895. Its local dips match single years: a cut just after a year with 0 or 3+ disasters assigns that year to the "wrong" regime, and the profile pays for it (slide

20). Far from the peak it falls 20–35 log-units. The dashed line, the single-rate fit, lies **below every two-regime cut**. That is unavoidable: one rate is the special case $\lambda_1 = \lambda_2$ of (3), so it can never fit better.

> **What the data can and cannot say (slides 19–20).** The maximum is at $\tau = 1891$ with weight 0.251. The years 1889–1892 together carry $0.154 + 0.194 + 0.251 + 0.098 = \mathbf{0.696}$. **The data locate the change around 1890 but cannot pin down the exact year.** Across all eleven cuts the rates barely move ($\hat{\lambda}_1 \approx 3$, $\hat{\lambda}_2 \approx 0.9$): **the rate fell by a factor of about 3.4.**

---

6 · C — the likelihood-ratio statistic (slide 19; B §6.5)

The single-rate fit. With $\hat{\lambda} = 191/112 = 1.70536$ and the same constant as (6):

$$\ell(\hat{\lambda}) = 191 \log 1.70536 - 191 = 191(0.53377) - 191 = 101.951 - 191 = -89.049.$$

The statistic. Twice the gap between the two maximised log-likelihoods:

$$\mathrm{LR} = 2\{\ell_p(1891) - \ell(\hat{\lambda})\} = 2\{-54.055 - (-89.049)\} = 2 \times 34.994 = 69.99 \approx \mathbf{70.0}. \tag{8}$$

A compact form. Write both fits with (6). The $-s_j$ terms and the single-rate -191 cancel, because $s_1 + s_2 = 191$:

$$\mathrm{LR} = 2\Big[s_1 \log \frac{\hat{\lambda}_1}{\hat{\lambda}} + s_2 \log \frac{\hat{\lambda}_2}{\hat{\lambda}}\Big] = 2\big[143.588 - 6.643 - 101.951\big] = 69.99. \tag{9}$$

(9) is the two-sample version of Bolfarine's Poisson GLRT (Ex. 6.5.5, p. 108), where the null value is fixed instead of estimated: $-2\log\lambda(x) = -2[-n(\lambda_0 - \bar{x}) + \sum x_i \log(\lambda_0/\bar{x})]$.

In Bolfarine's terms (6.5.1), p. 103, the GLRT statistic is $\lambda(x) = \sup_{\Theta_0} L / \sup_{\Theta} L$, and $\mathrm{LR} = -2\log\lambda(x)$ with $\Theta_0 = \{\lambda_1 = \lambda_2\}$. **Wilks** (Thm 6.5.1, p. 108) says that under regularity $-2\log\lambda(X) \to \chi^2$, with degrees of freedom equal to the difference in free parameters.

> **Why 70.0 is not referred to χ^2_1.** Wilks needs a regular problem. Here: (i) the cut τ was **chosen to maximise** the statistic over about 111 candidates, so 70.0 is a *maximum* of LR statistics, not one of them; (ii) under H_0 ($\lambda_1 = \lambda_2$) the cut τ **is not identified**, since every τ gives the same likelihood; (iii) τ is discrete. No standard χ^2 count of parameters applies. **It does not matter here.** Had 1890 been fixed in advance (LR 69.47 at $\tau = 1890$), χ^2_1 would be legitimate and $p \approx 8 \times 10^{-17}$. Even a χ^2_2 tail at 70.0 is 6×10^{-16}. The evidence is overwhelming on any reading.
>
> **Interactive companion.** The Change-Point Dial (`companions/companion-poisson-change-point.html`): drag the **cut year** τ and watch the two rate lines, the profile curve's marker and the **LR read-out** (70.0 at $\tau = 1891$; it is not referred to χ^2_1). Change the **weight window** h to see $w(1891)$ move from 0.251 at the slide's $h = 5$. At $\tau = 1890$ the per-segment dispersion read-outs reproduce §2's p-values, 0.774 and 0.168: each part is consistent with a Poisson.

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7 · C — what the slides conclude (slides 20–21)

1. **For iid Poisson data**, $\hat{\lambda}_{MLE} = \bar{y}$ is sufficient-based, unbiased, consistent and efficient (L01-02-poisson-mle-properties).
2. **A single Poisson does not fit the coal-mining series.** Overdispersion (§3) is the symptom of a change in rate.
3. **Splitting gives two well-fitting Poissons**, $\hat{\lambda}_1 \approx 3.1$ and $\hat{\lambda}_2 \approx 0.9$, and the profile likelihood points to a change around 1890–1891.
4. **Next step** (deck's last bullet): a Bayesian change-point model $p(\tau, \lambda_1, \lambda_2 \mid y)$ that **integrates rather than profiles** the rates.

---

8 · G — profiling against integrating (forward link to Classes 3+4)

The model on the slide's last bullet is Greenberg's switch-point model (§7.1.1, pp. 93–94): $p(y_i)$ is Poisson(θ_1) for $i \le k$ and Poisson(θ_2) after, as in his (7.3), with priors $\theta_1 \sim G(\alpha_{10}, \beta_{10})$, $\theta_2 \sim G(\alpha_{20}, \beta_{20})$ and $\pi(k = j) = 1/n$, as in his (7.4). There it is sampled by Gibbs. That belongs to Classes 3+4 and is **not developed here**.

The integrated likelihood of τ in closed form. Use L01-04-first-bayesian-poisson's prior, $\lambda_j \sim$ Exp(a) independently, and that note's Gamma integral (2). For each segment,

$$\int_0^\infty \lambda^{s_j} e^{-n_j\lambda} \cdot a e^{-a\lambda}\, d\lambda = a \int_0^\infty \lambda^{s_j} e^{-(n_j+a)\lambda} d\lambda = \frac{a\,\Gamma(s_j+1)}{(n_j+a)^{s_j+1}},$$

so, with a uniform prior on τ,

$$p(\tau \mid y) \;\propto\; \prod_{j=1}^{2} \frac{a\,\Gamma(s_j+1)}{(n_j+a)^{s_j+1}}. \tag{10}$$

Profiling keeps the **height** of each segment's likelihood at its peak. Integrating keeps its **area**: roughly height times width, where the width is about $\sqrt{2\pi\hat{\lambda}_j/n_j}$ and is weighted by the prior. **On the coal data with $a = 1/2$ the two give the same answer.** (10) normalised over 1885–1895 gives 0.008, 0.089, 0.104, 0.038, 0.153, 0.194, **0.251**, 0.099, 0.040, 0.017, 0.007, against the table's $w(\tau)$. Both segments are long, and the width factor hardly changes across the window. The two routes part ways when a segment is short or the prior is informative, which is when the uncertainty about a rate is itself large. That uncertainty is exactly what profiling ignores.

Links

- Prev: the MLE properties used in every column of §2, L01-02-poisson-mle-properties. Next: the first prior on λ (1851–1891 data), L01-04-first-bayesian-poisson. Formulas: L01-09-formulas; pitfalls (profile ≠ posterior, searched LR ≠ χ^2_1): L01-10-pitfalls; code: L01-11-python.
- Class 2: letting the rate depend on covariates (Poisson regression, mean–variance check), L02 §6.
- Rules: rules/08 (Gibbs, for the Bayesian version); rules/07; scope rules/09.
- Companion: The Change-Point Dial (`companions/companion-poisson-change-point.html`).
- Data provenance: `.agent/notes/coal-mining-data.md`; the site's R page `Aula/class-01/materials/class1-poisson.html` (1851–1891 only).

- Reading: B §3.4 (pp. 46–49, profile likelihood), §6.5 (pp. 103–114, GLRT and Wilks); G §7.1.1 (pp. 91–95, switch-point model).
- Practice (statements in the books; see exercises-index): **B 6.11** (p. 116, approximate GLRT for equal Poisson means of two samples, the fixed-cut version of (9)); **G 7.3** (p. 106, Gibbs for the coal switch point, a preview of Classes 3+4).

First Bayesian example · Exponential prior + Poisson likelihood = Gamma posterior, shrunk toward the prior

Claim. The prior $\lambda \sim \text{Exp}(a)$ with $a = 1/2$ encodes "about 2 disasters a year, variance 4". Multiplied by the Poisson likelihood $\lambda^{s_n} e^{-n\lambda}$, it gives a posterior with the kernel $\lambda^{s_n} e^{-(a+n)\lambda}$, which is **Gamma**$(s_n+1,\ a+n)$. On the 1851–1891 coal data ($n = 41$, $s_n = 127$) the posterior mean is $128/41.5 = \mathbf{3.084}$, the mode $127/41.5 = \mathbf{3.060}$ and the sd $\sqrt{128}/41.5 = \mathbf{0.273}$, against the MLE $127/41 = \mathbf{3.098}$. The prior is weak (it is worth half a year of data), but it still pulls both summaries **a little to the left of the MLE**. That pull is **shrinkage**.
Prev: L01-03-coal-mining-change-point · Next: L01-05-comparing-estimators-mse-bootstrap · Map: L01-00-map

Notation.

Symbol	Meaning	Range / value
λ	Poisson rate (disasters per year)	$(0, \infty)$
a	rate parameter of the exponential prior	$a > 0$; here $1/2$
$y_1, \dots, y_n$	yearly counts	$n = 41$ (1851–1891)
$s_n = \sum y_i$	total count	127
$p(\lambda)$, $p(\lambda \mid y)$	prior, posterior density (Greenberg writes $\pi(\cdot)$)	—
$G(\alpha, \beta)$	Gamma with shape α and **rate** β	$\alpha, \beta > 0$
$\Gamma(\cdot)$	gamma function, $\Gamma(k+1) = k!$	—

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1 · U — some pre-existing knowledge about λ (handwritten p. 1)

The page opens with a sketch. Before seeing the data, the researcher thinks λ is probably around 2–3, surely positive, and unlikely above 5. The Bayesian move is to **write that belief down as a density** and let the data update it. The handwritten notes pick a deliberately simple shape:

> "An easy to handle density is the Exponential density."

$$p(\lambda \mid a) = a\, e^{-a\lambda}, \qquad \lambda > 0. \tag{1}$$

(The handwritten page writes the kernel $e^{-a\lambda}$. The normalising factor a is visible in its p. 4 plot, where the prior starts at $0.5 = a$ at $\lambda = 0$.) The exponential's shape is not the hump of the sketch. Its mode is 0. The choice trades shape for algebra, and §5 shows the price is small here.

One integral does all the work in this note:

$$\int_0^\infty \lambda^k e^{-b\lambda}\, d\lambda = \frac{\Gamma(k+1)}{b^{k+1}}, \qquad k > -1,\ b > 0. \tag{2}$$

Proof. Substitute $u = b\lambda$, so $d\lambda = du/b$: $\int_0^\infty (u/b)^k e^{-u}\, du/b = b^{-(k+1)} \int_0^\infty u^k e^{-u} du = \Gamma(k+1)/b^{k+1}$.

Prior moments. Apply (2) with $b = a$, taking $k = 1$ for the mean and $k = 2$ for the second moment:

$$E(\lambda) = a \cdot \frac{\Gamma(2)}{a^2} = \frac{1}{a}, \qquad E(\lambda^2) = a \cdot \frac{\Gamma(3)}{a^3} = \frac{2}{a^2}, \qquad V(\lambda) = \frac{2}{a^2} - \frac{1}{a^2} = \frac{1}{a^2}. \tag{3}$$

Calibrating a (handwritten p. 1). "If we believe $E(\lambda) \approx 2$ and are happy with $V(\lambda) \approx 4$, then $a = 1/2$ is a reasonable choice." Indeed (3) gives $1/a = 2$ and $1/a^2 = 4$. The prior sd is 2, and the central 95% of the prior runs from $-2\log 0.975 = 0.051$ to $-2\log 0.025 = 7.38$. That is a vague statement about the rate.

---

2 · C — Bayes' theorem (Greenberg §2.2, pp. 13–14)

For a parameter θ and data y, Greenberg's (2.2) is

$$\pi(\theta \mid y) = \frac{f(y \mid \theta)\pi(\theta)}{f(y)}, \qquad f(y) = \int f(y \mid \theta)\pi(\theta)\, d\theta,$$

and, since $f(y)$ does not involve θ, his (2.3) is

$$\pi(\theta \mid y) \;\propto\; f(y \mid \theta)\, \pi(\theta). \tag{4}$$

Posterior ∝ likelihood × prior. The likelihood is "not a p.d.f. for θ" (Greenberg, p. 14). It is the data density read as a function of θ, the same object the MLE maximises in L01-02-poisson-mle-properties (5). The prior supplies what the data do not, and the normaliser $f(y)$ can be recovered at the end from the kernel's family.

---

3 · C — the posterior is Gamma (handwritten p. 2)

"The Bayesian approach allows us to combine different sources of information":

$$\text{Likelihood:} \quad L(\lambda \mid y_1, \dots, y_n) \;\propto\; \lambda^{s_n} e^{-n\lambda}, \tag{5}$$

$$\text{Prior:} \quad p(\lambda) \;\propto\; e^{-a\lambda}.$$

The likelihood is L01-02-poisson-mle-properties (5) without its constant; the prior is (1) without its constant a.

Substitute both into (4):

$$p(\lambda \mid y) \;\propto\; (\lambda^{s_n} e^{-n\lambda})(e^{-a\lambda}) = \lambda^{s_n} e^{-(a+n)\lambda}. \tag{6}$$

Normalise. By (2) with $k = s_n$ and $b = a + n$, the area under the kernel (6) is $\Gamma(s_n + 1)/(a+n)^{s_n+1}$. Divide by it:

$$p(\lambda \mid y) = \frac{(a+n)^{s_n+1}}{\Gamma(s_n+1)}\, \lambda^{(s_n+1)-1} e^{-(a+n)\lambda}. \tag{7}$$

Compare with the Gamma density on the handwritten p. 3, $X \sim G(\alpha, \beta) \iff p(x) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}$: (7) is that density with $\alpha = s_n + 1$ and $\beta = a + n$.

$$\boxed{(\lambda \mid y_1, \ldots, y_n) \ \sim \ G(s_n + 1, \ a + n)} \tag{8}$$

Conjugacy, read off the parameters. The prior $\text{Exp}(a) = G(1, a)$ is a Gamma. The posterior adds the **event count s_n to the shape** and the **exposure n to the rate**. So the prior acts like **1 disaster observed over $a = 0.5$ years** of earlier data. This is Greenberg's reading of the beta prior's α_0, β_0 as prior "heads" and "tails" (p. 17), transposed to counts.

---

4 · C — posterior summaries (handwritten p. 3)

The Gamma facts on the handwritten p. 3 follow from (2) exactly as (3) did:

$$E(X) = \frac{\alpha}{\beta}, \qquad V(X) = \frac{\alpha}{\beta^2}, \qquad \text{Mode}(X) = \frac{\alpha - 1}{\beta} \ \ (\alpha \geq 1). \tag{9}$$

The mode comes from setting the derivative of $\log p(x) = (\alpha-1)\log x - \beta x + \text{const}$ to zero: $(\alpha-1)/x - \beta = 0$.

Substitute $\alpha = s_n + 1$, $\beta = a + n$ from (8) into (9), then put $a = 1/2$:

$$E(\lambda \mid y) = \frac{s_n + 1}{a + n} \overset{a=1/2}{=} \frac{s_n + 1}{n + 1/2}, \tag{10}$$

$$\text{Mode}(\lambda \mid y) = \frac{(s_n + 1) - 1}{a + n} = \frac{s_n}{a + n} \overset{a=1/2}{=} \frac{s_n}{n + 1/2}, \tag{11}$$

$$V(\lambda \mid y) = \frac{s_n + 1}{(a + n)^2} \overset{a=1/2}{=} \frac{s_n + 1}{(n + 1/2)^2}. \tag{12}$$

---

5 · Worked — coal mining, 1851–1891 (handwritten pp. 3–4)

The data are the 41 counts on the site's R page (`class1-poisson.html`): $n = 41$, $s_n = 127$ (mean 3.097561). Substitute into (10)–(12):

$$E(\lambda \mid n, s_n) = \frac{127 + 1}{41 + 0.5} = \frac{128}{41.5} = 3.0843 \approx \mathbf{3.084},$$

$$\text{Mode}(\lambda \mid n, s_n) = \frac{127}{41.5} = 3.0602 \approx \mathbf{3.060},$$

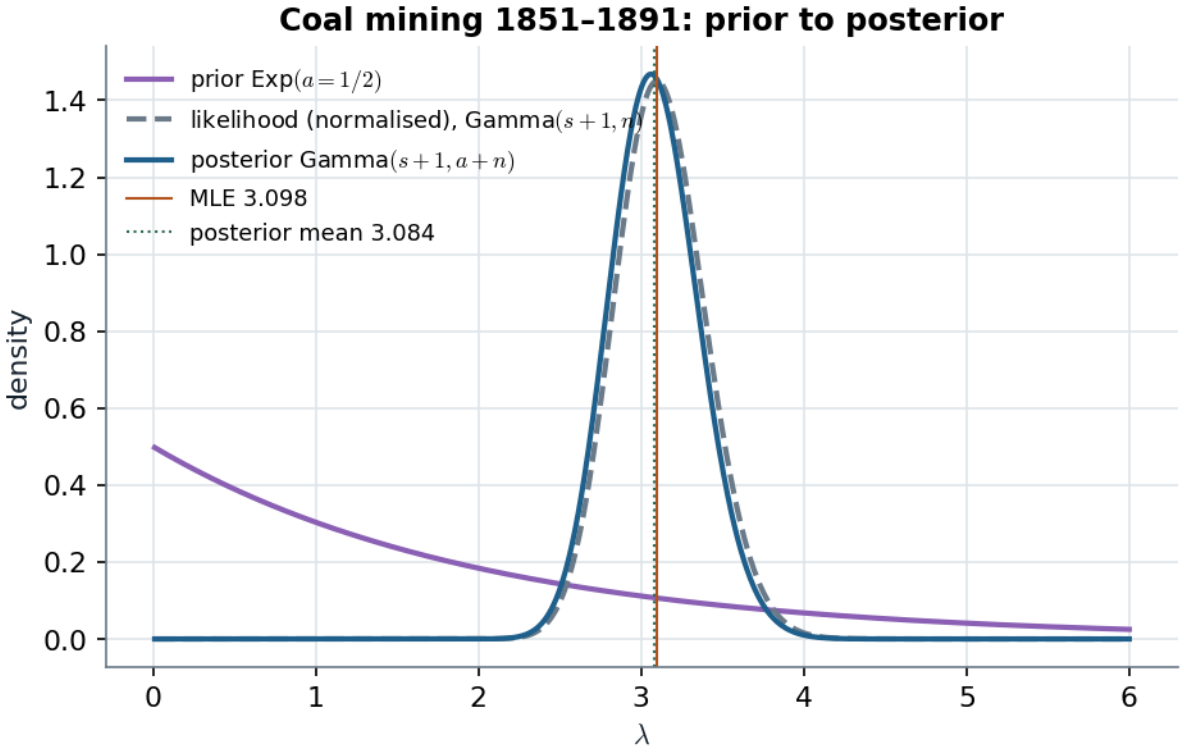
$$\sqrt{V(\lambda \mid n, s_n)} = \frac{\sqrt{128}}{41.5} = \frac{11.3137}{41.5} = 0.2726 \approx \mathbf{0.273}.$$

"Recall that"

$$\hat{\lambda}_{MLE} = \frac{s_n}{n} = \frac{127}{41} = \mathbf{3.098}.$$

	MLE	Posterior mode	Posterior mean
Formula	s_n/n	$s_n/(n+a)$	$(s_n+1)/(n+a)$
Value	3.098	3.060	3.084
Spread	SE $\sqrt{3.098/41} = 0.275$		posterior sd 0.273
95% interval	Wald (2.559, 3.636)		equal-tail (2.573, 3.641)

The last row is computed here, not on the handwritten page: Wald from L01-02-poisson-mle-properties (14), and the posterior interval from the 2.5% and 97.5% quantiles of $G(128, 41.5)$. **Numerically the two intervals almost coincide.** Their *readings* differ: a probability statement about λ given the data, against a coverage statement about the procedure.



Prior (1) with $a = 1/2$, likelihood (5) normalised to a density, and posterior (8) for $n = 41$, $s_n = 127$. Source: handwritten pp. 3–4; Greenberg Fig. 2.2 (p. 18) for the layout.

The posterior is a slightly shifted copy of the normalised likelihood. Over the region where the likelihood lives, roughly 2.3–4, the prior is almost flat (it falls from 0.16 to 0.07), so it can only tilt the likelihood a little toward smaller values. Compare Greenberg's Figure 2.2, where $n = 10$ leaves the prior visible and $n = 50$ buries it.

> **"In this example, the prior information is small when compared to the information in the likelihood. Still $E(\lambda \mid y_1, \ldots, y_n)$ and Mode$(\lambda \mid y_1, \ldots, y_n)$ are a little bit to the left of $\hat{\lambda}_{MLE}$. This is what we call shrinkage."** (handwritten p. 4)

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6 · C — shrinkage as a weighted average

Split (10) the way Greenberg splits the beta-binomial mean in his (2.5) (p. 17):

$$\frac{s_n+1}{n+a} = \underbrace{\frac{a}{n+a}}_{\text{prior weight}} \cdot \underbrace{\frac{1}{a}}_{\text{prior mean}} + \underbrace{\frac{n}{n+a}}_{\text{data weight}} \cdot \underbrace{\frac{s_n}{n}}_{\hat{\lambda}_{MLE}} . \tag{13}$$

Check: $\frac{a}{n+a}\cdot\frac{1}{a} = \frac{1}{n+a}$ and $\frac{n}{n+a}\cdot\frac{s_n}{n} = \frac{s_n}{n+a}$, which add to (10).

Coal numbers. The prior weight is $0.5/41.5 = 0.01205$ and the data weight $41/41.5 = 0.98795$:

$$0.01205 \times 2 + 0.98795 \times 3.09756 = 0.02410 + 3.06024 = 3.08434. \checkmark$$

The **mode** shrinks toward the prior's mode, which is 0:

$$\frac{s_n}{n+a} = \frac{n}{n+a}\cdot\frac{s_n}{n} = 0.98795 \times 3.09756 = 3.06024. \tag{14}$$

> **Both summaries sit left of the MLE because the prior centre (2 for the mean, 0 for the mode) is left of 3.098.** The pull is proportional to the prior weight $a/(n+a)$, which **vanishes like** $1/n$. With 41 years of data and a prior worth half a year, it is about 1%.

Why this matters later. The same "prior weight versus data weight" structure is behind every shrinkage estimator in the course: the normal-mean posterior, ridge, and the hierarchical models of Classes 3+4. There the prior weight is **not** negligible.

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7 · G — links outward (named, not examinable here)

- **Which summary to report?** The posterior mean (10) is the Bayes estimator under quadratic loss (Bolfarine Thm 4.4.1, p. 63). The mode (11) is the maximum a posteriori value. Loss functions and their estimators are Greenberg §3.2.1 (Classes 3+4).
- **General conjugate result.** With a Gama(a, b) prior (B (4.4.12), shape a, rate b), B Ex. 4.4.2 (p. 67) gets $\lambda \mid X \sim \text{Gama}(\sum x_i + a,\ n+b)$ and $E[\lambda \mid X] = (\sum x_i + a)/(n+b)$. Our case is shape 1 and rate a. (**Letter clash:** the handwritten a is Bolfarine's b.)
- **More data do not cure a wrong model.** On all 112 years ($s_n = 191$), (10) gives $192/112.5 = 1.707$, next to the MLE 1.705. Both are confident estimates of a single rate that L01-03-coal-mining-change-point showed does not exist. The prior regularises estimates; it does not test the likelihood.
- **Next.** The posterior predictive for next year's count (a negative binomial; Greenberg §3.2.3), and the change-point posterior that integrates the two rates with these same Gamma integrals (L01-03-coal-mining-change-point §8, (10)), are Classes 3+4.

Links

- Prev: the two-regime model this prior will later be attached to, L01-03-coal-mining-change-point. Next: judging estimators by MSE (bias is allowed if variance falls), L01-05-comparing-estimators-

mse-bootstrap. The Beta–Bernoulli analogue and the likelihood principle: L01-01-likelihood-sampling-rules §10. Formulas: L01-09-formulas; pitfalls: L01-10-pitfalls; code: L01-11-python.

- Rules: rules/07 (prior, posterior, conjugacy); scope rules/09.
- Reading: G §2.2 (pp. 12–18); B §4.3–4.4 (pp. 61–70) as the decision-theory view.
- Practice (statements in the books; see exercises-index): **G 2.2** (p. 19, gamma–Poisson; posterior mean as a weighted average), **G 3.2** (p. 39, Poisson–gamma predictive), **B 4.9** (p. 72), **B 5.11** (p. 89, Bayesian interval with an Exp(1) prior).

MSE, Monte Carlo and the bootstrap · four estimators of a uniform bound, and where resampling breaks

> **Claim. Mean squared error is variance plus squared bias**, so unbiasedness is one ingredient of a good estimator, not the criterion. For $x_i \sim U(0,\theta)$ the exact MSEs rank the four candidates $\hat\theta_4 < \hat\theta_2 < \hat\theta_3 < \hat\theta_1$ for every $n \geq 3$, and the two built on the maximum $x_{(n)}$ (the sufficient statistic and the MLE) converge at rate $1/n^2$ in MSE against $1/n$ for the two built on $\bar x$. **The nonparametric bootstrap reproduces the observed maximum with probability** $1-(1-1/n)^n \to 0.632$, a point mass the true continuous law does not have: it is inconsistent for $x_{(n)}$, and resampling from the fitted model (the parametric bootstrap) repairs it. Prev: L01-04-first-bayesian-poisson · Next: L01-06-gaussian-linear-regression · Map: L01-00-map

Notation.

Symbol	Meaning	Range
$x_1, \dots, x_n$	iid draws from $U(0,\theta)$	$0 < x_i < \theta$
θ	upper bound of the support (the parameter)	$\theta > 0$
n	sample size	$n \geq 1$
$\bar x = \frac{1}{n}\sum_i x_i$	sample mean	
$x_{(1)} \leq \dots \leq x_{(n)}$	order statistics; $x_{(n)} = \max_i x_i$	
$\hat\theta_1, \dots, \hat\theta_4$	$\bar x$, $x_{(n)}$, $2\bar x$, $\frac{n+1}{n}x_{(n)}$	
$E(\cdot), V(\cdot)$	mean and variance over repeated samples, θ fixed	
$B(\hat\theta) = E(\hat\theta) - \theta$	bias	
$\mathrm{MSE}(\hat\theta) = E[(\hat\theta-\theta)^2]$	mean squared error (B: *EQM*, "erro quadrático médio")	≥ 0
N	Monte Carlo replications (deck: 1000)	
B	bootstrap resamples (deck: 1000)	
$x_1^*, \dots, x_n^*$	a bootstrap sample; $x_{(n)}^*$ its maximum	
c	scale in the family $c\,x_{(n)}$ (§8 only)	$c > 0$

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1 · U — why a criterion is needed at all

Four reasonable-looking numbers can be computed from one uniform sample, and they disagree. To choose, we need a loss that does not reward a property for its own sake. The deck's opening question, "Is $\hat\theta_1 = \bar x$ a 'natural choice'?" (slide 7), has a blunt answer:

> **Not quite. Every x_i is at most θ, so $\bar x$ estimates $E(x) = \theta/2$, not θ.** Doubling it removes the bias but doubles the standard deviation. **A better route is to look at the likelihood** — which points to the maximum.

The rest of the note makes "better" precise (MSE), proves the ranking, checks it by simulation, and then asks what happens when only one sample is available (the bootstrap).

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2 · C — mean squared error = variance + bias² (B Def. 1.3.7, (1.3.2), p. 6)

$$\mathrm{MSE}(\hat{\theta}) = E[(\hat{\theta} - \theta)^2] \tag{1}$$

Add and subtract $E(\hat{\theta})$ inside the square:

$$\mathrm{MSE}(\hat{\theta}) = E\Big[\big(\{\hat{\theta} - E(\hat{\theta})\} + \{E(\hat{\theta}) - \theta\}\big)^2\Big]$$

Expand the square:

$$= E[\{\hat{\theta} - E(\hat{\theta})\}^2] + 2\{E(\hat{\theta}) - \theta\}\, E[\hat{\theta} - E(\hat{\theta})] + \{E(\hat{\theta}) - \theta\}^2$$

The middle expectation is $E(\hat{\theta}) - E(\hat{\theta}) = 0$, so

$$\boxed{\mathrm{MSE}(\hat{\theta}) = V(\hat{\theta}) + B(\hat{\theta})^2} \tag{2}$$

and for an unbiased estimator ($B = 0$)

$$\mathrm{MSE}(\hat{\theta}) = V(\hat{\theta}). \tag{3}$$

> **The sentence to keep: an unbiased estimator can lose to a biased one if the bias buys enough variance.** Section 8 shows a biased multiple of $x_{(n)}$ beating the unbiased one.

(The handwritten note writes the definition as $E\{(\hat{\theta} - E(\theta))^2\}$ on p. 2 — a slip; (1) is the definition, and its second line, $V + Bias^2$, is correct.)

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3 · C — the four estimators and their first two moments (slides 2–3)

3.1 · One uniform draw (B §1.1.5, p. 3)

The density is $f(x \mid \theta) = \frac{1}{\theta} I_{(0,\theta)}(x)$ — **its support depends on** θ, which is what makes this model unusual (B, p. 3: "o suporte da variável X … depende do parâmetro θ").

$$E(x) = \int_0^\theta x\, \frac{1}{\theta}\, dx = \frac{1}{\theta} \cdot \frac{\theta^2}{2} = \frac{\theta}{2} \tag{4}$$

$$E(x^2) = \int_0^\theta x^2\, \frac{1}{\theta}\, dx = \frac{1}{\theta} \cdot \frac{\theta^3}{3} = \frac{\theta^2}{3} \tag{5}$$

$$V(x) = E(x^2) - E(x)^2 = \frac{\theta^2}{3} - \frac{\theta^2}{4} = \frac{\theta^2}{12} \tag{6}$$

3.2 · The sample mean

By linearity and (4):

$$E(\bar{x}) = \frac{1}{n}\sum_{i=1}^{n} E(x_i) = \frac{1}{n}\cdot n \cdot \frac{\theta}{2} = \frac{\theta}{2} \tag{7}$$

By independence and (6):

$$V(\bar{x}) = \frac{1}{n^2}\sum_{i=1}^{n} V(x_i) = \frac{1}{n^2}\cdot n \cdot \frac{\theta^2}{12} = \frac{\theta^2}{12n} \tag{8}$$

3.3 · The maximum: $x_{(n)}/\theta \sim \mathrm{Beta}(n,1)$ (slide 3's "useful fact"; B (1.3.14), p. 11)

The maximum is below t exactly when every draw is:

$$P(x_{(n)} \le t) = \prod_{i=1}^{n} P(x_i \le t) = \Big(\frac{t}{\theta}\Big)^n, \qquad 0 \le t \le \theta \tag{9}$$

Differentiating (9) in t gives the density

$$p(x_{(n)} \mid \theta) = \frac{n\, x_{(n)}^{n-1}}{\theta^n}, \qquad 0 \le x_{(n)} \le \theta, \tag{10}$$

so $u = x_{(n)}/\theta$ has density nu^{n-1} on $[0,1]$ — the $\mathrm{Beta}(n,1)$ law. Its moments come from one integral:

$$E(u^k) = \int_0^1 u^k\, nu^{n-1}\, du = \frac{n}{n+k} \tag{11}$$

With $k=1$ and $k=2$ in (11):

$$E(u) = \frac{n}{n+1}, \qquad E(u^2) = \frac{n}{n+2} \tag{12}$$

$$V(u) = \frac{n}{n+2} - \Big(\frac{n}{n+1}\Big)^2 = \frac{n(n+1)^2 - n^2(n+2)}{(n+1)^2(n+2)} = \frac{n\,[\,n^2+2n+1-n^2-2n\,]}{(n+1)^2(n+2)} = \frac{n}{(n+1)^2(n+2)} \tag{13}$$

Scaling back by θ (B (1.3.15), p. 11):

$$E(x_{(n)}) = \frac{n}{n+1}\theta, \qquad V(x_{(n)}) = \frac{n\,\theta^2}{(n+1)^2(n+2)} \tag{14}$$

3.4 · The unbiased versions

$\hat{\theta}_3 = 2\bar{x}$: from (7) and (8), $E = 2 \cdot \frac{\theta}{2} = \theta$ and $V = 4 \cdot \frac{\theta^2}{12n} = \frac{\theta^2}{3n}$.

$\hat{\theta}_4 = \frac{n+1}{n} x_{(n)}$: from (14), $E = \frac{n+1}{n} \cdot \frac{n}{n+1}\theta = \theta$ and

$$V(\hat{\theta}_4) = \Big(\frac{n+1}{n}\Big)^2 \cdot \frac{n\theta^2}{(n+1)^2(n+2)} = \frac{\theta^2}{n(n+2)} \tag{15}$$

Slide 3's table, now derived:

	$\hat{\theta}_1 = \bar{x}$	$\hat{\theta}_2 = x_{(n)}$	$\hat{\theta}_3 = 2\bar{x}$	$\hat{\theta}_4 = \frac{n+1}{n} x_{(n)}$
$E(\hat{\theta})$	$\theta/2$	$\frac{n}{n+1}\theta$	θ	θ
$V(\hat{\theta})$	$\frac{\theta^2}{12n}$	$\frac{n\theta^2}{(n+1)^2(n+2)}$	$\frac{\theta^2}{3n}$	$\frac{\theta^2}{n(n+2)}$
Source	(7), (8)	(14)	§3.4	(15)

---

4 · C — the four MSEs (slide 4; B Example 1.3.8, p. 11)

$\hat{\theta}_1$. Substituting (7) and (8) into (2):

$$\mathrm{MSE}(\hat{\theta}_1) = \frac{\theta^2}{12n} + \Big(\frac{\theta}{2} - \theta\Big)^2 = \frac{\theta^2}{12n} + \frac{\theta^2}{4} = \frac{\theta^2 + 3n\theta^2}{12n} = \frac{1+3n}{12n}\,\theta^2 \tag{16}$$

$\hat{\theta}_2$. Substituting (14) into (2); the bias is $\frac{n}{n+1}\theta - \theta = -\frac{\theta}{n+1}$:

$$\mathrm{MSE}(\hat{\theta}_2) = \frac{n\theta^2}{(n+1)^2(n+2)} + \Big(-\frac{\theta}{n+1}\Big)^2 = \frac{n\theta^2 + (n+2)\theta^2}{(n+1)^2(n+2)} = \frac{2(n+1)\theta^2}{(n+1)^2(n+2)} = \frac{2\theta^2}{(n+1)(n+2)} \tag{17}$$

$\hat{\theta}_3, \hat{\theta}_4$. Unbiased, so by (3):

$$\mathrm{MSE}(\hat{\theta}_3) = \frac{\theta^2}{3n}, \qquad \mathrm{MSE}(\hat{\theta}_4) = \frac{\theta^2}{n(n+2)} \tag{18}$$

> **Two rates, visible in the formulas.** (17) and (18, right) fall like $1/n^2$; (18, left) falls like $1/n$; and (16) does not go to zero at all — it tends to $\theta^2/4$, the squared bias $(\theta/2)^2$ that no amount of data removes (B, p. 12: "quando $n \to \infty$, $EQM[\hat{\theta}_1] \to \theta^2/4$").

---

5 · C — the ranking, proved (slide 5)

All four MSEs are θ^2 times a function of n, so **the ranking does not depend on θ**. Compare them in pairs, dividing out θ^2.

(a) $\hat{\theta}_4$ vs $\hat{\theta}_2$. From (18) and (17):

$$\frac{1}{n(n+2)} < \frac{2}{(n+1)(n+2)} \iff (n+1) < 2n \iff n > 1 \tag{19}$$

(multiply both sides by $n(n+1)(n+2) > 0$).

(b) $\hat{\theta}_2$ vs $\hat{\theta}_3$. From (17) and (18):

$$\frac{2}{(n+1)(n+2)} < \frac{1}{3n} \iff 6n < (n+1)(n+2) = n^2+3n+2 \iff n^2-3n+2 > 0 \iff (n-1)(n-2) > 0 \tag{20}$$

For integer $n \geq 1$ this holds exactly when $n > 2$; at $n = 1$ and $n = 2$ the product is zero, so the two MSEs tie.

(c) $\hat{\theta}_3$ vs $\hat{\theta}_1$. From (18) and (16):

$$\frac{1}{3n} < \frac{1+3n}{12n} \iff 4 < 1+3n \iff n > 1 \tag{21}$$

Chaining (19), (20) and (21):

$$\boxed{n \geq 3: \quad \mathrm{MSE}(\hat{\theta}_4) < \mathrm{MSE}(\hat{\theta}_2) < \mathrm{MSE}(\hat{\theta}_3) < \mathrm{MSE}(\hat{\theta}_1)} \tag{22}$$

The small-n cases, in units of θ^2 (each entry is (16)–(18) evaluated):

n	MSE_1	MSE_2	MSE_3	MSE_4	Ordering
1	$4/12 = 0.3333$	$2/6 = 0.3333$	$1/3 = 0.3333$	$1/3 = 0.3333$	all equal ($\hat{\theta}_1 = \hat{\theta}_2 = x$, $\hat{\theta}_3 = \hat{\theta}_4 = 2x$, and both pairs have the same MSE)
2	$7/24 = 0.2917$	$2/12 = 0.1667$	$1/6 = 0.1667$	$1/8 = 0.1250$	$\hat{\theta}_4 < \hat{\theta}_2 = \hat{\theta}_3 < \hat{\theta}_1$ (slide 5: "At $n = 2$, $\mathrm{MSE}(\hat{\theta}_2) = \mathrm{MSE}(\hat{\theta}_3) = \theta^2/6$")
3	$10/36 = 0.2778$	$2/20 = 0.1000$	$1/9 = 0.1111$	$1/15 = 0.0667$	strict, as in (22)
10	$31/120 = 0.2583$	$2/132 = 0.0152$	$1/30 = 0.0333$	$1/120 = 0.0083$	strict

Watch the boundary. The handwritten note (p. 2) writes "for $n \geq 2$" with strict inequalities; the typeset deck's "$n \geq 3$" is the right statement, because of the $n = 2$ tie in (20).

Erratum in B Table 1.1 (p. 12). The table compares $\bar{x}$ and $x_{(n)}$ only. Its MSE$[\widehat{\theta}_1]$ column ($5\theta^2/18$, $4\theta^2/15$, $31\theta^2/120$, $61\theta^2/240$) agrees with (16), and the $n = 3, 5$ entries of MSE$[\widehat{\theta}_2]$ ($\theta^2/10$, $\theta^2/21$) agree with (17). But (17) gives $2/(11 \cdot 12) = \theta^2/66$ at $n = 10$ and $2/(21 \cdot 22) = \theta^2/231$ at $n = 20$, where the book prints $\theta^2/662$ and $\theta^2/2312$; and the ratio column should read 0.36, 0.18, 0.059, 0.017, not 0.27, 0.12, 0.04, 0.01. The book's conclusion ("$X_{(n)}$ é melhor que $\bar{X}$ para todo θ e $n > 1$") is unaffected. *(All entries recomputed in Python for this note.)*

---

6 · C — the likelihood points to the maximum (slide 8; B Examples 2.2.5, 3.1.3)

The joint density is a product of indicator-weighted constants:

$$p(x_1, \ldots, x_n \mid \theta) = \prod_{i=1}^{n} \frac{1}{\theta}\, \mathbb{1}(x_i \le \theta) = \frac{1}{\theta^n}\, \mathbb{1}(x_{(n)} \le \theta) \tag{23}$$

because all n indicators equal one exactly when the largest observation is below θ. Read as a function of θ:

$$L(\theta \mid x_1, \ldots, x_n) = \frac{1}{\theta^n}\, \mathbb{1}(\theta \ge x_{(n)}) \tag{24}$$

which is **zero for $\theta < x_{(n)}$ and strictly decreasing for $\theta \ge x_{(n)}$**. The maximum sits on the boundary:

$$\hat{\theta}_{MLE} = x_{(n)} = \hat{\theta}_2 \tag{25}$$

> **The score equation does not find it.** $\partial \log L/\partial\theta = -n/\theta$ never equals zero; B (p. 36) flags exactly this case — when the maximum is on the boundary, "o máximo é obtido a partir da inspeção da função de verossimilhança" — and works it as Example 3.1.3 (Figure 3.1, p. 38).

Sufficiency. (23) factorizes as $g(x_{(n)}, \theta) \cdot h(x)$ with $h \equiv 1$ on the sample space, so by the factorization criterion $x_{(n)}$ is sufficient for θ (B Example 2.2.5, p. 23). **That is why the two winners in (22) are functions of $x_{(n)}$:** $\bar{x}$ uses the whole sample but throws away exactly the information that matters — how close the data come to the edge.

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7 · Worked — Monte Carlo at θ = 3, n = 10, N = 1000 (slides 9–12)

Algorithm (slide 9). For $j = 1, \ldots, N$: draw $x_1^{(j)}, \ldots, x_n^{(j)} \sim U(0, \theta)$; compute $\widehat{\theta}_k^{(j)}$, $k = 1, \ldots, 4$. The N values approximate each sampling distribution, and

$$\widehat{\mathrm{MSE}}(\hat{\theta}_k) = \frac{1}{N} \sum_{j=1}^{N} \left(\widehat{\theta}_k^{(j)} - \theta\right)^2. \tag{26}$$

Theory at $\theta = 3$, $n = 10$ — every entry is (7)/(14)/§3.4 for the mean, the square root of (8)/(14)/§3.4/(15) for the SD, and (16)–(18) for the MSE:

	E theory	SD theory	MSE theory
$\hat{\theta}_1$	$3/2 = 1.500$	$3/\sqrt{120} = 0.274$	$\frac{31}{120} \cdot 9 = 2.325$
$\hat{\theta}_2$	$\frac{10}{11} \cdot 3 = 2.727$	$\sqrt{\frac{10 \cdot 9}{121 \cdot 12}} = 0.249$	$\frac{2 \cdot 9}{11 \cdot 12} = 0.136$
$\hat{\theta}_3$	3.000	$3/\sqrt{30} = 0.548$	$9/30 = 0.300$
$\hat{\theta}_4$	3.000	$3/\sqrt{120} = 0.274$	$9/120 = 0.075$

The deck's MC column (slide 12): means 1.497, 2.717, 2.994, 2.989; SDs 0.278, 0.264, 0.555, 0.290; MSEs 2.336, 0.149, 0.308, 0.084. **How close should they be?** The SD of a sample SD from N draws is roughly $\mathrm{SD}/\sqrt{2(N-1)} = \mathrm{SD}/\sqrt{1998} \approx 2.2\%$ of the SD — the deck's "accurate to roughly ±2%". The largest gap (0.290 vs 0.274 for $\hat{\theta}_4$) is 5.8%, about 2.6 such units; the skewed Beta shape makes the max-based SDs noisier than the normal approximation suggests.

Three readings from slide 12:

- $\mathrm{SD}(\hat{\theta}_3) = 2\,\mathrm{SD}(\hat{\theta}_1)$ and $\mathrm{SD}(\hat{\theta}_4) = \frac{n+1}{n}\mathrm{SD}(\hat{\theta}_2) = 1.1\,\mathrm{SD}(\hat{\theta}_2)$ — rescaling an estimator rescales its spread.
- **Curiosity:** at $n = 10$, $12n = 120 = n(n+2)$, so (8) and (15) coincide and $\mathrm{SD}(\hat{\theta}_1) = \mathrm{SD}(\hat{\theta}_4)$ exactly. **Same spread, but $\hat{\theta}_1$ is centred at 1.5:** its MSE is 31 times larger, all of the difference being bias.
- The ranking (22) is reproduced.

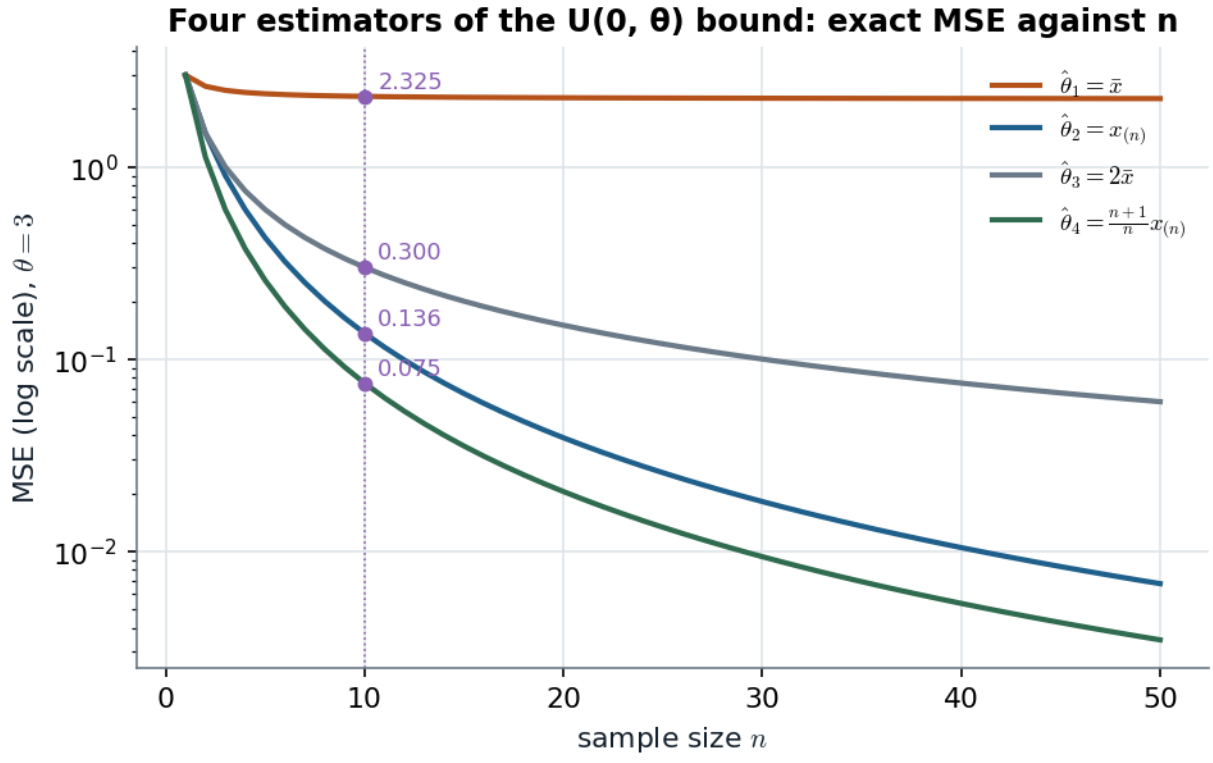


Figure — the four exact MSE curves (16)–(18) at θ = 3, log scale, with the deck's n = 10 values marked. Source: slides 5–6 and 12.

The top curve flattens at $\theta^2/4 = 2.25$: $\bar{x}$ converges, but to the wrong number. The other three fall, the two max-based curves visibly steeper (slope -2 on a log–log reading against -1 for $2\bar{x}$). All four start together at $n = 1$, and the $\hat{\theta}_2$ and $\hat{\theta}_3$ curves touch at $n = 2$ — the two ties of §5.

Interactive companion. The Four Estimators (`companions/companion-four-estimators-mse.html`) — move θ, n and a scale c on $x_{(n)}$; watch the four MSEs, the ranking (with its ties at n = 1 and n = 2) and the min-MSE scale $(n+2)/(n+1)$. Defaults reproduce 2.325, 0.136, 0.300, 0.075.

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8 · G — the best multiple of the maximum is biased

Slide 20 says $\hat{\theta}_4$ "wins for $n \geq 3$" among the four. Among all multiples $c\,x_{(n)}$ it does not. Expand (1) with $\hat{\theta} = c\,x_{(n)}$ and use (12):

$$\mathrm{MSE}(c) = c^2 E(x_{(n)}^2) - 2c\theta E(x_{(n)}) + \theta^2 = \theta^2\Big[c^2\frac{n}{n+2} - 2c\frac{n}{n+1} + 1\Big] \tag{27}$$

Setting $d\,\mathrm{MSE}/dc = 0$:

$$2c\frac{n}{n+2} - 2\frac{n}{n+1} = 0 \iff c^* = \frac{n+2}{n+1} \tag{28}$$

Substituting (28) into (27):

$$\mathrm{MSE}(c^*) = \theta^2\Big[\frac{(n+2)^2}{(n+1)^2}\cdot\frac{n}{n+2} - 2\cdot\frac{n+2}{n+1}\cdot\frac{n}{n+1} + 1\Big] = \theta^2\Big[1 - \frac{n(n+2)}{(n+1)^2}\Big] = \frac{\theta^2}{(n+1)^2} \tag{29}$$

At $\theta = 3$, $n = 10$: $9/121 = 0.0744 < 0.075$. The gain is small, but the lesson of (2) is exact: c^* **shrinks less than the unbiased** $\frac{n+1}{n}$ **would require (**$\frac{12}{11} = 1.0909 < \frac{11}{10}$**), accepting a bias of** $-\theta/(n+1)^2$ **in return for lower variance.** Related practice: B Exercise 1.12 (p. 14).

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9 · C — the nonparametric bootstrap (slides 13–14; I §5.2, pp. 212–214)

Monte Carlo needs θ; in practice we hold **one** sample. The bootstrap replaces the unknown $U(0,\theta)$ by the empirical distribution $\hat{F}_n$ (mass $1/n$ on each observation). For $b = 1, \ldots, B$: draw $x_1^{*b}, \ldots, x_n^{*b}$ **with replacement** from $\{x_1, \ldots, x_n\}$; compute $\hat{\theta}_k^{*b}$. The bootstrap standard error is the SD of the B replicates, ISLP (5.8), p. 214:

$$\mathrm{SE}_B(\hat{\theta}_k) = \sqrt{\frac{1}{B-1}\sum_{b=1}^{B}\Big(\hat{\theta}_k^{*b} - \frac{1}{B}\sum_{b'=1}^{B}\hat{\theta}_k^{*b'}\Big)^2} \tag{30}$$

Slide 14 (one sample of 100 draws from $U(0,3)$; first 10, first 50, all 100):

		$\hat{\theta}_1$	$\hat{\theta}_2$	$\hat{\theta}_3$	$\hat{\theta}_4$
$n = 10$	bootstrap SD	0.117	0.062	0.234	0.068
	theoretical SD	0.274	0.249	0.548	0.274
$n = 50$	bootstrap SD	0.113	0.041	0.226	0.042
	theoretical SD	0.122	0.058	0.245	0.059
$n = 100$	bootstrap SD	0.089	0.036	0.177	0.036

	$\hat{\theta}_1$	$\hat{\theta}_2$	$\hat{\theta}_3$	$\hat{\theta}_4$
theoretical SD	0.087	0.029	0.173	0.030

The theoretical row at $n = 50$ is (8) and (14) at $\theta = 3$: $3/\sqrt{600} = 0.122$; $\sqrt{50 \cdot 9/(51^2 \cdot 52)} = 0.058$; $3/\sqrt{150} = 0.245$; $3/\sqrt{2600} = 0.059$.

Two separate failures:

- $n = 10$**, all four too small.** The ten draws "happen to lie in [1.35, 2.33]" (slide 15): their spread is small, so resampling them gives small spreads. **"The bootstrap cannot recover what the sample does not contain."** This is a small-sample problem and fades with n for $\bar{x}$ (0.113 vs 0.122 at $n = 50$; 0.089 vs 0.087 at $n = 100$).
- $x_{(n)}$**-based, wrong at every** n**, in no fixed direction:** too small at $n = 50$ (0.041 vs 0.058), too large at $n = 100$ (0.036 vs 0.029). At $n = 100$ the boxes "collapse onto $x_{(n)}$, a warning sign" (slide 17). The failure is in the *shape* of the bootstrap law, a point mass at $x_{(n)}$, not in a consistent bias of its SD, and it does not fade: §10.

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10 · C — why the bootstrap fails for the maximum (slide 18)

A bootstrap sample reproduces the observed maximum whenever $x_{(n)}$ is drawn at least once. Each of the n draws misses it with probability $1 - 1/n$, independently:

$$P(x^*_{(n)} \neq x_{(n)}) = \left(1 - \frac{1}{n}\right)^n \tag{31}$$

$$\boxed{P(x^*_{(n)} = x_{(n)}) = 1 - \left(1 - \frac{1}{n}\right)^n \xrightarrow[n\to\infty]{} 1 - e^{-1} \approx 0.632} \tag{32}$$

using $\lim_{n\to\infty}(1 - 1/n)^n = e^{-1}$. The same calculation for any fixed observation j is ISLP Exercise 5.2 (pp. 224–225).

n	(32) exact	Deck's observed share ($B = 1000$)	Gap in MC standard errors $\sqrt{p(1-p)/1000} \approx 0.015$
10	$1 - 0.9^{10} = 0.651$	0.666	+1.0
50	$1 - 0.98^{50} = 0.636$	0.622	−0.9
100	$1 - 0.99^{100} = 0.634$	0.632	−0.1

> **So** $x^*_{(n)}$ **has a point mass of about 63% at** $x_{(n)}$**, while the true sampling law (10), a scaled** Beta$(n, 1)$**, is continuous: the nonparametric bootstrap is inconsistent for the maximum.** No sample size fixes it — the mass converges to 0.632, not to zero.

The root cause is that $\hat{F}_n$ never puts mass above $x_{(n)}$, while the true sampling law (10) puts positive probability on every interval below θ, including the stretch $(x_{(n)}, \theta]$ the data never reached: **the statistic depends on the edge of the support, and the empirical distribution has no information beyond its own edge.** For smooth statistics like $\bar{x}$ this does not matter, and the bootstrap works.

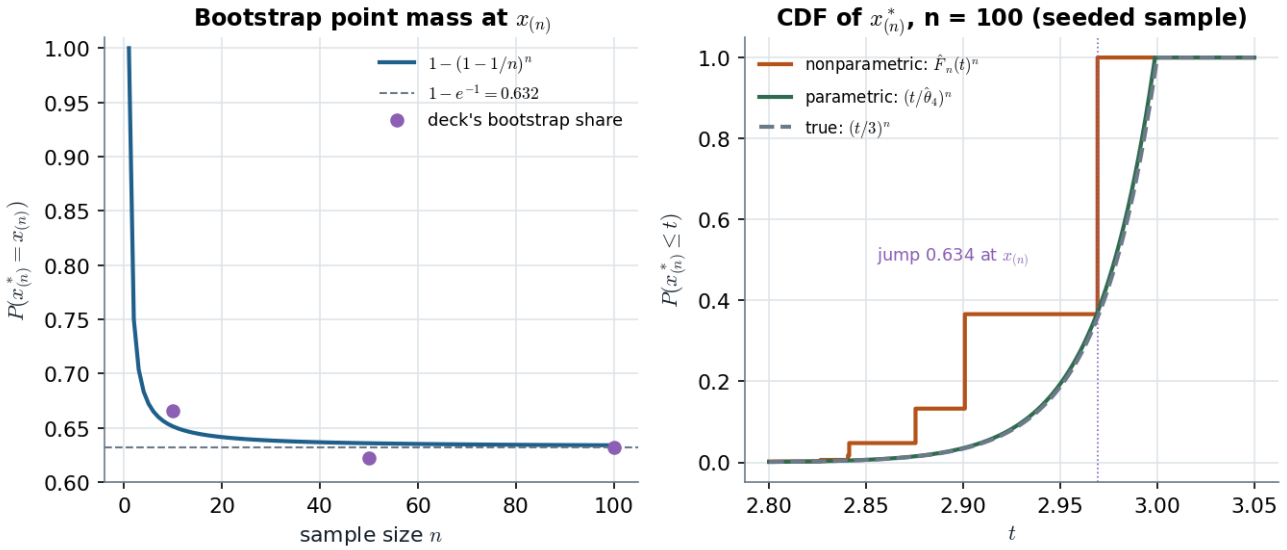


Figure — left: (32) against n with the deck's observed shares (slide 18). Right: exact CDFs of the resampled maximum for one seeded simulated sample ($n = 100$, $x_{(n)} = 2.969$, $\hat{\theta}_4 = 2.999$): nonparametric $\hat{F}_n(t)^n$, parametric $(t/\hat{\theta}_4)^n$, true $(t/3)^n$ — a simulation, not the deck's sample.

The left panel shows the mass settling at 0.632 almost immediately; the deck's three shares scatter around the curve by about one Monte Carlo standard error. On the right, the nonparametric CDF climbs in a handful of steps — one per top order statistic, with heights $((n-j)/n)^n - ((n-j-1)/n)^n$ — and leaps by 0.634 at $x_{(n)}$, whereas the parametric curve hugs the true one: **resampling from the fitted model gives the right shape.**

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11 · C — the fix: parametric bootstrap (slide 19)

Use the model: draw $x_1^*, \dots, x_n^*$ iid $U(0, \hat{\theta}_4)$ instead of resampling the data. With enough resamples this reproduces the **plug-in SD** — the theoretical SD formula evaluated at $\hat{\theta}_4$ instead of θ. **Worked check,** $n = 10$**,** $\hat{\theta}_4 = 2.568$ (slide 14's estimate):

$$\mathrm{SD}(\hat{\theta}_4; \hat{\theta}_4) = \frac{\hat{\theta}_4}{\sqrt{n(n+2)}} = \frac{2.568}{\sqrt{120}} = \frac{2.568}{10.954} = 0.234$$

$$\mathrm{SD}(\hat{\theta}_2; \hat{\theta}_4) = \hat{\theta}_4 \sqrt{\frac{n}{(n+1)^2(n+2)}} = 2.568 \sqrt{\frac{10}{121 \cdot 12}} = 2.568 \times 0.0830 = 0.213$$

		$\hat{\theta}_1$	$\hat{\theta}_2$	$\hat{\theta}_3$	$\hat{\theta}_4$
$n = 10$	parametric SD (deck, slide 19)	0.237	0.213	0.475	0.234
	exact plug-in (formulas at $\hat{\theta}_4 = 2.568$)	0.234	0.213	0.469	0.234
$n = 50$	parametric SD (deck)	0.119	0.054	0.238	0.055

		$\hat{\theta}_1$	$\hat{\theta}_2$	$\hat{\theta}_3$	$\hat{\theta}_4$
$n = 100$	exact plug-in ($\hat{\theta}_4 = 2.926$)	0.119	0.056	0.239	0.057
	parametric SD (deck)	0.084	0.031	0.167	0.032
	exact plug-in ($\hat{\theta}_4 = 3.029$)	0.087	0.030	0.175	0.030

The deck's parametric column is a $B = 1000$ simulation of the plug-in row, and matches it to Monte Carlo error. **At $n = 10$ the parametric SDs (0.213–0.234) are still below theory (0.249–0.274) because $\hat{\theta}_4 = 2.568$ is well below $\theta = 3$ — the deck: "it tracks the theory as long as $\hat{\theta}_4$ is close to θ."** At $n = 50$ and $n = 100$ the max-based SDs (0.054–0.057; 0.030–0.032) now sit on the theoretical 0.058/0.059 and 0.029/0.030, which the nonparametric 0.041 and 0.036 missed in opposite directions.

Bootstrap	Resamples from	Works for $\bar{x}$	Works for $x_{(n)}$	Needs
Nonparametric	$\hat{F}_n$ (the data)	yes, as n grows	**no** — 63% point mass	nothing about the model
Parametric	$U(0, \hat{\theta})$	yes	yes	the model to be right

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12 · C/U — the site's companion pages

12.1 · Mean, weighted mean, median, trimmed mean (`comparing-estimators.html`)

Same question, location instead of scale: estimate a population mean from $n = 21$ draws of $N(0, 1)$, t_4 (mean 0) and Beta(3, 9) (mean $3/12 = 0.25$), with $\hat{\theta}_1 = \bar{x}$; $\hat{\theta}_2 = \sum a_i x_i$ with triangular weights $w_i = i$ ($i \le 11$), $w_i = 22 - i$ ($i \ge 12$), $a_i = w_i / \sum w_j$; the trimmed mean with $k = 2$; and the median $x_{(11)}$. The page shows boxplots of 10,000 replications; two of its results can be computed exactly.

The weighted mean. The data are iid, so the weights attach to arbitrary positions: $\hat{\theta}_2$ is unbiased with $V = \sigma^2 \sum a_i^2$. Here $\sum w_i = 2(1 + \cdots + 10) + 11 = 121$ and $\sum w_i^2 = 2 \cdot 385 + 121 = 891$, so

$$\sum a_i^2 = \frac{891}{121^2} = \frac{891}{14641} = 0.0609 > \frac{1}{21} = 0.0476,$$

a relative efficiency of $0.0476/0.0609 = 0.78$. **Equal weights minimise $\sum a_i^2$ subject to $\sum a_i = 1$ — unequal weights on exchangeable data only add variance.**

The median under heavy tails. For t_4, $V(\bar{x}) = \frac{\nu/(\nu-2)}{n} = \frac{2}{n}$, while the median's large-sample variance is $\frac{1}{4nf(0)^2}$ with $f(0) = 0.375$, i.e. $\frac{1.78}{n}$: **the median beats the mean under t_4**, the reverse of the normal case where its efficiency is $2/\pi = 0.64$. Under Beta(3, 9) the median estimates the *median* (0.236), not the mean, so its MSE carries a bias^2 of $(0.236 - 0.25)^2 \approx 0.0002$ — formula (2) again. A seeded simulation run for this note (20,000 replications, not the page's own run) gives MSEs (mean, weighted, median, trimmed) of 0.047, 0.060, 0.073, 0.050 (normal); 0.093, 0.120, 0.085, 0.070 (t_4); 0.00069, 0.00088, 0.00131, 0.00079 (beta). **The trimmed mean is the robust compromise: near-best under normality, best under t_4.**

12.2 · Bootstrap vs exact intervals (`bootstrap-example.html`)

Fifty listed values, assumed $N(\mu, \sigma^2)$: $\hat{\mu} = \bar{x} = 0.10006$, $\hat{\sigma}^2_{MLE} = 0.8334424$, $\hat{\sigma}_{MLE} = 0.9129307$, MAD $= 0.7508248$ (all reproduced in Python). The exact t interval uses $t_{0.975,49} = 2.00958$:

$$0.10006 \pm 2.00958 \times \frac{0.91293}{\sqrt{50}} = 0.10006 \pm 0.25945 = [-0.15939,\ 0.35951]$$

and the χ^2 interval for σ^2 is $[50(0.83344)/\chi^2_{0.975,49},\ 50(0.83344)/\chi^2_{0.025,49}] = [0.593,\ 1.321]$ — **the page prints it as "1.3206221 0.5934305", upper limit first**, because its code divides by the 0.025 quantile first. (It also plugs $\hat{\sigma}_{MLE}$, divisor n, into the t interval where the textbook statistic uses s, divisor $n-1$ — a 1% difference at $n = 50$.) Its conclusion matches §9: **for μ the 10,000-resample bootstrap interval is almost identical to the exact one; for σ^2 the bootstrap distribution looks Gaussian while the exact one is a skewed χ^2**, and the gap closes as n grows.

12.3 · Correlation MLE (`correlation-bivariate-gaussian.html`) — named

For (x, y) standard bivariate normal with correlation ρ, the page maximises $L(\rho)$ numerically through $\rho = (e^\gamma - 1)/(e^\gamma + 1)$ (an unconstrained γ), simulates $n = 30$ pairs with $\rho = 0.95$, and bootstraps both the MLE and the sample correlation 10,000 times. No numbers are printed; the point is that the two estimators differ (the MLE uses the known unit variances) and that a bounded parameter near its edge gives a skewed bootstrap distribution — a milder cousin of §10.

Links

- Neighbours: L01-04-first-bayesian-poisson (the Poisson MLE whose efficiency §2's criterion formalises: L01-02-poisson-mle-properties) · L01-06-gaussian-linear-regression (MSE returns as prediction error and overfitting).
- Unit: L01-00-map · L01-09-formulas · L01-10-pitfalls · L01-11-python · L01-01-likelihood-sampling-rules · L01-03-coal-mining-change-point.
- Class 2: bias–variance for prediction, L02 §1; bootstrap confidence bands and CV, L02 §3.
- Rules: rules/08 (Monte Carlo); rules/07 for the Bayesian treatment of the same uniform model (Pareto prior, Classes 3+4).
- Companion: The Four Estimators (`companions/companion-four-estimators-mse.html`).
- **Practice (number and printed page only):** B 1.4 (p. 13), B 1.6 (p. 13, the density (10)), B 1.9 (p. 14), B 1.12 (p. 14); I 5.2 (pp. 224–225), I 5.9 (pp. 227–228).

Gaussian linear regression · OLS as a projection and as an MLE, and why R² cannot choose the model

Claim. Least squares is a projection: $\hat{\beta} = (X'X)^{-1}X'y$ solves the normal equations, and $\hat{y} = Py$ is the orthogonal projection of y on the columns of X. **Add Gaussian errors and the same $\hat{\beta}$ is the maximum likelihood estimator**; the MLE of σ^2 divides the residual sum of squares by n, the unbiased one by $n - q$. **R^2 can only rise as regressors are added, so it cannot select a model**: adjusted R^2, C_p, AIC and BIC subtract a price per parameter, and out-of-sample error is the honest referee — in the site's Boston example, the model with $R^2_{in} = 1.0000$ predicts eleven times worse than the plain one. The municipal-abstention regressions show the whole toolkit read on real output: coefficients, HC1 standard errors, joint F tests and a small R^2 gain. Prev: L01-05-comparing-estimators-mse-bootstrap · Next: L01-07-ar-var-review · Map: L01-00-map

Notation.

Symbol	Meaning	Range
n	observations	
p	slopes (predictors); deck slide 13 sets $q = p + 1$ columns with the intercept	
q	columns of X (parameters in β)	$q \le n$
y	$n \times 1$ response	
X	$n \times q$ design matrix, full column rank	
β	$q \times 1$ coefficients	
ε	errors, $E(\varepsilon) = 0$, $V(\varepsilon) = \sigma^2 I$ (Gaussian in §4)	
$\hat{\beta}$, $\hat{y} = X\hat{\beta}$, $e = y - \hat{y}$	OLS estimate, fitted values, residuals	
$P = X(X'X)^{-1}X'$	hat (projection) matrix	
$\mathrm{RSS} = e'e$, TSS	residual and total sums of squares	
d	predictors in a candidate model (§6)	$1 \le d \le p$
$\hat{\sigma}^2$	error-variance estimate (§4; in §6 from the full model)	
k	Wooldridge's slope count ($k = p$) in (6.21)–(6.22)	

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1 · U — one covariate is rarely enough (slides 4–12)

The deck starts from what the class "already studied": $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$ with $E(\varepsilon_i) = 0$, $V(\varepsilon_i) = \sigma^2$, $\mathrm{Cov}(x_i, \varepsilon_i) = 0$, and OLS estimates

$$\hat{\beta}_1 = \frac{\sum_i (y_i - \bar{y})(x_i - \bar{x})}{\sum_i (x_i - \bar{x})^2}, \qquad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} \tag{1}$$

(W §2.2, pp. 24–32). The houseprice data ($n = 128$ sales; price, size, offers, bedrooms, bathrooms, brick, neighbourhood) show why one covariate is not the end (slide 11): price on size alone gives $R^2 = 31\%$ with residual SD 71.4; letting the line shift by brick/non-brick, 47% and 62.1; by neighbourhood, **69% and 48.1**. **The same size–price slope explains far more once the groups it was averaging over are separated** — the multiple-regression idea, "holding other factors fixed" (W §3.1, pp. 61–64).

Centring and standardising (slide 5). Replace y_i, x_i by $\tilde{y}_i = (y_i - \bar{y})/s_y$, $\tilde{x}_i = (x_i - \bar{x})/s_x$ with $s^2 = \frac{1}{n}\sum(\cdot - \text{mean})^2$. The intercept in (1) becomes $\hat{\beta}_0 = 0 - \hat{\beta}_1 \cdot 0 = 0$, and since $\sum_i \tilde{x}_i^2 = n$,

$$\hat{\beta}_1 = \frac{\sum_i \tilde{y}_i \tilde{x}_i}{\sum_i \tilde{x}_i^2} = \frac{\tilde{x}'\tilde{y}}{\tilde{x}'\tilde{x}} = \frac{1}{n}\sum_i \tilde{x}_i \tilde{y}_i = r_{xy} : \tag{2}$$

on standardised data the slope is the sample correlation, and its form $(\tilde{x}'\tilde{x})^{-1}\tilde{x}'\tilde{y}$ already previews the matrix formula.

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2 · C — matrix form and the normal equations (slides 12–14; W App. E)

Stack the n equations $y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip} + \varepsilon_i$:

$$y = X\beta + \varepsilon, \qquad \varepsilon \sim (0, \sigma^2 I_n). \tag{3}$$

OLS minimises the residual sum of squares

$$\text{RSS}(\beta) = (y - X\beta)'(y - X\beta) = y'y - 2\beta' X'y + \beta' X'X\beta \tag{4}$$

(the two cross terms $\beta' X'y$ and $y'X\beta$ are the same scalar). Slide 14 writes the β-dependent part, $\beta' X'X\beta - 2\beta' X'y$. Differentiate (4), using $\partial(\beta' a)/\partial\beta = a$ and $\partial(\beta' A\beta)/\partial\beta = 2A\beta$ for symmetric A:

$$\frac{\partial \,\text{RSS}}{\partial \beta} = -2X'y + 2X'X\beta \tag{5}$$

Setting (5) to zero gives the **normal equations**

$$X'X\hat{\beta} = X'y, \tag{6}$$

and, when X has full column rank so that $X'X$ is invertible,

$$\boxed{\hat{\beta}_{ols} = (X'X)^{-1}X'y} \tag{7}$$

It is the global minimum, without second derivatives. For any β write $y - X\beta = (y - X\hat{\beta}) + X(\hat{\beta} - \beta)$ and expand:

$$\text{RSS}(\beta) = \text{RSS}(\hat{\beta}) + 2(\hat{\beta} - \beta)'X'(y - X\hat{\beta}) + (\hat{\beta} - \beta)'X'X(\hat{\beta} - \beta)$$

The middle term is zero by (6) ($X'(y - X\hat{\beta}) = X'y - X'X\hat{\beta} = 0$), so

$$\mathrm{RSS}(\beta) = \mathrm{RSS}(\hat{\beta}) + \|X(\hat{\beta} - \beta)\|^2 \;\geq\; \mathrm{RSS}(\hat{\beta}), \tag{8}$$

with equality only at $\beta = \hat{\beta}$ (full rank).

2.1 · The projection (slide 14)

Substituting (7) into $\hat{y} = X\hat{\beta}$:

$$\hat{y} = X(X'X)^{-1}X'y = Py, \qquad P = X(X'X)^{-1}X'. \tag{9}$$

Property of P	One-line proof	Meaning
$P' = P$	$(X(X'X)^{-1}X')' = X(X'X)^{-1}X'$	orthogonal (not oblique) projection
$PP = P$	$X(X'X)^{-1}\underbrace{X'X(X'X)^{-1}}_{I}X' = P$	projecting twice changes nothing
$PX = X$	$X(X'X)^{-1}X'X = X$	the column space is left fixed
$\mathrm{tr}(P) = q$	$\mathrm{tr}(X(X'X)^{-1}X') = \mathrm{tr}((X'X)^{-1}X'X) = \mathrm{tr}(I_q)$	q fitted dimensions

Residuals and their sum of squares (slide 14):

$$e = y - Py = (I - P)y, \qquad X'e = X'(I - P)y = (X' - X')y = 0, \tag{10}$$

$$\mathrm{RSS} = e'e = y'(I - P)'(I - P)y = y'(I - P)y, \tag{11}$$

because $I - P$ is also symmetric and idempotent. **(10) says the residuals are orthogonal to every column of** X; with an intercept column, they sum to zero (W §2.3, pp. 32–36). Two consequences used in §7 and §8: since $(I - P)X = 0$,

$$e = (I - P)(X\beta + \varepsilon) = (I - P)\varepsilon, \tag{12}$$

and the fitted plane passes through the sample means.

> **The geometry to hold: y splits into Py (in the plane spanned by the regressors) plus $(I - P)y$ (perpendicular to it), and OLS is nothing more than dropping the perpendicular.**

---

3 · C — sampling properties without normality (W §3.3, §5.1–5.2)

Substituting (3) into (7):

$$\hat{\beta} = (X'X)^{-1}X'(X\beta + \varepsilon) = \beta + (X'X)^{-1}X'\varepsilon \tag{13}$$

so $E(\hat{\beta} \mid X) = \beta$ whenever $E(\varepsilon \mid X) = 0$ (W Theorem 3.1, §3.3), and

$$V(\hat{\beta} \mid X) = (X'X)^{-1}X'\,(\sigma^2 I)\,X(X'X)^{-1} = \sigma^2(X'X)^{-1}. \tag{14}$$

Consistency and asymptotic normality need only $E(\varepsilon) = 0$ and $\mathrm{Cov}(x, \varepsilon) = 0$ plus finite moments (W §5.1, pp. 150–154; §5.2, pp. 154–161): **Gaussianity is not what makes OLS work; it is what makes it the MLE and makes small-sample t and F exact.** If $V(\varepsilon \mid X)$ is not $\sigma^2 I$, (14) is wrong but (13) still centres on β — the robust-SE fix of §8.

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4 · C — Gaussian errors: OLS = MLE (slide 6)

Assume $\varepsilon \sim N(0, \sigma^2 I_n)$, so $y \mid X \sim N(X\beta, \sigma^2 I_n)$. The log-likelihood is

$$\ell(\beta, \sigma^2) = -\frac{n}{2}\log(2\pi) - \frac{n}{2}\log\sigma^2 - \frac{(y - X\beta)'(y - X\beta)}{2\sigma^2} = -\frac{n}{2}\log(2\pi\sigma^2) - \frac{\mathrm{RSS}(\beta)}{2\sigma^2}. \tag{15}$$

Step 1 — β. For every fixed $\sigma^2 > 0$, β enters (15) only through $-\mathrm{RSS}(\beta)/(2\sigma^2)$, which is largest where RSS is smallest: by (8),

$$\hat{\beta}_{MLE} = \hat{\beta}_{ols}. \tag{16}$$

Step 2 — σ^2. Substitute (16) into (15) and differentiate in σ^2:

$$\frac{\partial \ell}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{\mathrm{RSS}(\hat{\beta})}{2\sigma^4} = 0 \iff \hat{\sigma}^2_{MLE} = \frac{\mathrm{RSS}}{n}. \tag{17}$$

Step 3 — its bias. By (12), $\mathrm{RSS} = \varepsilon'(I - P)\varepsilon$. A scalar equals its trace, and trace is invariant to cycling:

$$E(\mathrm{RSS}) = E\,\mathrm{tr}\big[(I - P)\varepsilon\varepsilon'\big] = \mathrm{tr}\big[(I - P)\,\sigma^2 I\big] = \sigma^2(n - \mathrm{tr}P) = \sigma^2(n - q), \tag{18}$$

using $\mathrm{tr}P = q$ from §2.1. Hence $E(\hat{\sigma}^2_{MLE}) = \sigma^2(n - q)/n < \sigma^2$, and the unbiased estimator is

$$\hat{\sigma}^2 = \frac{\mathrm{RSS}}{n - q}. \tag{19}$$

(Slide 14 writes $\hat{\sigma}^2 = \varepsilon'\varepsilon/(n - p)$: with slide 13's convention — y centred, intercept dropped, p columns — the divisor is the column count, which is (19).)

"OLS estimates and maximum likelihood estimates are the same. ML is usually more suitable for formal inference. The paid price is more modeling assumptions." (slide 6). Under (15), $\hat{\beta} \sim N(\beta, \sigma^2(X'X)^{-1})$ exactly, and $\mathrm{RSS}/\sigma^2 \sim \chi^2_{n-q}$ independent of $\hat{\beta}$ (W §4.1) — which is where exact t and F come from.

The site's companion script `normallinearregression-R.txt` runs this on the Koop–Tobias NLSY data (N = 2,178; log hourly wage on education, potential experience, ability, mother's education): four simple regressions with their adjusted R^2 in the titles, the correlation matrix, then `lm(logwage~educ+exper+ability)`. The file reads the data from the web and prints no output, so no numbers are reproduced here; the Python port is in L01-11-python.

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5 · C — R^2 and adjusted R^2 (slide 15; W §6.3a, p. 181)

$$R^2 = 1 - \frac{\mathrm{RSS}}{\mathrm{TSS}} = 1 - \frac{y'(I-P)y}{y'y} \tag{20}$$

(slide 15, with y centred so that $\mathrm{TSS} = y'y$): the share of the variance of y explained by the predictors.

Why it can never fall. Nest model M_1 inside M_2 (the columns of X_1 are among those of X_2). Any β_1 for M_1 is a β_2 for M_2 with the extra coefficients set to zero, so the minimum over the larger set is no larger:

$$\mathrm{RSS}_2 = \min_{\beta_2} \|y - X_2\beta_2\|^2 \le \min_{\beta_1} \|y - X_1\beta_1\|^2 = \mathrm{RSS}_1 \implies R_2^2 \ge R_1^2. \tag{21}$$

Adding pure noise cannot lower R^2, and with $q \ge n$ and $\operatorname{rank} X = n$, $\hat{y} = y$ and $R^2 = 1$.

Adjusted R^2. Wooldridge replaces both "variances" in (20) by unbiased versions (6.21):

$$\bar{R}^2 = 1 - \frac{\mathrm{RSS}/(n-k-1)}{\mathrm{TSS}/(n-1)} = 1 - \frac{\hat{\sigma}^2}{\mathrm{TSS}/(n-1)}, \tag{22}$$

equivalently (6.22):

$$\bar{R}^2 = 1 - (1-R^2)\,\frac{n-1}{n-k-1}. \tag{23}$$

Slide 15 uses the centred-data version, $1 - \frac{y'(I-P)y}{y'y} \cdot \frac{n}{n-p}$; for houseprice "with $n = 128$ and $p = 6$, $n/(n-6) = 1.049$" ($128/122 = 1.0492$).

Wooldridge's numbers (p. 181), in (23): $R^2 = 0.30$, $n = 51$, $k = 10$:

$$\bar{R}^2 = 1 - (1 - 0.30) \cdot \frac{50}{40} = 1 - 0.70 \times 1.25 = 1 - 0.875 = 0.125;$$

and $R^2 = 0.10$ gives $1 - 0.90 \times 1.25 = -0.125$ — **adjusted R^2 can be negative.**

An exact fact about the penalty (W, p. 181): adding one regressor raises $\bar{R}^2$ if and only if its $|t| > 1$; adding a group raises it iff the group's $F > 1$. Since $|t| = 1$ is nowhere near significance, $\bar{R}^2$ is a lenient selector — it tends to keep too much (§6's figure).

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6 · C — C_p, AIC, BIC and best subset (slides 16–22; AME §1.5.1.1, p. 21)

For a model with d predictors and $\hat{\sigma}^2$ estimated from the largest model (slide 16):

$$C_p = \frac{1}{n}(\mathrm{RSS} + 2d\hat{\sigma}^2), \qquad \mathrm{AIC} = \frac{1}{n\hat{\sigma}^2}(\mathrm{RSS} + 2d\hat{\sigma}^2), \qquad \mathrm{BIC} = \frac{1}{n}(\mathrm{RSS} + \log(n)\, d\hat{\sigma}^2). \tag{24}$$

Read (24) as "training error + price per parameter". AME (p. 21) writes the same idea as $R(g) \approx \mathrm{EQM}(g) + P(g)$ with $P(g) = 2p\hat{\sigma}^2/n$ (AIC) or $\log(n)p\hat{\sigma}^2/n$ (BIC). In the deck's Gaussian form AIC is $C_p/\hat{\sigma}^2$, so the two rank models identically. BIC charges $\log(n)\hat{\sigma}^2$ per parameter instead of $2\hat{\sigma}^2$, heavier whenever $\log n > 2$, i.e. $n > e^2 = 7.39$; for houseprice $\log 128 = 4.85$, **so BIC penalises each extra predictor about 2.4 times as hard as C_p.**

Why $2d\hat{\sigma}^2$ is the right price for C_p. By (18), $E(\mathrm{RSS}) = \sigma^2(n-q)$ understates the error a new sample would show, $\sigma^2(n+q)$ (derived in §7), by $2q\sigma^2$: C_p adds that optimism back, so it is an (approximately) unbiased estimate of test error per observation.

Best subset (slides 17–20). With p candidates there are 2^p models; slide 17 counts "$2^6 = 64$" for houseprice's six raw variables, while `regsubsets` sees seven columns (neighbourhood becomes the two dummies N1, N2), i.e. $2^7 = 128$. The exhaustive search keeps the best model of each size (slide 20): size 1 is SqFt; 2 adds Offers; 3 adds Brick; size 4 swaps Offers for the two neighbourhood dummies; 5 restores Offers; 6 adds Bathrooms; 7 adds Bedrooms. **The best models are not nested** (size 4 drops a variable size 3 had), which is why stepwise search can miss them (AME §3.1, pp. 33–35). Slides 21–22 then plot R^2, R^2_{adj}, C_p and BIC against size: the R^2 curve rises with size by (21).

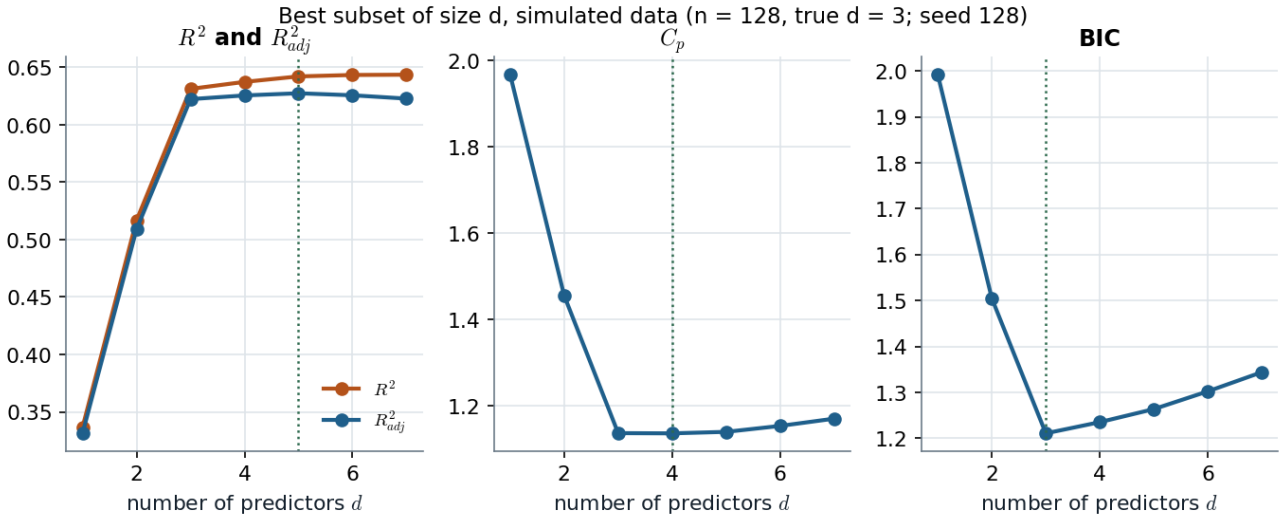


Figure — best-subset curves on a reproducible simulation (seed 128; n = 128 as houseprice, 7 candidates as in slide 20, 3 with nonzero slopes 1, 0.7, 0.5 and unit noise), each criterion exactly as in (22) and (24); dotted line = chosen size.

R^2 climbs from 0.337 to 0.644 and never turns down — by (21) it cannot. Adjusted R^2 peaks at $d = 5$ (0.627), keeping two noise predictors, as the $|t| > 1$ rule predicts; C_p chooses $d = 4$, by a hair over $d = 3$ (1.1356 vs 1.1361); **BIC's heavier price picks the true $d = 3$ (1.211).** One simulated data set proves nothing general — but the ordering of the three penalties is exactly what (23)–(24) imply.

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7 · C — overfitting, measured (`overfitting_a_linear_model.html`)

The experiment. Boston housing, $n = 506$ tracts, response `medv`, $d = 13$ standardised predictors, split in halves (253 training, 253 test). Model M_k contains all interactions of up to k distinct predictors, so it has

$$p_k = 1 + \sum_{j=1}^{k} \binom{13}{j} \tag{25}$$

coefficients: $p_1 = 1 + 13 = 14$; $p_2 = 14 + 78 = 92$; $p_3 = 92 + 286 = 378$; $p_4 = 378 + 715 = 1093$; $p_5 = 1093 + 1287 = 2380$. From $k = 3$ on, $p > n = 253$: $X'X$ is singular, and the page uses a ridge solve with $\lambda = 0.001$ purely so the linear system has a solution ("not to regularize in any meaningful way"; the ridge estimator itself is Classes 3+4).

k	p	R^2_{in}	RMSE_{in}	MAD_{in}	R^2_{out}	RMSE_{out}	MAD_{out}
1	14	0.7303	4.97	3.51	0.7469	4.48	3.41
2	92	0.9307	2.52	1.82	0.5854	6.83	4.06
3	378	0.9946	0.70	0.36	0.1407	25.41	13.66
4	1093	0.9998	0.13	0.05	0.1381	36.82	16.02
5	2380	1.0000	0.03	0.01	0.0519	47.13	16.71

Why the gap opens — the page's formulas, derived. Fix X, assume the linear model is correct with $p < n$ columns.

In sample. By (18), $E(\text{RSS}) = \sigma^2(n-p)$, so

$$E(\text{MSE}_{in}) = \frac{E(\text{RSS})}{n} = \sigma^2\Big(1 - \frac{p}{n}\Big). \tag{26}$$

Out of sample (new errors ε^* at the same design): the prediction error is $y^* - X\hat{\beta} = X\beta + \varepsilon^* - X\beta - P\varepsilon = \varepsilon^* - P\varepsilon$, using (13). Since ε^* and ε are independent,

$$E\|\varepsilon^* - P\varepsilon\|^2 = E\|\varepsilon^*\|^2 + E(\varepsilon' P' P\varepsilon) = n\sigma^2 + \sigma^2 \,\text{tr}P = \sigma^2(n+p),$$

$$E(\text{MSE}_{out}) = \sigma^2\Big(1 + \frac{p}{n}\Big). \tag{27}$$

Subtracting (26) from (27): the training error is optimistic by $2\sigma^2 p/n$, linear in the number of parameters (for random designs the page notes the out-of-sample error blows up like $\sigma^2(n-1)/(n-p-1)$ as $p \to n$).

Reading the table. M_1 is honest: RMSE 4.97 in vs 4.48 out (the test half happens to be easier — (26)–(27) are expectations, and one split is one draw). M_2 already loses out of sample (6.83) while R^2_{in} jumps to 0.931. From M_3 the training data are interpolated and the test RMSE explodes to 47.13 — $47.13/4.48 = 10.5$**, "roughly 11 times worse than the simplest model"**, with $R^2_{out} = 0.052$.

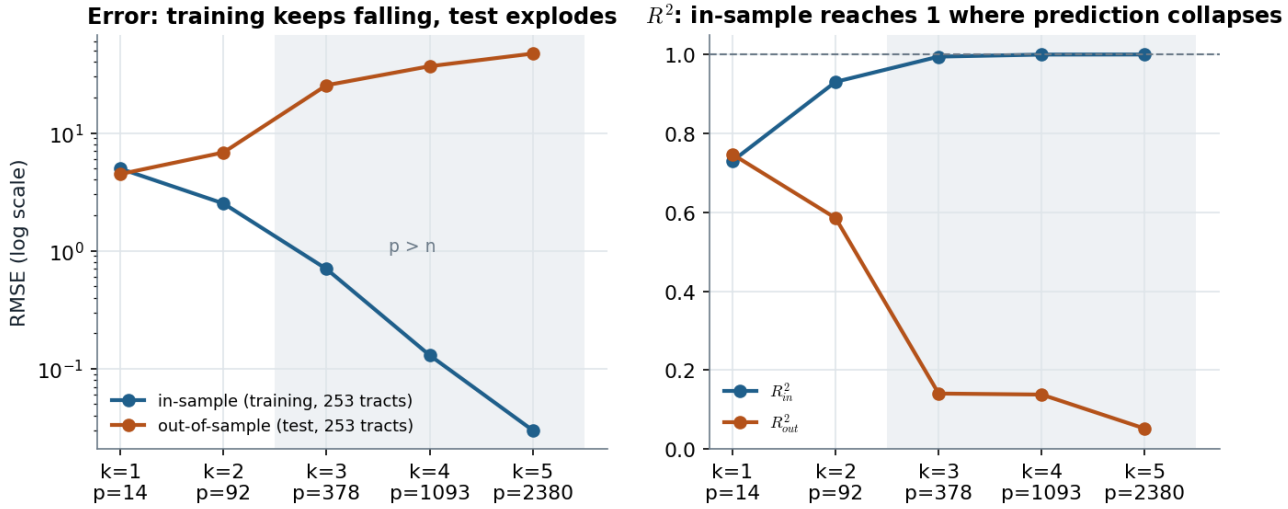


Figure — the page's table plotted: in-sample vs out-of-sample RMSE (log scale) and R^2; shaded where $p > n = 253$. Source: `overfitting_a_linear_model.html`*, table "In-sample versus out-of-sample performance".*

The two lines cross between $k = 1$ and $k = 2$ and then diverge for good: every coefficient beyond the first fourteen fits noise that is not in the test half. The right panel is the page's headline — **"The 'holy grail' $R^2 = 1$ is reached exactly where prediction collapses."** Class 2 turns this into a procedure (validation set, k-fold CV, LOOCV): L02 §3.

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8 · Worked — municipal abstention in the 2026 presidential first round (`regressao_abstencao_2026.pdf`)

Data and model (slides 1, 7–8). Unit: municipality, 5,565 of 5,570 (five created in 2013 lack IDHM 2010). Response y_i = "% de abstenção no 1º turno de 2026" (preliminary ballot-box totals of 05/10). Specification (slide 8):

$$y_i = \alpha + \beta_1\,\text{Abst22}_i + \beta_2\,\text{BN26}_i + \beta_3\,\text{IDHM}_i + \sum_{r \neq SE} \gamma_r D_{ri}^{reg} \left(\text{or} \sum_{s \neq SP} \delta_s D_{si}^{UF}\right) + \sum_{k \neq \leq 10\text{mil}} \theta_k D_{ki}^{porte} + \varepsilon_i \quad (28)$$

"IDHM em décimos (β_3 = efeito de +0,1). MQO com erros-padrão HC1." References: Sudeste, SP, municipalities up to 10 thousand eligible voters ("aptos"). Models: (0) no dummies; (1) region; (2) UF; (3) region + size ("porte"); (4) UF + size.

The results table (slide 11), coefficient (HC1 SE):

	(0)	(1) Região	(2) UF	(3) Região+porte	(4) UF+porte
Intercepto	−2.200 (0.354)	9.025 (0.555)	5.578 (0.641)	9.817 (0.638)	7.378 (0.756)
% Abstenção 2022	0.717 (0.007)	0.649 (0.008)	0.686 (0.009)	0.651 (0.008)	0.684 (0.009)

	(0)	(1) Região	(2) UF	(3) Região+porte	(4) UF+porte
% Brancos+Nulos 2026	0.225 (0.024)	0.286 (0.026)	0.228 (0.030)	0.276 (0.026)	0.196 (0.031)
IDHM 2010 (+0,1)	1.013 (0.042)	−0.356 (0.064)	−0.010 (0.069)	−0.473 (0.076)	−0.246 (0.085)
10 a 45 mil aptos				−0.130 (0.067)	0.069 (0.067)
45 a 100 mil aptos				0.225 (0.107)	0.439 (0.112)
100 a 500 mil aptos				0.635 (0.140)	0.960 (0.152)
Acima de 500 mil aptos				1.005 (0.304)	1.140 (0.317)
R^2	0.682	0.727	0.755	0.728	0.757
R^2 ajustado	0.682	0.727	0.754	0.728	0.756

(Region coefficients in (1)/(3): Norte −1.872/−1.959, Nordeste −2.750/−2.821, Sul −0.271/−0.236, Centro-Oeste 0.693/0.687.) Decimal commas of the slide are written as points here.

8.1 · Persistence — a coefficient, its t and its interval

"1 p.p. de abstenção em 2022 se traduz em 0,65–0,72 p.p. em 2026" (slide 13). In model (4):

$$t = \frac{0.684}{0.009} = 76.0, \qquad 0.684 \pm 1.96 \times 0.009 = 0.684 \pm 0.018 = [0.666,\ 0.702].$$

The interval excludes 1 by a wide margin: **abstention persists but regresses toward the mean** — a municipality 10 p.p. above average in 2022 is expected about 6.8 p.p. above in 2026, other regressors fixed.

A check on the arithmetic of OLS. With an intercept, (10) makes the fitted plane pass through the sample means (slide 7: 20.72, 21.27, 4.45, IDHM 0.659 → 6.59 in tenths). Model (0) at the means:

$$-2.200 + 0.717(21.27) + 0.225(4.45) + 1.013(6.59) = -2.200 + 15.251 + 1.001 + 6.676 = 20.728 \approx 20.72 \ \checkmark$$

8.2 · IDHM changes sign: omitted-variable bias in action

Model	(0)	(1)	(2)	(3)	(4)
IDHM (+0,1)	1.01	−0.36	−0.01	−0.47	−0.25

In (0), IDHM proxies for *where* a municipality is: the Nordeste combines low IDHM with low abstention (−2.75 p.p. in (1)), so across Brazil higher IDHM goes with higher abstention. Once region dummies absorb that, the within-region association is negative. The omitted-variable bias formula (W §3.3, pp. 73–81) gives the sign: bias = (effect of omitted variable) × (slope of omitted on included). **Size**

works the same way: "Cidades grandes têm IDHM alto; separando o porte, o IDHM volta a ter efeito negativo" (slide 13) — size raises abstention ($\theta_k > 0$) and correlates positively with IDHM, so omitting it biases β_3 upward: -0.010 in (2) becomes -0.246 in (4).

8.3 · Size: individual ts and the joint test

Model (4), relative to municipalities up to 10 thousand voters (slide 12):

Size	Coef.	SE	t	95% interval
10–45 mil	0.069	0.067	1.03	[−0.062, 0.200]
45–100 mil	0.439	0.112	3.92	[0.219, 0.659]
100–500 mil	0.960	0.152	6.32	[0.662, 1.258]
> 500 mil	1.140	0.317	3.60	[0.519, 1.761]

"+0,07 p.p. (não significativo)" is the first row: $|t| = 1.03 < 1.96$. **Conditional on 2022 abstention, the gradient in size appears although the raw means by size are flat (20.0%–21.1%, slide 5).** The joint robust Wald test that all four are zero gives $F = 13.22$ with UF and 10.84 with region, $p < 0.0001$ (the 5% critical value of $F_{4,5531}$ is 2.37).

Is $F = 13.22$ consistent with the R^2 gain? Under homoskedasticity the R^2 form (W (4.41), p. 133) with $q = 4$ restrictions, 34 parameters and $n - 34 = 5531$ residual df:

$$F = \frac{(R^2_{ur} - R^2_r)/q}{(1 - R^2_{ur})/(n-34)} = \frac{(0.757 - 0.755)/4}{(1 - 0.757)/5531} = \frac{0.000500}{0.0000439} = 11.4$$

Since each R^2 is rounded to three decimals, $R^2_{ur} - R^2_r$ lies anywhere in $[0.001, 0.003]$ and the formula in [5.7, 17.1]: **the robust 13.22 sits inside what the printed R^2s allow.** (The robust statistic need not equal the R^2 form; the two coincide only under homoskedasticity.)

And with the adjusted R^2. (23) with $k = 29$ slopes for (2) and $k = 33$ for (4):

$$\bar{R}^2_{(2)} = 1 - 0.245 \cdot \frac{5564}{5535} = 1 - 0.24628 = 0.754, \qquad \bar{R}^2_{(4)} = 1 - 0.243 \cdot \frac{5564}{5531} = 1 - 0.24445 = 0.756,$$

reproducing slide 11. The rise matches Wooldridge's rule: the size group's $F > 1$, so $\bar{R}^2$ must go up. **"O ganho de ajuste é pequeno (R² ajustado de 0,754 para 0,756): a abstenção passada já captura quase tudo"** — a variable can be highly significant and add almost nothing to fit; significance and explanatory share are different questions.

8.4 · UF or region? A test of 22 restrictions (slide 15)

The region model is the UF model with all UFs of a region forced to share one coefficient: 26 UF dummies collapse to 4 region dummies, $26 - 4 = 22$ **linear restrictions**. Robust Wald: $F = 31.08$ (without size, (1) vs (2)) and $F = 32.09$ (with size, (3) vs (4)), both $p < 0.0001$: rejected — e.g. in the Centro-Oeste MS and GO are positive, MT negative.

8.5 · HC1 standard errors and the caveats

HC1 is White's heteroskedasticity-robust variance (W (8.4), pp. 245–246) times the degrees-of-freedom factor $n/(n-k-1)$ mentioned on p. 246:

$$\hat{V}_{HC1}(\hat{\beta}) = \frac{n}{n-q}\,(X'X)^{-1}\Big(\sum_{i=1}^{n} e_i^2\, x_i x_i'\Big)(X'X)^{-1} \tag{29}$$

— (14) with $\sigma^2 I$ replaced by $\mathrm{diag}(e_i^2)$. It is the `vcovHC(m, type = "HC1")` of the slides' R code.

The deck's own "Ressalvas" (slide 17), each a lesson in reading a regression: 2026 data are preliminary; IDHM is from 2010; "Brancos+nulos de 2026 é determinado junto com a abstenção, então o coeficiente descreve associação" (not a causal effect); all municipalities weigh the same (an electorate-weighted version would favour large cities); and **errors may be correlated within UF — with 27 clusters, UF-clustered SEs "seriam uma verificação natural"**, since HC1 handles heteroskedasticity, not within-cluster correlation.

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9 · G — named, not developed

- **Gauss–Markov** (W §3.5, pp. 89–90): under $V(\varepsilon) = \sigma^2 I$, OLS has the smallest variance among linear unbiased estimators — §12.1 of L01-05-comparing-estimators-mse-bootstrap (equal weights beat triangular ones) is its one-parameter case.
- **Shrinkage**: ridge, lasso, elastic net, normal–gamma and horseshoe priors — the later pages of the same deck, after the best-subset and Credit examples — trade bias for variance exactly as §8 of the previous note did for $c\,x_{(n)}$. **Classes 3+4.**
- **OLS as the best linear predictor** even when $E(y \mid x)$ is not linear (AME Ch. 2, pp. 27–31; Teorema 2): $\widehat{\beta} \to \beta^* = \arg\min E[(Y - \beta' X)^2]$.

Links

- Neighbours: L01-05-comparing-estimators-mse-bootstrap (MSE = variance + bias²) · L01-07-ar-var-review (the AR(p) is this regression with lagged y as regressors).
- Unit: L01-00-map · L01-09-formulas · L01-10-pitfalls · L01-11-python · L01-08-p-hacking-evidence (specification search on regressions like §8's).
- Class 2: L02 §1 (bias–variance behind (26)–(27)) · L02 §2 (reading level/log coefficients) · L02 §3 (CV and the LOOCV hat-matrix shortcut, which uses P from §2.1).
- Rules: rules/06 (validation, model selection); rules/01 (omitted-variable bias, §8.2).
- **Practice (number and printed page only):** W 11.C8 (p. 370) uses this regression with a lagged regressor; I 3.7 (p. 129), I 3.9 (pp. 129–130).

AR(1), unit root and VAR(1) · the regression with a lagged response, and the distribution it settles into

> **Claim. An AR(1) is a linear regression of y_t on y_{t-1}**, and with Gaussian shocks OLS on it is the conditional MLE. Its behaviour hangs on one number: **for $|\phi| < 1$ the process forgets its start at rate ϕ^t and settles into the equilibrium (stationary) distribution $N(\mu,\ \sigma^2/(1-\phi^2))$; at $\phi = 1$ (a unit root) the variance grows like $t\sigma^2$ and there is no equilibrium at all.** The AR(p) is the same regression with p lags. **The VAR(1) is its matrix version**: stationary when all eigenvalues of Φ lie inside the unit circle, with mean $(I-\Phi)^{-1}\phi_0$ and covariance solving $\Gamma_0 = \Phi\Gamma_0\Phi' + \Sigma$, i.e. $\mathrm{vec}(\Gamma_0) = (I-\Phi\otimes\Phi)^{-1}\mathrm{vec}(\Sigma)$. Prev: L01-06-gaussian-linear-regression · Next: L01-08-p-hacking-evidence · Map: L01-00-map

Notation.

Symbol	Meaning	Range
r_t or y_t	the series at time t (Tsay writes r_t for returns; Wooldridge y_t)	
a_t	shock (innovation), white noise with variance σ_a^2 (written σ^2 where no confusion)	
μ	$E(r_t)$ under stationarity	
$\gamma_\ell = \mathrm{Cov}(r_t, r_{t-\ell})$	lag-ℓ autocovariance; $\gamma_0 = V(r_t)$	
$\rho_\ell = \gamma_\ell/\gamma_0$	lag-ℓ autocorrelation (ACF)	$[-1, 1]$
ϕ_0, ϕ_1	AR(1) intercept and slope (persistence)	
ψ_i	weights of the linear (MA(∞)) representation, $\psi_0 = 1$	
T	sample length	
m	number of lags in the portmanteau test	
k	dimension of the vector series (§9)	
ϕ_0, Φ, Σ	VAR(1) intercept ($k \times 1$), coefficient matrix ($k \times k$), shock covariance ($k \times k$)	
Γ_ℓ	lag-ℓ cross-covariance matrix of r_t	
$\mathrm{vec}(A)$, $\otimes$	columns of A stacked; Kronecker product	

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1 · C — stationarity (deck pp. 5–7; T §2.1, p. 30)

Strict stationarity: the joint distribution of $(r_{t_1}, \ldots, r_{t_k})$ equals that of $(r_{t_1+t}, \ldots, r_{t_k+t})$ for every shift t, every k and every choice of times — "a very strong condition that is hard to verify empirically."

Weak stationarity: (a) $E(r_t) = \mu$, constant; (b) $\mathrm{Cov}(r_t, r_{t-\ell}) = \gamma_\ell$, depending only on the lag ℓ. In a time plot: **the values fluctuate with constant variation around a fixed level.** It is what makes past data informative about the future.

Two properties of γ_ℓ (deck p. 7): $\gamma_0 = V(r_t)$, and $\gamma_{-\ell} = \gamma_\ell$, because with $t_1 = t + \ell$

$$\gamma_{-\ell} = \mathrm{Cov}(r_t, r_{t+\ell}) = \mathrm{Cov}(r_{t+\ell}, r_t) = \mathrm{Cov}(r_{t_1}, r_{t_1-\ell}) = \gamma_\ell. \tag{1}$$

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2 · C — the ACF and the Ljung–Box test (deck pp. 8–11; T §2.2, pp. 30–36)

Under weak stationarity $V(r_t) = V(r_{t-\ell}) = \gamma_0$, so

$$\rho_\ell = \frac{\mathrm{Cov}(r_t, r_{t-\ell})}{\sqrt{V(r_t)V(r_{t-\ell})}} = \frac{\gamma_\ell}{\gamma_0}, \qquad \hat\rho_\ell = \frac{\sum_{t=\ell+1}^{T}(r_t - \bar r)(r_{t-\ell} - \bar r)}{\sum_{t=1}^{T}(r_t - \bar r)^2}. \tag{2}$$

Portmanteau tests of $H_0 : \rho_1 = \cdots = \rho_m = 0$ against "some $\rho_i \neq 0$": Box–Pierce $Q^*(m) = T\sum_{\ell=1}^{m}\hat\rho_\ell^2$ and its finite-sample improvement, Ljung–Box,

$$Q(m) = T(T+2)\sum_{\ell=1}^{m}\frac{\hat\rho_\ell^2}{T-\ell} \;\overset{a}{\sim}\; \chi^2_m \text{ under } H_0; \tag{3}$$

reject when $Q(m)$ exceeds the $1-\alpha$ quantile of χ^2_m.

Worked — IBM monthly simple returns (deck p. 11). `Box.test(sibm, lag=30, type="Ljung")` gives $Q(30) = 38.241$; with 30 df, $P(\chi^2_{30} > 38.241) = 0.1437$ (reproduced in Python): **no evidence of linear autocorrelation in returns.** The same test on squared returns gives $Q(30) = 182.12$, $p < 2.2 \times 10^{-16}$ ($\approx 10^{-23}$): **returns are serially uncorrelated but not independent** — their size is predictable. That is volatility clustering, the subject of ARCH/GARCH and stochastic volatility in Class 8.

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3 · C — white noise and linear time series (deck pp. 12–14; T §2.3, pp. 36–37)

White noise: iid with finite mean and variance; all ACFs zero. Gaussian white noise: $N(0, \sigma^2)$.

Linear time series: $r_t = \mu + \sum_{i=0}^{\infty}\psi_i a_{t-i}$, $\psi_0 = 1$, $\{a_t\}$ white noise. By independence of the a's, $E(r_t) = \mu$ and $V(r_t) = \sigma_a^2\sum_i \psi_i^2$. The autocovariance (deck p. 14):

$$\gamma_\ell = E\Big[\Big(\sum_{i=0}^{\infty}\psi_i a_{t-i}\Big)\Big(\sum_{j=0}^{\infty}\psi_j a_{t-\ell-j}\Big)\Big] = \sum_{i,j}\psi_i\psi_j E(a_{t-i}a_{t-\ell-j})$$

Only terms with $t - i = t - \ell - j$, i.e. $i = j + \ell$, survive, each contributing σ_a^2:

$$\gamma_\ell = \sigma_a^2\sum_{j=0}^{\infty}\psi_j\psi_{j+\ell}, \qquad \rho_\ell = \frac{\sum_{i\geq 0}\psi_i\psi_{i+\ell}}{1 + \sum_{i\geq 1}\psi_i^2}. \tag{4}$$

Every model in this note is a recipe for the ψ weights.

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4 · C — AR(1): mean, variance, ACF (deck pp. 15–17; T eqs. 2.8–2.11, pp. 37–40)

$$r_t = \phi_0 + \phi_1 r_{t-1} + a_t \tag{5}$$

"in the same form as the well-known simple linear regression model" (T, p. 37), with $E(r_t \mid r_{t-1}) = \phi_0 + \phi_1 r_{t-1}$ and $V(r_t \mid r_{t-1}) = \sigma_a^2$.

Mean. Take expectations of (5), $E(a_t) = 0$: $E(r_t) = \phi_0 + \phi_1 E(r_{t-1})$. Under stationarity $E(r_t) = E(r_{t-1}) = \mu$:

$$\mu = \phi_0 + \phi_1 \mu \iff \mu = \frac{\phi_0}{1-\phi_1} \quad (\phi_1 \neq 1). \tag{6}$$

Substituting $\phi_0 = (1-\phi_1)\mu$ from (6) into (5):

$$r_t - \mu = \phi_1 (r_{t-1} - \mu) + a_t. \tag{7}$$

MA(∞) form. Substitute (7) into itself repeatedly:

$$r_t - \mu = a_t + \phi_1(\phi_1(r_{t-2} - \mu) + a_{t-1}) = a_t + \phi_1 a_{t-1} + \phi_1^2 (r_{t-2} - \mu) = \cdots = \sum_{i=0}^{\infty} \phi_1^i a_{t-i}, \tag{8}$$

the linear form of §3 with $\psi_i = \phi_1^i$ (T eq. 2.11). So r_{t-1} depends only on $a_{t-1}, a_{t-2}, \ldots$, and $\mathrm{Cov}(r_{t-1}, a_t) = 0$.

Variance. Square (7) and take expectations; the cross term vanishes because $\mathrm{Cov}(r_{t-1}, a_t) = 0$:

$$V(r_t) = \phi_1^2 V(r_{t-1}) + \sigma_a^2.$$

Under stationarity $V(r_t) = V(r_{t-1}) = \gamma_0$:

$$\gamma_0 = \phi_1^2 \gamma_0 + \sigma_a^2 \iff \gamma_0 = \frac{\sigma_a^2}{1-\phi_1^2}, \tag{9}$$

which is a positive finite number only if $\phi_1^2 < 1$: weak stationarity implies $|\phi_1| < 1$; conversely $|\phi_1| < 1$ makes (8) converge with finite, time-invariant moments. **Necessary and sufficient:** $|\phi_1| < 1$ (T, p. 39; W (11.3), p. 347).

ACF. Multiply (7) by $(r_{t-\ell} - \mu)$, $\ell \geq 1$, and take expectations; a_t is uncorrelated with the past:

$$\gamma_\ell = \phi_1 E[(r_{t-1} - \mu)(r_{t-\ell} - \mu)] + E[a_t (r_{t-\ell} - \mu)] = \phi_1 \gamma_{\ell-1} + 0. \tag{10}$$

Iterating (10) down to γ_0: $\gamma_\ell = \phi_1^\ell \gamma_0$, so

$$\boxed{\rho_\ell = \phi_1^\ell, \quad \ell = 0, 1, 2, \ldots} \tag{11}$$

exponential decay at rate ϕ_1 (alternating in sign when $\phi_1 < 0$; T Figure 2.3, p. 40). (The deck's p. 16 writes the $\ell = 0$ case as $\gamma_0 = \phi_1 \gamma_1 + \sigma_a^2$; with $\gamma_1 = \phi_1 \gamma_0$ this is (9) again.) **Wooldridge's number (W (11.4), p. 347):** with $\phi_1 = 0.9$, $\mathrm{Corr}(y_t, y_{t+5}) = 0.9^5 = 0.59049 \approx .591$ — high persistence, yet the correlation fades.

5 · C — estimation: OLS is the conditional MLE (T pp. 49–50)

Assume Gaussian shocks. By (5), $r_t \mid r_{t-1} \sim N(\phi_0 + \phi_1 r_{t-1}, \sigma_a^2)$, and by the Markov property the joint density factors as

$$p(r_1, \dots, r_T) = p(r_1) \prod_{t=2}^{T} p(r_t \mid r_{t-1}). \tag{12}$$

Conditioning on r_1 drops the first factor. The conditional log-likelihood is

$$\ell_c(\phi_0, \phi_1, \sigma_a^2) = -\frac{T-1}{2} \log(2\pi\sigma_a^2) - \frac{1}{2\sigma_a^2} \sum_{t=2}^{T} (r_t - \phi_0 - \phi_1 r_{t-1})^2, \tag{13}$$

which is (15) of L01-06-gaussian-linear-regression with $y = (r_2, \dots, r_T)'$ and X having rows $(1, r_{t-1})$. By the argument there, $(\hat{\phi}_0, \hat{\phi}_1)$ **is the OLS regression of** r_t **on** r_{t-1} — Tsay's "conditional least-squares method" — and "if the conditional-likelihood method is used, the estimates of ϕ_i remain unchanged" (T, p. 50); only the divisor of $\hat{\sigma}_a^2$ changes.

The exact likelihood keeps $p(r_1)$, which under stationarity is the equilibrium law of §6, $N(\mu, \sigma_a^2/(1-\phi_1^2))$. That factor depends on ϕ_1, so the exact MLE differs slightly from OLS — negligibly for large T — and it **imposes** $|\phi_1| < 1$, while OLS can return any value.

6 · C — the equilibrium (stationary) distribution

Start the Gaussian AR(1) at a fixed y_0 and iterate (7) t times, stopping instead of going to infinity as in (8):

$$y_t - \mu = \phi^t (y_0 - \mu) + \sum_{j=0}^{t-1} \phi^j a_{t-j}. \tag{14}$$

A fixed number plus a sum of independent normals is normal, with

$$E(y_t \mid y_0) = \mu + \phi^t (y_0 - \mu), \qquad V(y_t \mid y_0) = \sigma^2 \sum_{j=0}^{t-1} \phi^{2j} = \sigma^2 \, \frac{1 - \phi^{2t}}{1 - \phi^2} \tag{15}$$

(geometric sum, $\phi^2 \neq 1$). Let $t \to \infty$ with $|\phi| < 1$: $\phi^t \to 0$ and $\phi^{2t} \to 0$, so

$$\boxed{y_t \mid y_0 \;\xrightarrow{d}\; N\Big(\mu,\ \frac{\sigma^2}{1-\phi^2}\Big) \quad \text{whatever } y_0} \tag{16}$$

The start is forgotten geometrically. (16) is also the only *invariant* normal law: if $y_{t-1} \sim N(\mu, v)$ then by (7) $y_t \sim N(\mu, \phi^2 v + \sigma^2)$, which equals $N(\mu, v)$ exactly when $v = \phi^2 v + \sigma^2$, i.e. $v = \sigma^2/(1-\phi^2)$ — (9) again.

Half-life (T, p. 56): the horizon h at which the expected deviation from μ has halved, $\phi^h = \frac{1}{2}$:

$$h = \frac{\ln 0.5}{\ln |\phi|}. \tag{17}$$

Worked, $\phi = 0.9$**,** $\sigma = 1$**,** $\mu = 0$**,** $y_0 = 8$**.** Equilibrium variance $1/(1 - 0.81) = 1/0.19 = 5.263$ (sd 2.294). Half-life $\ln 0.5/\ln 0.9 = -0.6931/-0.1054 = 6.58$ periods. At $t = 10$, from (15): mean $0.9^{10} \times 8 = 0.3487 \times 8 = 2.79$; variance $(1 - 0.9^{20})/0.19 = (1 - 0.1216)/0.19 = 4.62$. For $\phi = 0.5$ the half-life is exactly 1 and the equilibrium variance $1/0.75 = 1.333$.

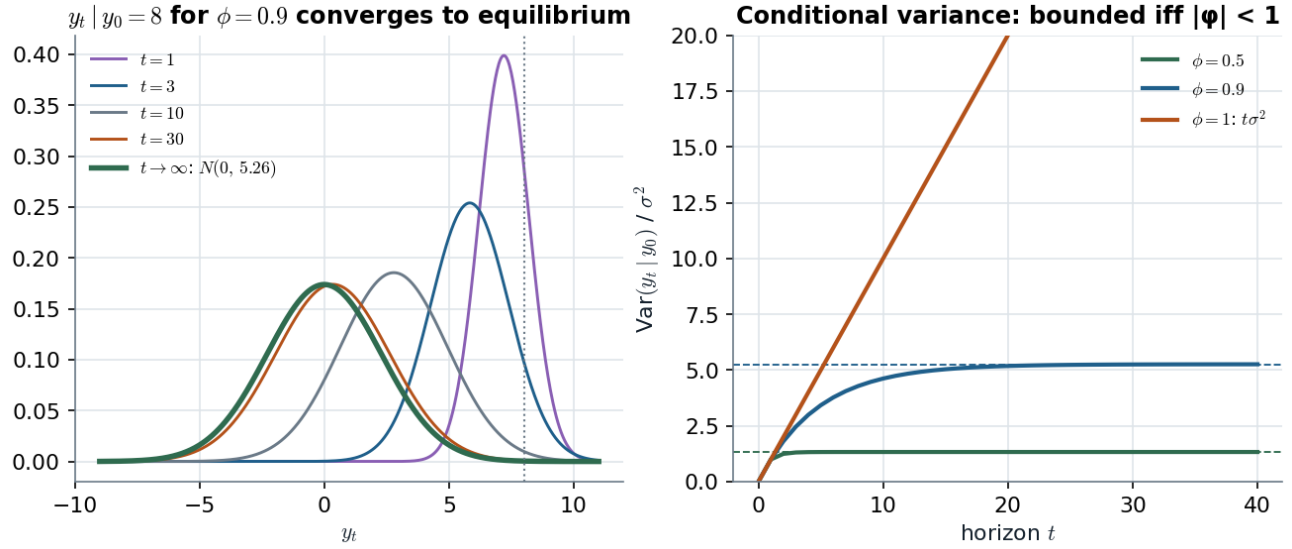


Figure — exact laws (15)–(16): left, $y_t \mid y_0 = 8$ *for* $\phi = 0.9$*,* $\sigma = 1$*; right,* $V(y_t \mid y_0)/\sigma^2$ *for* $\phi = 0.5, 0.9$ *(dashed: limits* $1/(1 - \phi^2)$*) and* t *for* $\phi = 1$ *(W (11.21)).*

On the left the density slides from near $y_0 = 8$ toward zero and spreads; by $t = 30$ it is indistinguishable from the equilibrium curve ($0.9^{30} \times 8 = 0.34$). On the right, the stationary variances level off — fast for $\phi = 0.5$, after a few half-lives for $\phi = 0.9$ — **while the unit-root line never does.**

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7 · C — the unit root, $\phi = 1$ (W §11.3, pp. 354–355)

With $\phi_1 = 1$ and $\phi_0 = 0$, (5) is the **random walk** $y_t = y_{t-1} + e_t$ (W (11.20)). Repeated substitution with y_0 fixed:

$$y_t = y_0 + e_1 + e_2 + \cdots + e_t, \qquad E(y_t) = y_0, \qquad V(y_t) = t\,\sigma_e^2 \tag{18}$$

(W (11.21), p. 354): **the variance grows linearly in** t**, so the process cannot be stationary**, and (16) has no limit — $1 - \phi^2 = 0$ in the denominator. Forecasts never revert:

$$E(y_{t+h} \mid y_t) = y_t \text{ for all } h \geq 1 \quad \text{(W (11.22), p. 355)}, \qquad \text{vs} \quad E(y_{t+h} \mid y_t) = \phi^h y_t \to 0 \text{ when } |\phi| < 1. \tag{19}$$

Persistence does not fade either: $\mathrm{Corr}(y_t, y_{t+h}) = \sqrt{t/(t+h)}$ (W, p. 355), which depends on t (not stationary) and, for $h = 100$, exceeds 0.95 once $t > 1000$: $\sqrt{1000/1100} = 0.9535$.

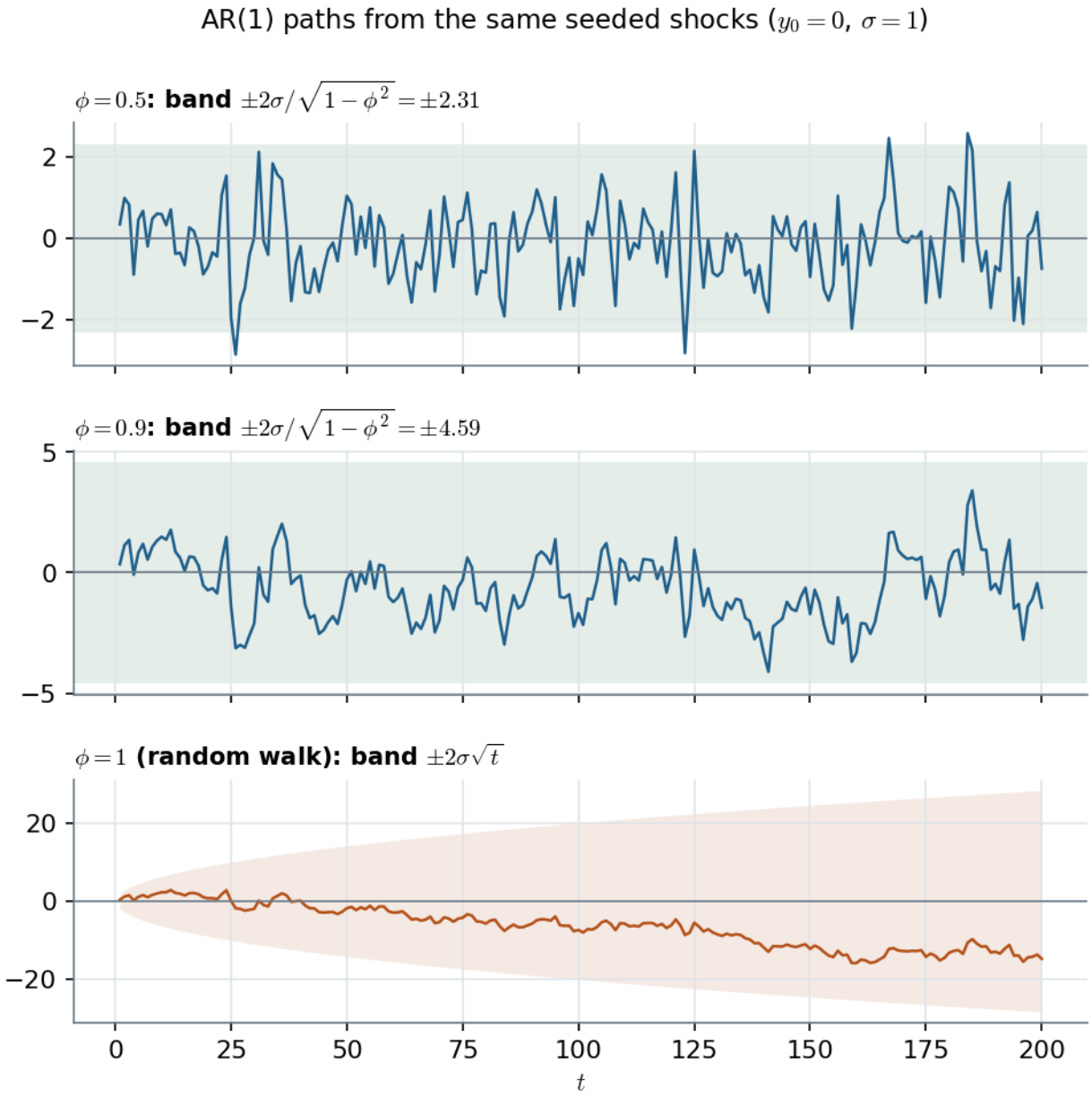


Figure — one seeded simulation of 200 standard normal shocks fed through (5) with $\phi = 0.5, 0.9, 1$ and $y_0 = 0$; shaded: ± 2 equilibrium sd from (9), or $\pm 2\sqrt{t}$ from (18) for the random walk.

Same shocks, three behaviours: at $\phi = 0.5$ the path crosses zero constantly; at $\phi = 0.9$ it makes long excursions but keeps returning (the half-life of 6.6 periods is visible as runs of that length); at $\phi = 1$ every shock is permanent and the path drifts to about -15 with nothing pulling it back.

Why it matters for estimation (named; Class 8). At $\phi = 1$, $\hat{\phi}$ from §5 is still consistent (faster, at rate T), but its t-statistic for $H_0 : \phi = 1$ no longer has a $N(0,1)$ limit — the Dickey–Fuller distribution — and regressing one random walk on another produces spurious significance (W §11.3 for the persistence, §18.2–18.3 for the tests and spurious regressions; `rules/04`).

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8 · C — AR(p) as a regression (T eq. 2.9, pp. 49–50)

$$r_t = \phi_0 + \phi_1 r_{t-1} + \cdots + \phi_p r_{t-p} + a_t, \qquad t = p+1, \ldots, T \tag{20}$$

"is in the form of a multiple linear regression and can be estimated by the least-squares method" (T, p. 49): $y = (r_{p+1}, \ldots, r_T)'$, rows of X equal to $(1, r_{t-1}, \ldots, r_{t-p})$, and $\hat{\beta} = (X'X)^{-1}X'y$ from L01-06-gaussian-linear-regression (7). There are $T-p$ observations and $p+1$ parameters, hence Tsay's $\hat{\sigma}_a^2 = \sum \hat{a}_t^2/(T-2p-1)$ — the divisor $n-q$ of that note's (19).

Stationary mean. Taking expectations as in (6):

$$\mu = \frac{\phi_0}{1 - \phi_1 - \cdots - \phi_p}. \tag{21}$$

Worked — Tsay's AR(3) for monthly value-weighted index returns (T, p. 50): $r_t = 0.0091 + 0.116 r_{t-1} - 0.019 r_{t-2} - 0.104 r_{t-3} + \hat{a}_t$, $\hat{\sigma}_a = 0.054$. By (21):

$$\mu = \frac{0.0091}{1 - 0.116 - (-0.019) - (-0.104)} = \frac{0.0091}{1.007} = 0.00904,$$

about 0.9% a month: with these small coefficients the implied mean barely differs from the intercept. AR(2) moments and forecasts are T Exercise 2.2 (p. 104).

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9 · C — VAR(1) in matrix form (T §8.2, pp. 399–403)

$$r_t = \phi_0 + \Phi r_{t-1} + a_t, \qquad a_t \text{ serially uncorrelated, } E(a_t) = 0, \ \mathrm{Cov}(a_t) = \Sigma \tag{22}$$

(T eq. 8.8). For $k = 2$:

$$r_{1t} = \phi_{10} + \Phi_{11} r_{1,t-1} + \Phi_{12} r_{2,t-1} + a_{1t}, \qquad r_{2t} = \phi_{20} + \Phi_{21} r_{1,t-1} + \Phi_{22} r_{2,t-1} + a_{2t}.$$

Φ_{12} is the effect of $r_{2,t-1}$ on r_{1t} given $r_{1,t-1}$. $\Phi_{12} = 0, \Phi_{21} \neq 0$: one-way dependence from r_1 to r_2; both zero: uncoupled; both nonzero: **feedback**. Contemporaneous links live only in Σ_{12} — the **reduced form**; a Cholesky factor of Σ turns it into a recursive structural form (T §8.2.1; structural VARs and impulse responses are Class 9). **Each equation is an OLS regression on the same lagged regressors**, so equation-by-equation OLS estimates Φ.

Stationarity (T, p. 402). Repeated substitution as in (8) gives $\tilde{r}_t = r_t - \mu = a_t + \Phi a_{t-1} + \Phi^2 a_{t-2} + \ldots$; the weights Φ^j must vanish, which happens **iff all eigenvalues of Φ are less than 1 in modulus** — the matrix version of $|\phi_1| < 1$.

Mean. Expectations of (22) under stationarity:

$$\mu = \phi_0 + \Phi\mu \iff (I - \Phi)\mu = \phi_0 \iff \mu = (I - \Phi)^{-1}\phi_0. \tag{23}$$

Covariance. With $\tilde{r}_t = \Phi \tilde{r}_{t-1} + a_t$ and $\mathrm{Cov}(\tilde{r}_{t-1}, a_t) = 0$:

$$\Gamma_0 = E(\tilde{r}_t \tilde{r}_t') = \Phi E(\tilde{r}_{t-1}\tilde{r}_{t-1}')\Phi' + E(a_t a_t') = \Phi \Gamma_0 \Phi' + \Sigma. \tag{24}$$

Apply vec and the identity $\operatorname{vec}(ABC) = (C' \otimes A)\operatorname{vec}(B)$ with $A = \Phi$, $B = \Gamma_0$, $C = \Phi'$:

$$\operatorname{vec}(\Gamma_0) = (\Phi \otimes \Phi)\operatorname{vec}(\Gamma_0) + \operatorname{vec}(\Sigma)$$

$$\boxed{\operatorname{vec}(\Gamma_0) = (I_{k^2} - \Phi \otimes \Phi)^{-1}\operatorname{vec}(\Sigma)} \tag{25}$$

($I - \Phi \otimes \Phi$ is invertible because the eigenvalues of $\Phi \otimes \Phi$ are the products $\lambda_i \lambda_j$, all of modulus below 1.) Post-multiplying $\tilde{r}_t = \Phi \tilde{r}_{t-1} + a_t$ by $\tilde{r}'_{t-\ell}$ and taking expectations gives T (8.12):

$$\Gamma_\ell = \Phi \Gamma_{\ell-1} \implies \Gamma_\ell = \Phi^\ell \Gamma_0, \quad \ell > 0. \tag{26}$$

For $k = 1$, (23), (25) and (26) collapse to (6), (9) and (10).

9.1 · Worked — T Example 8.3 (p. 400)

$$\phi_0 = \begin{pmatrix} 0.2 \\ 0.4 \end{pmatrix}, \qquad \Phi = \begin{pmatrix} 0.2 & 0.3 \\ -0.6 & 1.1 \end{pmatrix}, \qquad \Sigma = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}.$$

Stationary? $\operatorname{tr}\Phi = 1.3$, $\det\Phi = 0.2(1.1) - 0.3(-0.6) = 0.22 + 0.18 = 0.40$; the eigenvalues solve $\lambda^2 - 1.3\lambda + 0.40 = 0$:

$$\lambda = \frac{1.3 \pm \sqrt{1.69 - 1.60}}{2} = \frac{1.3 \pm 0.3}{2} \in \{0.8,\ 0.5\}$$

both inside the unit circle — **stationary, although** $\Phi_{22} = 1.1 > 1$: a single coefficient above one does not make a VAR explosive.

Mean, by (23). $I - \Phi = \begin{pmatrix} 0.8 & -0.3 \\ 0.6 & -0.1 \end{pmatrix}$, $\det = 0.8(-0.1) - (-0.3)(0.6) = -0.08 + 0.18 = 0.10$, so

$$(I - \Phi)^{-1} = \frac{1}{0.10}\begin{pmatrix} -0.1 & 0.3 \\ -0.6 & 0.8 \end{pmatrix} = \begin{pmatrix} -1 & 3 \\ -6 & 8 \end{pmatrix}, \qquad \mu = \begin{pmatrix} -1(0.2) + 3(0.4) \\ -6(0.2) + 8(0.4) \end{pmatrix} = \begin{pmatrix} 1.0 \\ 2.0 \end{pmatrix}.$$

Covariance, by (25). $\Phi \otimes \Phi$ has blocks $\Phi_{ij}\Phi$; for instance the top-left block is $0.2\Phi = \begin{pmatrix} 0.04 & 0.06 \\ -0.12 & 0.22 \end{pmatrix}$. Then

$$I_4 - \Phi \otimes \Phi = \begin{pmatrix} 0.96 & -0.06 & -0.06 & -0.09 \\ 0.12 & 0.78 & 0.18 & -0.33 \\ 0.12 & 0.18 & 0.78 & -0.33 \\ -0.36 & 0.66 & 0.66 & -0.21 \end{pmatrix}, \qquad \operatorname{vec}(\Sigma) = \begin{pmatrix} 2 \\ 1 \\ 1 \\ 1 \end{pmatrix},$$

and solving gives $\operatorname{vec}(\Gamma_0) = (25/9,\ 20/9,\ 20/9,\ 40/9)'$:

$$\Gamma_0 = \begin{pmatrix} 25/9 & 20/9 \\ 20/9 & 40/9 \end{pmatrix} = \begin{pmatrix} 2.778 & 2.222 \\ 2.222 & 4.444 \end{pmatrix}.$$

Check one entry of (24). First, $\Gamma_0\binom{0.2}{0.3} = \binom{0.2(25/9)+0.3(20/9)}{0.2(20/9)+0.3(40/9)} = \binom{11/9}{16/9}$; then $(\Phi\Gamma_0\Phi')_{11} = (0.2, 0.3)\binom{11/9}{16/9} = \frac{2.2+4.8}{9} = \frac{7}{9}$; and $\frac{7}{9} + \Sigma_{11} = \frac{7}{9} + \frac{18}{9} = \frac{25}{9}$ ✓.

Lag 1, by (26): $\Gamma_1 = \Phi\Gamma_0 = \begin{pmatrix} 11/9 & 16/9 \\ 7/9 & 32/9 \end{pmatrix}$. The contemporaneous correlation is $\frac{20/9}{\sqrt{(25/9)(40/9)}} = \frac{20}{\sqrt{1000}} = 0.632$. *(All entries computed exactly and by* ***`numpy.linalg.solve`*** *for this note.)*

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10 · U — the site's R script (`ar-var-R.txt`)

Taken from Penn State's STAT510: AR(1) and AR(2) fits by `sarima` to three LA pollution–mortality series (`cmort`, `tempr`, `part` from `astsa`), a VAR(1) with constant and trend on the three jointly (`VAR(x, p=1, type="both")`), and a VAR(1) on the `vars` package's Canada data (productivity, employment, real wage, unemployment) with 52-step impulse responses and their bands. The file prints no output, so no numbers are taken from it. Python equivalents: `statsmodels.tsa.ar_model.AutoReg` (OLS = §5), `statsmodels.tsa.api.VAR` with `.irf(52)`; see L01-11-python. **The impulse responses and their identification belong to Class 9.**

Links

- Neighbours: L01-06-gaussian-linear-regression (the regression machinery used in §5, §8, §9) · L01-08-p-hacking-evidence.
- Unit: L01-00-map · L01-09-formulas · L01-10-pitfalls · L01-11-python · L01-02-poisson-mle-properties (conditional vs full likelihood, again).
- Class 2: L02 §1 (forecasting as prediction).
- Rules: rules/04 (ARIMA, unit roots — Class 8) · rules/05 (VAR, IRF — Class 9).
- **Practice (number and printed page only):** T 2.2 (p. 104); W 11.C8 (p. 370); W 12.1 (p. 396); T 8.1 and T 8.2 (p. 462).

p-hacking and evidence · why 32 tries make "significance" likely, and why p = 0.036 can favour the null

> **Claim. A p-value is calibrated for one pre-specified test.** Search over 32 specifications of a null effect and, if they were independent, the chance that at least one gives $p < 0.05$ is $1 - 0.95^{32} \approx 0.81$; Bonferroni restores the 5% family-wise rate by demanding $p < 0.05/32 \approx 0.0016$. **Even a single honest $p < 0.05$ is not evidence for H_1**: with $X \sim N(\theta, 1)$, $H_0 : \theta = 0$ vs $H_1 : \theta = 4$ and $X = 1.8$, $p = 0.036$, yet the data are $e^{0.8} = 2.23$ times likelier under H_0, and with equal priors $P(H_1 \mid x) = 0.31$. **To reach $P(H_1 \mid x) = 0.9$ from these data one must start from prior odds of about 20:1 for H_1** — the hidden prior that reading $p < 0.05$ as strong evidence assumes. Prev: L01-07-ar-var-review · Next: L01-09-formulas · Map: L01-00-map

Notation.

Symbol	Meaning	Range
α	nominal size of each test	deck: 0.05
m	number of specifications (tests) tried	deck: 32
p	p-value of one test	$[0, 1]$
X	the single observation, $X \sim N(\theta, 1)$	
H_0, H_1	$\theta = 0$ and $\theta = 4$	
Φ, ϕ	standard normal CDF and density	
$f(x \mid H_j)$	density of X under H_j	
$BF_{10} = f(x \mid H_1)/f(x \mid H_0)$, $BF_{01} = 1/BF_{10}$	Bayes factors (likelihood ratio for two point hypotheses)	> 0
$P(H_j)$, $P(H_j \mid x)$	prior and posterior probabilities	

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1 · U — what p-hacking is (part1 slide 3; economics deck slides 3–4)

> "P-hacking (or data dredging) is the misuse of data analysis to produce statistically significant results (P < 0.05) where none may exist, often by trying multiple methods until a desired result appears." (part1, slide 3, quoting Wikipedia)

The practices (economics deck, slide 4): **specification searching** (many regressions, report the cleanest); **selective controls**; **sample slicing** (outliers, windows, subsamples); **outcome switching**; **optional stopping** (collect until significant); **HARKing** — hypothesising after the results are known. None requires bad faith; each is a fork in what Gelman and Loken (2014) call "the garden of forking paths". The formal symptom (slide 3): **under H_0 p-values are uniform on $[0, 1]$, and selective search produces a discontinuity at $p = 0.05$** (Brodeur et al., 2016).

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2 · C — why a single p-value has size α

Let T be a test statistic with continuous CDF F_0 under H_0, rejecting for large T, so $p = 1 - F_0(T)$. For $0 < \alpha < 1$:

$$P_{H_0}(p \le \alpha) = P_{H_0}(F_0(T) \ge 1-\alpha) = 1 - P_{H_0}(F_0(T) < 1-\alpha) = 1-(1-\alpha) = \alpha, \tag{1}$$

because $F_0(T) \sim U(0,1)$ under H_0 (probability integral transform). **(1) is a statement about one test, chosen before seeing the data.** It says nothing about the smallest of several p-values — which is what a reader is shown after a search.

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3 · C — the combinatorial explosion (part1 slides 4–6)

The setup (slide 4). Does a new teaching method raise test scores? $n = 30$ students, treatment vs control, **true effect zero**. The researcher explores: outlier rule (all / exclude >2 SD / exclude >1.5 SD); covariates (none / GPA / GPA + gender); sample (all / exclude late enrollees / exclude absentees); one- or two-tailed; outcome at week 4, 6 or 8.

The count (slide 5). "With 5 binary decisions, there are up to $2^5 = 32$ specifications." Let A_j be "specification j rejects". If the m tests were independent, each with size α by (1):

$$P(\text{no rejection}) = \prod_{j=1}^{m} P(A_j^c) = (1-\alpha)^m \tag{2}$$

$$P(\text{at least one rejection}) = 1-(1-\alpha)^m \tag{3}$$

Substituting $\alpha = 0.05$, $m = 32$ into (3):

$$1 - 0.95^{32} = 1 - 0.1937 = 0.806 \approx 0.81.$$

> **"There is roughly an 81% chance of finding at least one 'significant' result purely by chance, if the researcher reports only the best-looking specification."** (slide 5). A seeded check run for this note — 200,000 replications of 32 independent size-0.05 tests — gives 0.808.

m	1	5	10	32	162
$1-0.95^m$	0.050	0.226	0.401	**0.806**	0.9998

Two qualifications the slide's arithmetic hides.

- **The count.** The table's options are not binary: $3 \times 3 \times 3 \times 2 \times 3 = 162$ combinations, for which (3) gives 0.9998. "Up to 32" is the binary simplification.
- **The independence.** All specifications use the *same 30 students*, so their test statistics are strongly positively correlated, and (2) does not hold. For two-sided tests on jointly normal statistics, Šidák's inequality gives $P(\text{no rejection}) \ge (1-\alpha)^m$, so **0.81 is an upper bound**; at the other extreme, 32 perfectly correlated tests are one test, with rate 0.05. The true family-wise rate lies between — usually far above 0.05, which is the point.

Bonferroni (slide 6). Test each specification at α/m. By the union bound, with no independence assumption at all,

$$P\Big(\bigcup_{j=1}^{m} A_j\Big) \le \sum_{j=1}^{m} P(A_j) = m \cdot \frac{\alpha}{m} = \alpha, \tag{4}$$

so the family-wise rate is at most α: here $p < 0.05/32 = 0.0015625 \approx 0.0016$. (Without the correction the same bound reads $32 \times 0.05 = 1.6$ — useless, which is why (3) was needed.)

What the frequentist correction cannot fix (slide 6). "From the likelihood principle: inference should depend only on the data observed, not on the hypothetical tests that could have been run" — yet a p-value's meaning depends on exactly those unrun tests, and "the unreported search path is invisible to the reader." The same tension drives the stopping-rule example of L01-01-likelihood-sampling-rules: identical likelihoods, different p-values. **Remedy (slide 6): Bayesian model averaging or a multiverse analysis** — report the estimate across all specifications, so fragility is visible. The abstention regressions of L01-06-gaussian-linear-regression §8 are a small honest instance: five specifications side by side, with persistence stable at 0.65–0.72 in all.

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4 · C — a p-value below 0.05 that favours H_0 (part1 slides 7–9)

$X \sim N(\theta, 1)$, $H_0 : \theta = 0$ vs $H_1 : \theta = 4$, observed $X = 1.8$.

The p-value (slide 7):

$$p = P(X \ge 1.8 \mid \theta = 0) = 1 - \Phi(1.8) = 1 - 0.96407 = 0.0359 < 0.05. \tag{5}$$

"The result is 'statistically significant', so reject H_0. Surprisingly, nothing is said about H_1!" Swap the roles ($H_0 : \theta = 4$, $H_1 : \theta = 0$, reject for small X) and the p-value is

$$P(X \le 1.8 \mid \theta = 4) = \Phi(1.8 - 4) = \Phi(-2.2) = 0.0139 : \tag{6}$$

1.8 is "significant" against either hypothesis — it sits in the tail of both ("an even more extreme tail of H_1", slide 8).

The likelihood ratio (slide 8). With $\phi(z) = e^{-z^2/2}/\sqrt{2\pi}$, the constants cancel:

$$BF_{01} = \frac{f(x \mid H_0)}{f(x \mid H_1)} = \frac{\phi(1.8-0)}{\phi(1.8-4)} = \frac{\exp(-1.8^2/2)}{\exp(-(-2.2)^2/2)} = \frac{\exp(-1.62)}{\exp(-2.42)} = \exp(0.80) = 2.2255. \tag{7}$$

> **"The data are more than twice as likely under H_0 than under H_1. The observation $X = 1.8$ is closer to the null than to the alternative"** — 1.8 from 0, 2.2 from 4.

For every x. Expanding the squares,

$$\log BF_{10}(x) = -\frac{(x-4)^2}{2} + \frac{x^2}{2} = \frac{x^2 - (x^2 - 8x + 16)}{2} = 4x - 8, \tag{8}$$

so $BF_{10} = 1$ at $x = 2$, the midpoint. **Any x in $(1.645,\ 2)$ is significant at 5% one-sided (since $1 - \Phi(1.645) = 0.05$) and yet favours H_0.** At $x = 1.8$, (8) gives $4(1.8) - 8 = -0.8$, i.e. $BF_{10} = e^{-0.8} = 0.449$, the reciprocal of (7).

The posterior (slide 9). Bayes' theorem over the two hypotheses:

$$P(H_1 \mid x) = \frac{f(x \mid H_1)P(H_1)}{f(x \mid H_1)P(H_1) + f(x \mid H_0)P(H_0)}. \tag{9}$$

With $P(H_0) = P(H_1) = 0.5$ the priors cancel; divide through by $f(x \mid H_1)$ and substitute (7):

$$P(H_1 \mid x) = \frac{1}{1 + \frac{f(x|H_0)}{f(x|H_1)}} = \frac{1}{1 + BF_{01}} = \frac{1}{1 + 2.2255} = \frac{1}{3.2255} = 0.310. \tag{10}$$

"Despite a p-value of 0.036, the posterior probability of H_1 is only 31%." The Bayes factor $BF_{10} \approx 0.45$ "constitutes evidence in favour of H_0". A clean instance "of the Jeffreys–Lindley paradox in spirit": **a p-value measures distance from H_0 and is completely blind to where H_1 is.**

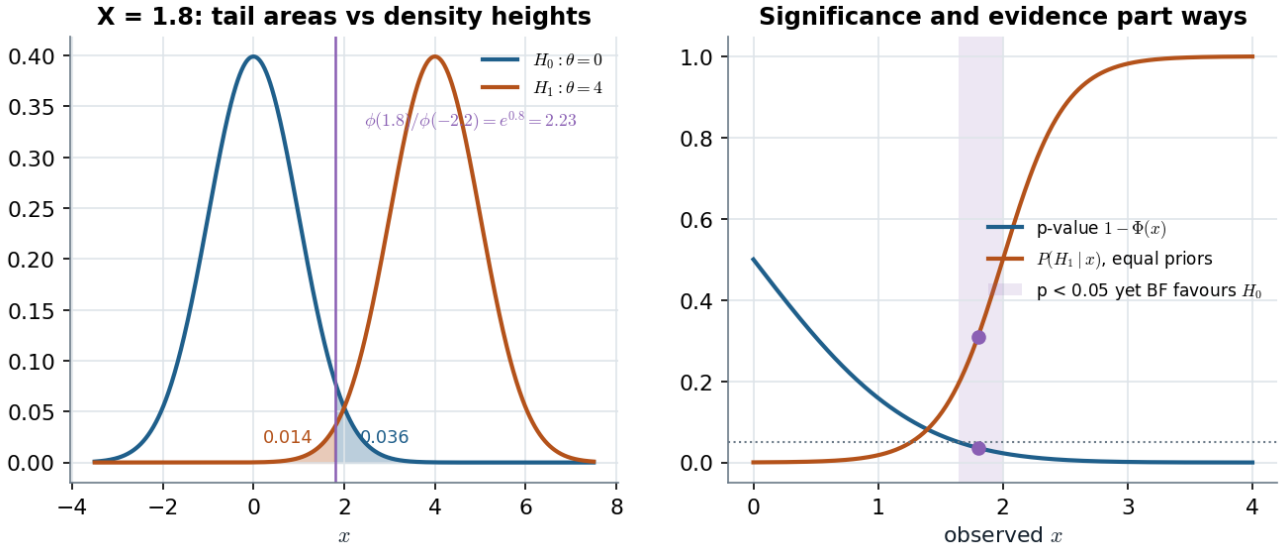


Figure — left: the two sampling densities at $x = 1.8$, tail areas (5)–(6) shaded; right: $p(x) = 1 - \Phi(x)$ and $P(H_1 \mid x) = 1/(1 + e^{-(4x-8)})$ from (8) and (10). Source: part1-p-hacking.pdf slides 7–9.

The left panel separates the two questions: the p-value is a *tail area* (shaded), the Bayes factor a *ratio of heights* at the observed point (the vertical line meets the blue curve higher than the orange one). On the right the two curves move in opposite directions, and the shaded band is where they disagree about which hypothesis the data support; $x = 1.8$ sits inside it.

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5 · C — the prior that would make $p = 0.036$ convincing (part1 slides 10–13)

Keep (9) but leave the priors free. Dividing numerator and denominator by $f(x \mid H_1)P(H_1)$:

$$P(H_1 \mid X) = \frac{1}{1 + BF_{01} \cdot \frac{P(H_0)}{P(H_1)}}. \tag{11}$$

Set (11) equal to 0.9 (slide 10):

$$1 + BF_{01}\frac{P(H_0)}{P(H_1)} = \frac{1}{0.9} \iff BF_{01}\frac{P(H_0)}{P(H_1)} = \frac{1}{0.9} - 1 = \frac{1}{9}. \tag{12}$$

Substitute $BF_{01} = 2.2255$ from (7) and solve (slide 11):

$$\frac{P(H_0)}{P(H_1)} = \frac{1}{9 \times 2.2255} = \frac{1}{20.03}, \qquad \boxed{\frac{P(H_1)}{P(H_0)} = 20.03 \approx 20} \tag{13}$$

(The deck, rounding BF_{01} to 2.23, prints 1/20.07.) Odds of 20:1 mean

$$P(H_1) = \frac{20}{21} = 0.952, \qquad P(H_0) = \frac{1}{21} = 0.048.$$

Equivalently, in odds form — posterior odds = Bayes factor × prior odds — $9 = 0.449 \times 20.03$.

Scenario (slide 13)	p-value	$P(H_1 \mid X)$
Equal priors, $X = 1.8$, $H_1 : \theta = 4$	0.036	0.31
Prior odds 20:1 in favour of H_1	0.036	0.90

> **"To obtain $P(H_1 \mid X) = 0.9$ from data that actually favour H_0 by a factor of 2.23, one must enter the analysis believing H_1 is 20 times more probable than H_0 a priori."** That is the implicit prior behind treating $p < 0.05$ as strong evidence here — "Making the prior explicit is not a weakness of Bayesian analysis, it is in fact its central virtue." (slide 12)

The three lessons (slide 13): p-hacking inflates false positives multiplicatively across specifications (§3); a significant p-value can coexist with a Bayes factor favouring H_0 (§4); a high posterior for H_1 despite unfavourable data needs a heavily loaded prior (§5).

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6 · U — the replication crisis and economics (economics deck)

All figures and studies below are the deck's, cited as the deck cites them.

Replication (slide 5). Open Science Collaboration (2015): only 36% of 100 psychology studies replicated. Camerer et al. (2016, 2018): 61% in experimental economics, 62% for social-science papers in *Nature/Science*. Effect sizes in successful replications are typically about 50% of the originals. Drivers: p-hacking, publication bias, low power, weak pre-registration.

Why it compounds (slide 6). Journals reward significant findings → incentives to reach $p < 0.05$ → forking paths push the real Type I error far above 5% → published effects biased upward, standard errors downward → meta-analyses and policy inherit the bias.

Evidence in economics (slide 7). Brodeur, Lé, Sangnier & Zylberberg (2016, *AEJ: Applied*; "Star Wars: The Empirics Strike Back") examined 50,000+ test statistics from top journals: a two-humped distribution of z around 1.96, with roughly 10–20% of marginally significant results attributable to p-hacking and selective reporting. Brodeur, Cook & Heyes (2020, *AER*): **IV and difference-in-differences show the most inflation near $p = 0.05$; RCTs and RDD less.** *(The deck's text lists the 2016 authors as "Brodeur, Lé, Sánchez, & Cook"; its reference list repeats that.)*

Why economics is vulnerable (slide 8) — observational data with many defensible specifications, a weak replication culture, significance stars as gatekeepers, long lags weakening pre-specification, and policy stakes. **Examples (slide 9):** minimum-wage employment effects (Neumark & Wascher vs

Card & Krueger); Sala-i-Martin's (1997) two million growth regressions; Reinhart & Rogoff's 90% debt threshold reversed after coding errors (Herndon et al., 2014); behavioural headline effects weakened by Many Labs.

Responses (slide 10): pre-registration and pre-analysis plans (AEA RCT Registry, since 2013); mandatory data and code archives; replication journals; multiple-testing corrections (Bonferroni — (4) — Romano–Wolf, sharpened FDR by Anderson, 2008); specification curves and multiverse analyses; and Benjamin et al. (2018): redefine "significant" as $p < 0.005$.

> **"The fundamental fix is cultural: rewarding credible research, not just significant research."** (slide 11)

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7 · G — where this goes

Bayes factors for composite hypotheses (priors over θ under H_1, marginal likelihoods), posterior model probabilities and Bayesian model averaging are Classes 3+4 (G §3.2.4; rules/07). There the Jeffreys–Lindley paradox appears in full: with a diffuse prior under H_1, a fixed $p = 0.05$ supports H_0 ever more strongly as n grows.

Links

- Neighbours: L01-07-ar-var-review · L01-09-formulas.
- Unit: L01-00-map · L01-10-pitfalls · L01-11-python · L01-01-likelihood-sampling-rules (same likelihood, different p-values; the likelihood principle) · L01-06-gaussian-linear-regression (§6 model selection, §8 reporting several specifications) · L01-04-first-bayesian-poisson (the first prior-to-posterior update).
- Class 2: L02 §3 (choosing among many models without fooling yourself).
- Rules: rules/07 (Bayes factors, model probabilities).
- **Practice (forward, Classes 3+4; number and printed page only):** G 3.7 and G 3.8 (p. 40), Bayes factors.

Class 1 · Formulas — the exam specification, every result, every number

Prev: L01-08-p-hacking-evidence · Next: L01-10-pitfalls · Map: L01-00-map

Equations are cited by the **section note's own numbering**: "L01-02 (9)" is equation (9) of L01-02-poisson-mle-properties. Every number in §2 is asserted in L01-11-python (script `L01_python_checks.py`).

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0 · The exam specification, mapped

0.1 · The site's description of Class 1 (`Aula/COURSE.md`)

"Brief review of basic ingredients", in five items:

#	Site item (verbatim)	Where it is settled
i	"Parametric models, likelihoods, estimators and their sampling distributions"	L01-01-likelihood-sampling-rules (one likelihood, three sampling distributions); L01-02-poisson-mle-properties (the Poisson MLE checklist); L01-03-coal-mining-change-point (a two-regime likelihood); L01-05-comparing-estimators-mse-bootstrap (exact, MC and bootstrap sampling distributions)
ii	"Gaussian linear regression: estimation and variable selection"	L01-06-gaussian-linear-regression §2–§4 (estimation), §5–§7 (selection, overfitting)
iii	"AR(1) model: estimation, unit root, equilibrium distribution"	L01-07-ar-var-review §5 (estimation), §7 (unit root), §6 (equilibrium)
iv	"AR(p) model: connection to Gaussian linear regression"	L01-07-ar-var-review §8
v	"VAR(1) model: multivariate estimation, matrix notation"	L01-07-ar-var-review §9

0.2 · The goals in `Map/class-plan.md`, item by item

#	Goal	Note and section
1	Write a likelihood and derive the MLE	L01-01 §2–§3 (Bernoulli, binomial, negative binomial); L01-02 §3–§4 (Poisson); L01-05 §6 (uniform: boundary MLE); L01-06 §4 (Gaussian regression); L01-07 §5 (AR(1), conditional)

#	Goal	Note and section
2	Sufficiency, unbiasedness, consistency, efficiency (Cramér–Rao)	L01-02 §3 (sufficiency, by definition and by factorisation), §5 (unbiased, consistent), §6 (CRLB, efficiency); L01-05 §6 ($x_{(n)}$ sufficient)
3	Check them by Monte Carlo	L01-02 §8 (slide 10's table, ± 0.25 coverage); L01-05 §7 (four estimators at $\theta = 3$, $n = 10$)
4	Compare estimators by MSE (the four $U(0, \theta)$ estimators)	L01-05 §2–§5 (MSE = Var + bias^2, ranking for $n \geq 3$), §8 (G: the best $c\, x_{(n)}$ is biased)
5	Nonparametric vs parametric bootstrap, and why the nonparametric one fails for the maximum	L01-05 §9 (nonparametric), §10 (the point mass $1 - (1 - 1/n)^n \to 0.632$), §11 (parametric fix)
6	Overdispersion and a Poisson change point (coal-mining disasters, profile likelihood, LR test)	L01-03 §2–§3 (dispersion test), §4–§5 (profile likelihood, $w(\tau)$), §6 (LR = 70.0 and why not χ^2_1)
7	Gaussian linear regression in matrix form and variable selection	L01-06 §2 (normal equations, projection), §4 (OLS = MLE), §5–§6 (R^2, $\bar{R}^2$, C_p, AIC, BIC, best subset), §7 (overfitting), §8 (reading real output)
8	AR(1) estimation, the stationary (equilibrium) distribution and the unit-root case	L01-07 §4 (moments, ACF), §5 (OLS = conditional MLE), §6 (equilibrium law, half-life), §7 (random walk)
9	AR(p) as a regression	L01-07 §8
10	VAR(1) in matrix notation	L01-07 §9 (stationarity by eigenvalues, mean, vec covariance; Tsay Ex. 8.3)
11	p-hacking and reproducibility	L01-08 §2–§3 (size, family-wise rate, Bonferroni), §4–§5 (p-value vs Bayes factor vs posterior), §6 (the evidence in economics)
—	Also on the Class 1 site list: the first Bayesian example	L01-04 (Exp prior → Gamma posterior, shrinkage)
—	Also on the Class 1 site list: sampling rules and the likelihood principle	L01-01 (same likelihood, different p-values; Beta posterior unchanged)

Exam format (`Map/class-plan.md`, "Final exam"): no coding. Expect **derivations** (likelihoods, MLEs, information, posteriors, AR moments) and **reading printed output** (the abstention table of L01-06 §8 is the model).

1 · Every definition and result

1.1 · Likelihood and sampling rules → L01-01-likelihood-sampling-rules

Eq.	Statement	Where
(1)–(3)	Bernoulli, binomial and negative-binomial data all give $L(\theta) \propto \theta^s(1-\theta)^{n-s}$	§2
Def.	**Likelihood principle**: proportional likelihoods give the same likelihood-based and Bayesian inference	§2 (slide 1)
(4)–(6)	$\ell = s\log\theta + (n-s)\log(1-\theta)$; score $s/\theta - (n-s)/(1-\theta)$; $\hat\theta = s/n$	§3
(7)–(8)	$J(\theta) = \frac{s}{\theta^2} + \frac{n-s}{(1-\theta)^2}$; $J(\hat\theta) = \frac{n}{\hat\theta(1-\hat\theta)}$, the same under every rule	§3
(9)–(11)	Binomial: $P(\hat\theta = k/n) = \binom{n}{k}\theta^k(1-\theta)^{n-k}$; $E(\hat\theta) = \theta$, $V = \theta(1-\theta)/n$; $I(\theta) = n/[\theta(1-\theta)]$ (CRLB attained)	§4
(12)	$\hat\theta \overset{a}{\sim} N(\theta, \theta(1-\theta)/n)$	§4
(13)–(15)	Negative binomial: $P(N=m) = \binom{m-1}{s-1}\theta^s(1-\theta)^{m-s}$; $I(\theta) = s/[\theta^2(1-\theta)]$; $\hat\theta \overset{a}{\sim} N(\theta, \theta^2(1-\theta)/s)$	§5
(16)–(18)	Delta method on $N \overset{a}{\sim} N(s/\theta, s(1-\theta)/\theta^2)$ gives (15); plug-in variances coincide: $\hat\theta(1-\hat\theta)/n$	§6
(19)–(20)	$E(s/N) > \theta$ (Jensen); for $s=1$, $E(1/N) = -\theta\log\theta/(1-\theta)$	§7
Haldane	$(s-1)/(N-1)$ is exactly unbiased for $s \geq 2$	§7
Def.	p-value = *nível descritivo* (B Def. 6.3.1, p. 100); binomial $P(S \leq s)$, negative binomial $P(N \geq n)$	§9
(21)–(22)	Beta(a,b) prior → Beta$(a+s, b+n-s)$ under every rule; posterior mean = weighted average of prior mean and s/n	§10
(23)	$n \geq z^2\theta(1-\theta)/E^2$; worst case $\theta = 1/2$	§12

1.2 · The Poisson MLE → L01-02-poisson-mle-properties

Eq.	Statement	Where
(1)–(3)	$\Pr(y \mid \lambda) = \lambda^y e^{-\lambda}/y!$; $E(y) = V(y) = \lambda$ (equidispersion)	§1
(4)–(5)	$L(\lambda) = c\,\lambda^{s_n} e^{-n\lambda}$	§3
Def. 2.2.1 / Thm 2.2.1	**Sufficiency**: the law of the sample given T is free of λ; the factorisation criterion	§3
(6)	$Y \mid T = t \sim$ multinomial$(t; 1/n, \dots, 1/n)$, free of λ	§3
(7)–(10)	$\ell = \log c + s_n \log\lambda - n\lambda$; $\hat\lambda = \bar y$; $\ell'' = -s_n/\lambda^2 < 0$ (no MLE if $s_n = 0$)	§4
(11)–(12)	$E(\hat\lambda) = \lambda$; $V(\hat\lambda) = \lambda/n \to 0$	§5
(13)	Chebyshev: $P(\lvert\hat\lambda - \lambda\rvert > \epsilon) \le \lambda/(n\epsilon^2) \to$ **consistency** (B Def. 3.7.1)	§5
(14)	$\hat\lambda \overset{a}{\sim} N(\lambda, \lambda/n)$; SE $= \sqrt{\hat\lambda/n}$	§5
(15)–(18)	**CRLB** $= 1/I(\lambda)$, $I(\lambda) = n/\lambda$; $e(\hat\lambda) = 1$: **efficient** (B Def. 2.1.1, Thm 2.1.1)	§6
(19)	$V(s^2) = \lambda/n + 2\lambda^2/(n-1)$	§7
(20)–(21)	$e(s^2) = \frac{1}{1+2\lambda n/(n-1)} \to \frac{1}{1+2\lambda}$	§7

1.3 · Coal mining, change point, LR → L01-03-coal-mining-change-point

Eq.	Statement	Where
(1)	Dispersion test $D = (n-1)s^2/\bar y \overset{a}{\sim} \chi^2_{n-1}$; upper tail = overdispersion	§3
(2)	Law of total variance: $V(y) = E(\lambda) + \pi(1-\pi)(\lambda_1 - \lambda_2)^2$ — a mixture is overdispersed	§3
(3)–(4)	Two-regime model and its log-likelihood	§4
(5)	$\hat\lambda_j(\tau) = s_j/n_j$	§4
(6)	**Profile likelihood** $\ell_p(\tau) = s_1 \log\hat\lambda_1 + s_2 \log\hat\lambda_2 - 191$ (+ const.) (B §3.4, p. 47)	§4
(7)	$w(\tau) = e^{\ell_p(\tau)} / \sum_W e^{\ell_p(\tau')} =$ posterior of τ under a uniform prior **with rates profiled out**	§4

Eq.	Statement	Where
(8)–(9)	LR $= 2\{\ell_p(\hat{\tau}) - \ell(\hat{\lambda})\} = 2[s_1 \log(\hat{\lambda}_1/\hat{\lambda}) + s_2 \log(\hat{\lambda}_2/\hat{\lambda})]$	§6
GLRT / Wilks	$\lambda(x) = \sup_{\Theta_0} L / \sup_{\Theta} L$ (B (6.5.1)); $-2\log\lambda \to \chi^2$ under regularity (B Thm 6.5.1) — **fails here**: τ searched, unidentified under H_0, discrete	§6
(10)	G: integrated $p(\tau \mid y) \propto \prod_j a\,\Gamma(s_j+1)/(n_j+a)^{s_j+1}$	§8

1.4 · The first Bayesian example → L01-04-first-bayesian-poisson

Eq.	Statement	Where
(1), (3)	Prior $p(\lambda) = ae^{-a\lambda}$; $E = 1/a$, $V = 1/a^2$	§1
(2)	$\int_0^\infty \lambda^k e^{-b\lambda} d\lambda = \Gamma(k+1)/b^{k+1}$	§1
(4)	Bayes: posterior ∝ likelihood × prior (G (2.3))	§2
(6)–(8)	Kernel $\lambda^{s_n} e^{-(a+n)\lambda} \Rightarrow$ $\lambda \mid y \sim G(s_n+1, a+n)$ (shape, **rate**)	§3
(9)	Gamma: $E = \alpha/\beta$, $V = \alpha/\beta^2$, mode $(\alpha-1)/\beta$	§4
(10)–(12)	$E(\lambda \mid y) = \frac{s_n+1}{a+n}$; mode $\frac{s_n}{a+n}$; $V = \frac{s_n+1}{(a+n)^2}$	§4
(13)–(14)	**Shrinkage**: $E(\lambda \mid y) = \frac{a}{n+a} \cdot \frac{1}{a} + \frac{n}{n+a} \cdot \bar{y}$; mode $= \frac{n}{n+a}\bar{y}$	§6

1.5 · MSE, Monte Carlo, bootstrap → L01-05-comparing-estimators-mse-bootstrap

Eq.	Statement	Where
(1)–(3)	MSE $= E[(\hat{\theta}-\theta)^2] = V(\hat{\theta}) + B(\hat{\theta})^2$ (B Def. 1.3.7, (1.3.2))	§2
(4)–(8)	$U(0,\theta)$: $E(x) = \theta/2$, $V(x) = \theta^2/12$; $E(\bar{x}) = \theta/2$, $V(\bar{x}) = \theta^2/(12n)$	§3
(9)–(14)	$x_{(n)}/\theta \sim \text{Beta}(n,1)$; $E(x_{(n)}) = \frac{n}{n+1}\theta$, $V(x_{(n)}) = \frac{n\theta^2}{(n+1)^2(n+2)}$	§3
(15)–(18)	MSEs: $\frac{1+3n}{12n}\theta^2$, $\frac{2\theta^2}{(n+1)(n+2)}$, $\frac{\theta^2}{3n}$, $\frac{\theta^2}{n(n+2)}$	§3–§4

Eq.	Statement	Where
(19)–(22)	**Ranking** $\hat{\theta}_4 < \hat{\theta}_2 < \hat{\theta}_3 < \hat{\theta}_1$ for $n \geq 3$; all equal at $n = 1$; $\hat{\theta}_2 = \hat{\theta}_3$ at $n = 2$	§5
(23)–(25)	$L(\theta) = \theta^{-n}\mathbb{1}(\theta \geq x_{(n)})$; **boundary MLE** $x_{(n)}$ (score never zero); $x_{(n)}$ sufficient	§6
(26)	MC: $\widehat{\text{MSE}} = \frac{1}{N}\sum_j(\hat{\theta}^{(j)} - \theta)^2$	§7
(27)–(29)	G: $\text{MSE}(c\,x_{(n)})$ minimised at $c^* = \frac{n+2}{n+1}$, value $\theta^2/(n+1)^2$	§8
(30)	Bootstrap SE (ISLP (5.8))	§9
(31)–(32)	$P(x^*_{(n)} = x_{(n)}) = 1-(1-1/n)^n \to 1-e^{-1} = 0.632$: **nonparametric bootstrap inconsistent for the max**	§10
—	Parametric bootstrap: resample $U(0, \hat{\theta}_4)$; reproduces the plug-in SD	§11

1.6 · Gaussian linear regression → L01-06-gaussian-linear-regression

Eq.	Statement	Where
(1)–(2)	Simple OLS; on standardised data the slope is r_{xy}	§1
(3)–(7)	$y = X\beta + \varepsilon$; normal equations $X'X\hat{\beta} = X'y$; $\hat{\beta} = (X'X)^{-1}X'y$	§2
(8)	$\text{RSS}(\beta) = \text{RSS}(\hat{\beta}) + \|X(\hat{\beta}-\beta)\|^2$: global minimum	§2
(9)–(12)	$P = X(X'X)^{-1}X'$ symmetric, idempotent, $\text{tr}P = q$; $X'e = 0$; $\text{RSS} = y'(I-P)y$; $e = (I-P)\varepsilon$	§2.1
(13)–(14)	$\hat{\beta} = \beta + (X'X)^{-1}X'\varepsilon$; $V(\hat{\beta} \mid X) = \sigma^2(X'X)^{-1}$	§3
(15)–(17)	Gaussian log-likelihood; $\hat{\beta}_{MLE} = \hat{\beta}_{ols}$; $\hat{\sigma}^2_{MLE} = \text{RSS}/n$	§4
(18)–(19)	$E(\text{RSS}) = \sigma^2(n-q)$; unbiased $\hat{\sigma}^2 = \text{RSS}/(n-q)$	§4
(20)–(21)	$R^2 = 1 - \text{RSS}/\text{TSS}$; never falls when regressors are added	§5
(22)–(23)	$\bar{R}^2 = 1-(1-R^2)\frac{n-1}{n-k-1}$ (W (6.21)–(6.22)); rises iff the added $\|t\| > 1$ (group: $F > 1$)	§5
(24)	$C_p = \frac{1}{n}(\text{RSS} + 2d\hat{\sigma}^2)$, $\text{AIC} = C_p/\hat{\sigma}^2$, $\text{BIC} = \frac{1}{n}(\text{RSS} + \log(n)d\hat{\sigma}^2)$	§6

Eq.	Statement	Where
(25)	Interaction model sizes $p_k = 1 + \sum_{j \le k} \binom{13}{j}$	§7
(26)–(27)	$E(\text{MSE}_{in}) = \sigma^2(1 - p/n)$, $E(\text{MSE}_{out}) = \sigma^2(1 + p/n)$: optimism $2p\sigma^2/n$	§7
(28)	Municipal abstention specification	§8
W (4.41)	$F = \frac{(R^2_{ur} - R^2_r)/q}{(1 - R^2_{ur})/(n-k-1)}$ (homoskedastic form)	§8.3
(29)	HC1: $\frac{n}{n-q}(X'X)^{-1}(\sum e_i^2 x_i x_i')(X'X)^{-1}$	§8.5

1.7 · AR, unit root, VAR → L01-07-ar-var-review

Eq.	Statement	Where
Def.	Strict vs **weak** stationarity (constant mean, γ_ℓ depends on the lag only)	§1
(1)–(2)	$\gamma_{-\ell} = \gamma_\ell$; $\rho_\ell = \gamma_\ell/\gamma_0$ and its sample version	§1–§2
(3)	Ljung–Box $Q(m) = T(T+2)\sum_{\ell \le m} \hat\rho_\ell^2/(T-\ell) \overset{a}{\sim} \chi^2_m$	§2
(4)	Linear series: $\gamma_\ell = \sigma_a^2 \sum_j \psi_j \psi_{j+\ell}$	§3
(5)–(9)	AR(1): $\mu = \phi_0/(1-\phi_1)$; MA(∞) weights ϕ_1^i; $\gamma_0 = \sigma_a^2/(1-\phi_1^2)$; **stationary iff** $\lvert\phi_1\rvert < 1$	§4
(10)–(11)	$\gamma_\ell = \phi_1 \gamma_{\ell-1}$; $\rho_\ell = \phi_1^\ell$	§4
(12)–(13)	Conditional Gaussian likelihood = OLS of r_t on $(1, r_{t-1})$; exact MLE adds $p(r_1)$	§5
(14)–(16)	$y_t \mid y_0 \sim N(\mu + \phi^t(y_0 - \mu),\ \sigma^2\frac{1-\phi^{2t}}{1-\phi^2}) \to N(\mu, \frac{\sigma^2}{1-\phi^2})$: **equilibrium distribution**	§6
(17)	Half-life $h = \ln 0.5/\ln\lvert\phi\rvert$	§6
(18)–(19)	Random walk: $V(y_t) = t\sigma^2$; $E(y_{t+h} \mid y_t) = y_t$; $\text{Corr}(y_t, y_{t+h}) = \sqrt{t/(t+h)}$	§7
(20)–(21)	AR(p) as a multiple regression; $\mu = \phi_0/(1 - \sum \phi_i)$; $\hat\sigma_a^2 = \sum \hat a_t^2/(T - 2p - 1)$	§8
(22)	VAR(1): $r_t = \phi_0 + \Phi r_{t-1} + a_t$, $\text{Cov}(a_t) = \Sigma$; equation-by-equation OLS	§9
—	Stationary iff all eigenvalues of Φ have modulus < 1	§9

Eq.	Statement	Where
(23)	$\mu = (I - \Phi)^{-1}\phi_0$	§9
(24)–(25)	$\Gamma_0 = \Phi\Gamma_0\Phi' + \Sigma$; $\mathrm{vec}(\Gamma_0) = (I_{k^2} - \Phi \otimes \Phi)^{-1}\mathrm{vec}(\Sigma)$	§9
(26)	$\Gamma_\ell = \Phi^\ell \Gamma_0$	§9

1.8 · p-hacking and evidence → L01-08-p-hacking-evidence

Eq.	Statement	Where
(1)	$P_{H_0}(p \le \alpha) = \alpha$ — for **one** pre-specified test	§2
(2)–(3)	Independent tests: $P(\text{at least one rejection}) = 1 - (1-\alpha)^m$	§3
(4)	Bonferroni: test at α/m; family-wise rate $\le \alpha$ with no independence	§3
(5)–(6)	$p = 1 - \Phi(1.8)$; the reversed test's $p = \Phi(-2.2)$	§4
(7)–(8)	$BF_{01} = \phi(1.8)/\phi(-2.2) = e^{0.8}$; $\log BF_{10}(x) = 4x - 8$	§4
(9)–(11)	$P(H_1 \mid x) = 1\big/(1 + BF_{01}\, P(H_0)/P(H_1))$	§4–§5
(12)–(13)	Prior odds needed for $P(H_1 \mid x) = 0.9$: $P(H_1)/P(H_0) = 9\, BF_{01}$	§5

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2 · Every number and worked case

2.1 · Sampling rules (L01-01)

Case	Numbers
$s = 1$, $n = 10$	$\hat\theta = 0.1$; $J(\hat\theta) = 1000/9 = 111.1$; asymptotic se 0.095; Wald 95% $(-0.086, 0.286)$ exact $E(s/N) = 0.1 \times 2.302585/0.9 =$ **0.256**, sd **0.281** (binomial: 0.100, 0.095) p-values for $H_0 : \theta = 0.5$ vs $<$: binomial $11/1024 =$ **0.0107**; negative binomial $0.5^9 =$ **0.0020**
$s = 4$, $n = 10$	$\hat\theta = 0.4$; se 0.155; Wald $(0.096, 0.704)$; exact NegBin mean **0.461**, sd **0.178** p-values: binomial $386/1024 =$ **0.377**; negative binomial $130/512 =$ **0.254**
Uniform prior	Beta(2,10), mean **0.167**; Beta(5,7), mean **0.417**

Case	Numbers
Site Bernoulli code	$s = 7$, $n = 10$: Wald 90% (0.462, 0.938), 95% (0.416, 0.984), 99% (0.327, **1.073**); Beta(3,3) → Beta(10,6): mean **0.625**, sd 0.117, 95% (0.384, 0.837); logistic part: $\hat{\alpha} = 2.877$, $\hat{\beta} = 17.995$ (file's comments)
Sample size	$E = 0.05$: $n \geq 400$ with $z \approx 2$; $n = 385$ with $z = 1.96$ (384.1); $E = 0.01$: 10,000

2.2 · Poisson MLE (L01-02)

Case	Numbers
pmf	$\Pr(y = 2 \mid 2) =$ **0.271**; $\Pr(y \leq 3 \mid 2) =$ **0.857**; $\Pr(y \leq 3 \mid 1) = 0.981$; $\Pr(y = 3 \mid 2) = 0.18045$ (handwritten 0.181)
Efficiency of s^2, $\lambda = 2$	theory $V(s^2)$: 1.089, 0.203, 0.050 at $n = 10, 50, 200$; efficiency **0.184, 0.197, 0.199** → **0.2**
Deck MC (slide 10)	$\bar{y}$ mean 2.003/1.994/1.998, var 0.193/0.038/0.011 (theory 0.200/0.040/0.010); s^2 var 1.245/0.187/0.051, efficiency 0.161/0.214/0.197
±0.25 coverage	deck MC 0.420/0.799/0.984; exact **0.424/0.789/0.987**; normal 0.424/0.789/0.988; Chebyshev bound −2.2/0.36/0.84
Figure seeds	MC variances 0.182, 0.041, 0.011 (figure `L01-poisson-mle-sampling`)

2.3 · Coal mining (L01-03)

Case	Numbers
Data	$n = 112$, $s = 191$; first 40 years 125, first 41 years 127; 33 zero years; 20 years with ≥ 4
1851–1962	$\hat{\lambda} =$ **1.705**, SE 0.123, CI (1.464, 1.947); $s^2 = 2.714$; $s^2/\bar{y} = 1.592$; $D = 176.7$, $p < 0.001$
1851–1890	**3.125**, SE 0.280, CI (2.577, 3.673); $s^2 = 2.574$; index 0.824; $D = 32.1$, $p = 0.774$
1891–1962	**0.917**, SE 0.113, CI (0.696, 1.138); $s^2 = 1.063$; index 1.160; $D = 82.4$, $p = 0.168$
Mixture	$V = 1.705 + 1.120 = 2.825$; sample split $301.3 = 175.9 + 125.4$
Fit of one rate	zeros 0.182 fitted vs 0.295 observed; ≥ 4: 0.094 vs 0.179
Relative log-lik at 1.705	−18.92 (1851–1890), −15.81 (1891–1962)
Profile at $\tau = 1891$	$n_1/n_2 = 41/71$, $\hat{\lambda}_1 = 3.098$, $\hat{\lambda}_2 = 0.901$, ratio **3.436**; $\ell_p = -54.055$; $\ell(\hat{\lambda}) = -89.049$

Case	Numbers
LR	**69.99 ≈ 70.0**; at fixed $\tau = 1890$: 69.47, χ^2_1 $p \approx 8 \times 10^{-17}$; χ^2_2 tail at 70.0: 6×10^{-16}
$w(\tau)$, 1885–1895	0.008, 0.091, 0.105, 0.038, 0.154, 0.194, **0.251**, 0.098, 0.039, 0.016, 0.007; 1889–1892 together **0.696**; normaliser 3.983 (rounded gaps) / 3.985 (exact)
Integrated, $a = 1/2$ (G)	0.008, 0.089, 0.104, 0.038, 0.153, 0.194, **0.251**, 0.099, 0.040, 0.017, 0.007

2.4 · First Bayesian example (L01-04)

Case	Numbers
Prior $a = 1/2$	$E = 2$, $V = 4$, sd 2; central 95%: (0.051, 7.38)
1851–1891, $n = 41$, $s_n = 127$	posterior $G(128, 41.5)$: mean $128/41.5 =$ **3.084**, mode $127/41.5 =$ **3.060**, sd $\sqrt{128}/41.5 =$ **0.273**; MLE $127/41 =$ **3.098**
Intervals	Wald (2.559, 3.636), SE 0.275; equal-tail posterior (2.573, 3.641)
Shrinkage	prior weight $0.5/41.5 = 0.01205$; $0.01205 \times 2 + 0.98795 \times 3.09756 = 3.08434$
All 112 years	posterior mean $192/112.5 = 1.707$ vs MLE 1.705

2.5 · MSE and the bootstrap (L01-05)

Case	Numbers
MSE in units of θ^2	$n = 1$: all 1/3; $n = 2$: 7/24, 1/6, 1/6, 1/8; $n = 3$: 10/36, 1/10, 1/9, 1/15; $n = 10$: 31/120, 1/66, 1/30, 1/120
$\theta = 3$, $n = 10$, theory	means 1.500, 2.727, 3.000, 3.000; SDs 0.274, 0.249, 0.548, 0.274; MSEs **2.325, 0.136, 0.300, 0.075**
Deck MC (slide 12)	means 1.497, 2.717, 2.994, 2.989; SDs 0.278, 0.264, 0.555, 0.290; MSEs 2.336, 0.149, 0.308, 0.084
Limit of MSE($\bar{x}$)	$\theta^2/4 = 2.25$
Best $c\,x_{(n)}$ (G)	$c^* = 12/11 = 1.0909$; MSE $= 9/121 = 0.0744 < 0.075$
Bootstrap point mass	$n = 10$: **0.651** (deck 0.666); $n = 50$: 0.636 (0.622); $n = 100$: 0.634 (0.632); limit **0.632**
Bootstrap SDs (slide 14)	$n = 10$: 0.117, 0.062, 0.234, 0.068 vs theory 0.274, 0.249, 0.548, 0.274; $n = 50$: 0.113, 0.041, 0.226, 0.042 vs 0.122, 0.058, 0.245, 0.059; $n = 100$: 0.089, 0.036, 0.177, 0.036 vs 0.087, 0.029, 0.173, 0.030
Parametric (slide 19)	$n = 10$, $\hat{\theta}_4 = 2.568$: 0.237, 0.213, 0.475, 0.234 (plug-in 0.234, 0.213, 0.469, 0.234); $n = 50$, 2.926: 0.119, 0.054, 0.238, 0.055; $n = 100$, 3.029: 0.084, 0.031, 0.167, 0.032

Case	Numbers
Weighted mean, $n = 21$	$\sum a_i^2 = 891/14641 = 0.0609$ vs $1/21 = 0.0476$; efficiency 0.78
Median under t_4	$f(0) = 0.375$, asymptotic variance $1.78/n$ vs $2/n$ for $\bar{x}$; normal-case efficiency $2/\pi = 0.64$; Beta(3,9) median 0.236 vs mean 0.25
`bootstrap-example.html`	$\bar{x} = 0.10006$, $\hat{\sigma}_{MLE} = 0.9129307$; t interval $[-0.15939, 0.35951]$; σ^2 interval $[0.593, 1.321]$
B Table 1.1 (corrected)	$\text{MSE}[x_{(n)}] = \theta^2/66$ at $n = 10$, $\theta^2/231$ at $n = 20$; ratio column 0.36, 0.18, 0.059, 0.017

2.6 · Regression (L01-06)

Case	Numbers
Houseprice ($n = 128$)	size alone $R^2 = 31\%$, resid. SD 71.4; + brick 47%, 62.1; + neighbourhood **69%, 48.1**; $n/(n-6) = 1.049$; $2^6 = 64$ (or $2^7 = 128$) subsets; $\log 128 = 4.85$
Wooldridge p. 181	$R^2 = 0.30$, $n = 51$, $k = 10 \to \bar{R}^2 = 0.125$; $R^2 = 0.10 \to -0.125$
Best-subset simulation (seed 128)	R^2 0.337 → 0.644; $\bar{R}^2$ peaks at $d = 5$ (0.627); C_p picks $d = 4$ (1.1356 vs 1.1361); BIC picks $d = 3$ (1.211)
Boston overfitting	$p_k = 14, 92, 378, 1093, 2380$; R^2_{in} 0.7303 → 1.0000; R^2_{out} 0.7469 → 0.0519; RMSE_{out} 4.48 → 47.13 (×10.5)
Abstention, model (4)	persistence 0.684 (SE 0.009), $t = 76.0$, CI [0.666, 0.702]; size dummies 0.069/0.439/0.960/1.140, $t = 1.03/3.92/6.32/3.60$
Abstention, tests	size: robust $F = 13.22$ (UF), 10.84 (region); R^2-form 11.4, range [5.7, 17.1]; $F_{4,5531}$ 5% critical 2.37; UF vs region: 22 restrictions, $F = 31.08$ / 32.09
Abstention, fit	R^2 0.682, 0.727, 0.755, 0.728, 0.757; $\bar{R}^2$ 0.754 → 0.756; model (0) at the means = 20.728

2.7 · AR and VAR (L01-07)

Case	Numbers
IBM returns	Ljung–Box $Q(30) = 38.241$, $p = 0.1437$; squared returns $Q(30) = 182.12$, $p \approx 10^{-23}$
Wooldridge	$\phi = 0.9$: $\text{Corr}(y_t, y_{t+5}) = 0.59049$; random walk $\sqrt{1000/1100} = 0.9535$
$\phi = 0.9$, $\sigma = 1$, $y_0 = 8$	equilibrium variance **5.263** (sd 2.294); half-life **6.58**; at $t = 10$: mean 2.79, variance 4.62; ±2 sd band ±4.59

Case	Numbers
$\phi = 0.5$	half-life 1; equilibrium variance 1.333; band ± 2.31
Tsay AR(3)	$\mu = 0.0091/1.007 = 0.00904$
Tsay Ex. 8.3	eigenvalues 0.8, 0.5 (stationary although $\Phi_{22} = 1.1$); $\mu = (1, 2)'$; $\Gamma_0 = \frac{1}{9}\begin{pmatrix} 25 & 20 \\ 20 & 40 \end{pmatrix}$; $\Gamma_1 = \frac{1}{9}\begin{pmatrix} 11 & 16 \\ 7 & 32 \end{pmatrix}$; correlation 0.632

2.8 · p-hacking (L01-08)

Case	Numbers
Family-wise rate	$m = 1, 5, 10, 32, 162$: 0.050, 0.226, 0.401, **0.806**, 0.9998; Bonferroni $0.05/32 = 0.0016$
$X = 1.8$	$p =$ **0.0359**; reversed $p = 0.0139$; $BF_{01} = e^{0.8} =$ **2.2255** ($BF_{10} = 0.449$); $P(H_1 \mid x) =$ **0.310**
Prior odds needed	$9 \times 2.2255 =$ **20.03** (deck 20.07); $P(H_1) = 20/21 = 0.952$
Disagreement band	$x \in (1.645, 2)$: $p < 0.05$ yet $BF_{10} < 1$
Replication (deck)	36% psychology; 61% experimental economics; 62% *Nature/Science*; replicated effects ≈ 50% of originals

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Links

Map: L01-00-map · Pitfalls: L01-10-pitfalls · Code: L01-11-python · Class 2 formulas: L02 §7 · Scope: rules/09 · Course plan: Class 1.

Class 1 · Pitfalls — the confusions, each with its minimal counterexample, and an original True/False self-test

Prev: L01-09-formulas · Next: L01-11-python · Map: L01-00-map

Every number below is asserted in L01-11-python. Equations are cited by the section note's own numbering.

Part I — the confusions this material produces

1 · Unbiased vs low MSE

The confusion. "Unbiased" is treated as "best".

Counterexample ($U(0,\theta)$, $\theta = 3$, $n = 10$; L01-05 (16)–(18)). The **biased** MLE $x_{(n)}$ has MSE 0.136. The **unbiased** $2\bar{x}$ has MSE 0.300. Within the family $c\,x_{(n)}$, the unbiased choice $c = 11/10$ gives $\theta^2/120$, and the **biased** $c^* = 12/11$ gives $\theta^2/121$ (L01-05 (29)).

> **MSE = variance + bias² (L01-05 (2)). Unbiasedness removes one term. It does not minimise the sum.** → L01-05-comparing-estimators-mse-bootstrap §2, §8.

2 · Same likelihood vs same sampling distribution

The confusion. If two experiments give proportional likelihoods, every inference must agree.

Counterexample ($s = 1$ success in $n = 10$ trials; L01-01 §8–§9). Both give $\hat{\theta} = 0.1$, the same $J(\hat{\theta})$, the same Wald interval and the same Beta posterior. But:

	Binomial (n fixed)	Negative binomial (s fixed)
$E(\hat{\theta})$ at $\theta = 0.1$	0.100	**0.256**
p-value, $H_0 : \theta = 0.5$ vs $<$	0.0107	**0.0020**

> **What depends only on the observed likelihood agrees: the MLE, $J(\hat{\theta})$, the Wald interval and the posterior. What averages over unobserved outcomes disagrees: bias, exact variance, coverage and p-values.** → L01-01-likelihood-sampling-rules §3, §10.

3 · Observed vs expected information

The confusion. "The information" is treated as a property of the data alone.

Counterexample. $J(\theta) = -\ell''(\theta)$ is computed from the realised likelihood, and it is the same function under every sampling rule (L01-01 (7)–(8)). $I(\theta) = E[J(\theta)]$ averages over the design. At $\theta = 0.5$, a binomial with $n = 10$ has $I = 10/0.25 = 40$, so the asymptotic sd is 0.158. A negative binomial with $s = 1$ has $I = 1/(0.25 \times 0.5) = 8$, so the sd is 0.354 (L01-01 (11), (14)). The **plug-ins** coincide at $\hat{\theta}$, (18). The functions of θ do not.

> **J is random and design-free. I is fixed and design-dependent. Using $J(\hat{\theta})$ for the standard error is what makes the Wald interval blind to the stopping rule.**

4 · p-value vs $P(H_0 \mid x)$

The confusion. "$p = 0.036$" is read as "a 3.6% chance that H_0 is true".

Counterexample ($X \sim N(\theta, 1)$, $H_0 : \theta = 0$ vs $H_1 : \theta = 4$, $x = 1.8$; L01-08 (5)–(10)). $p = 0.036$, yet with equal priors $P(H_0 \mid x) = 1 - 0.310 = \mathbf{0.690}$. The data favour H_0: $BF_{01} = e^{0.8} = 2.23$.

> **The p-value is a tail area under H_0. It never looks at H_1, so it cannot be a posterior probability.** Every $x \in (1.645, 2)$ is "significant" and also favours H_0, by (8). → L01-08-p-hacking-evidence §4.

5 · Profiling vs integrating

The confusion. $w(\tau)$ in L01-03 (7) is treated as "the posterior of the change point".

Counterexample (toy series $y = (6, 1, 1, 1, 1, 1, 1, 1)$, uniform prior on the cut, Exp(1/2) priors on both rates for the integrated version, L01-03 (10)). The weight on "the break is after year 1" is **0.783 profiled** and **0.653 integrated**. The one-year segment fits its own count exactly ($\hat{\lambda}_1 = 6$). Profiling keeps that perfect fit and ignores how uncertain a rate estimated from one observation is. Integrating charges for that uncertainty. On the coal data both routes give 0.251 at 1891, because both segments are long (L01-03 §8).

> **Profiling plugs in the peak height. Integrating keeps the area. They agree only when every segment is long and the prior is weak.**

6 · Overdispersion vs a change point

The confusion. "$s^2/\bar{y} = 1.59$, $p < 0.001$, so the rate changed."

Counterexample. $s^2/\bar{y}$ does not depend on the **order** of the years. Shuffle the 112 coal counts and the dispersion index is still exactly 1.592, with $p < 0.001$. But the best-cut LR of the shuffled series has a 95% quantile of about **14** across permutations, against **70.0** in calendar order. Overdispersion also arises from year-to-year rate heterogeneity with no break at all.

> **Overdispersion says "not one Poisson rate". Only the time structure (profile likelihood, segment fits) says "a change around 1890".** → L01-03-coal-mining-change-point §3, §5.

7 · A searched LR referred to χ^2_1

The confusion. LR = 70.0, so $p = P(\chi^2_1 > 70)$.

Counterexample. Simulate iid Poisson(1.705) series of length 112, so there is **no** change, and maximise the LR over all 111 cuts. The 95% quantile is **9.9**, not χ^2_1's 3.84. A nominal 5% test with the χ^2_1 cut-off would reject far too often. Here 70.0 lies above every one of 4,000 null and 4,000 permutation draws, so the conclusion survives. The χ^2_1 p-value does not.

> **Wilks needs a regular problem. A searched, discrete, unidentified-under-H_0 τ is not one.** → L01-03-coal-mining-change-point §6.

8 · R^2 vs out-of-sample fit

The confusion. A higher R^2 means a better model.

Counterexample (Boston, interaction order $k = 5$, $p = 2380 > n = 253$; L01-06 §7). $R^2_{in} = 1.0000$, $R^2_{out} = 0.052$, and the test RMSE is 47.13 against 4.48 for the plain model.

R^2 cannot fall when a column is added (L01-06 (21)), so it cannot choose between nested models. The training error is optimistic by $2p\sigma^2/n$ (L01-06 (26)–(27)).

9 · Significant vs important

The confusion. A joint F of 13.22 ($p < 0.0001$) means the variables matter a lot.

Counterexample (abstention, model (4) vs (2); L01-06 §8.3). The size dummies are jointly significant, yet $\bar{R}^2$ moves from 0.754 to **0.756**. With $n = 5{,}565$, a tiny share of variance is still precisely estimated.

10 · Stationary vs unit root (and "persistent" vs "nonstationary")

The confusion. A very persistent AR(1) is treated as nonstationary. Or a VAR with a coefficient above 1 is treated as explosive.

Counterexamples. $\phi = 0.9$ is stationary, with equilibrium variance $5.26\sigma^2$ and half-life 6.6. Only $\phi = 1$ has $V(y_t) = t\sigma^2$ and no equilibrium (L01-07 (9), (17), (18)). Tsay's Example 8.3 has $\Phi_{22} = 1.1$, yet its eigenvalues are 0.8 and 0.5, so it is **stationary** (L01-07 §9.1).

Stationarity is decided by $|\phi| < 1$, or by the eigenvalues of Φ. It is not decided by how slowly the ACF decays, or by any single coefficient.

11 · Consistency vs unbiasedness

The confusion. The two are treated as the same property, or one is assumed to imply the other.

Counterexamples ($U(0, \theta)$):

Estimator	Unbiased?	Consistent?	Why
$x_{(n)}$	no (bias $-\theta/(n+1)$)	**yes**	MSE $2\theta^2/[(n+1)(n+2)] \to 0$
$2x_1$ (double the first draw)	**yes**	no	MSE $= \theta^2/3$ for every n
$\bar{x}$	no (for θ)	no (it converges to $\theta/2$)	MSE $\to \theta^2/4$

Also s/N in inverse sampling is biased (0.256 at $\theta = 0.1$, $s = 1$) but consistent as $s \to \infty$ (L01-01 (15)).

Unbiasedness is about the centre at a fixed n. Consistency is about the limit. MSE $\to 0$ gives consistency, with or without bias.

12 · The nonparametric bootstrap for the maximum

The confusion. "The bootstrap always works if n is large."

Counterexample (L01-05 (32)). The resampled maximum equals the observed one with probability $1-(1-1/n)^n$: 0.651 at $n = 10$, **0.634 at** $n = 100$, and $\to 0.632$. The true law of $x_{(n)}$ is continuous. In the deck the bootstrap SD of $x_{(n)}$ is too small at $n = 50$ (0.041 against 0.058) and too large at $n = 100$ (0.036 against 0.029). **The shape is wrong at every n.**

The empirical distribution has no mass beyond its own maximum. A statistic that lives on the edge of the support cannot be bootstrapped from it. Resample from the fitted $U(0, \hat{\theta})$ instead (L01-05 §11).

13 · Uncorrelated vs independent

Counterexample (IBM monthly returns, L01-07 §2). Ljung–Box on returns gives $Q(30) = 38.2$, $p = 0.144$, so no autocorrelation. On **squared** returns it gives $Q(30) = 182.1$, $p \approx 10^{-23}$. The returns are uncorrelated but not independent: their size is predictable (volatility clustering, Class 8).

14 · Asymptotics where they do not apply

Counterexamples. With $s = 1$, $n = 10$, the Wald interval is $(-0.086,\ 0.286)$, which leaves $[0, 1]$ (L01-01 §8). The uniform MLE is **not** a root of the score: $\partial \log L/\partial\theta = -n/\theta$ never vanishes, and the maximum is at the boundary $x_{(n)}$ (L01-05 (25)). For iid Poisson with $s_n = 0$, the MLE **does not exist** on $\lambda > 0$ (L01-02 §4).

15 · A confident posterior vs a right model

Counterexample (L01-04 §7). On all 112 coal years the posterior mean is 1.707, with a tight posterior around it. It is a precise estimate of a single rate that L01-03 showed does not exist. **The prior regularises the estimate. It does not test the likelihood.**

16 · 0.81 as the false-positive rate of a specification search

The confusion. "32 tries means an 81% false-positive rate."

Counterexample (L01-08 §3). $1 - 0.95^{32} = 0.806$ assumes **independent** tests. With 32 one-sided z-tests sharing a common component (pairwise correlation 0.5, like re-analyses of the same 30 students) the family-wise rate is **0.405**. It is still eight times the nominal 0.05, but it is not 0.81. The real binary-choice count of the deck's setup is 162, not 32.

Part II — True/False self-test (original)

Scoring (rules/00): V/F score = $\max\{0,\ \text{correct} - \text{wrong}/2\}\times$ item value; blank items do not count as wrong. Difficulty: [**E**] easy · [**M**] medium · [**P**] hard. Numbers are illustrative unless a note is cited. Answers follow each block.

Block A — likelihood and sampling rules

Setting (items 1–5). Two Anvisa inspectors estimate the defect rate θ of a vaccine line. Inspector A audits a **fixed** batch of 20 vials and finds 2 defective. Inspector B audits vials **until the 2nd defective** appears, which happens at the 20th vial.

1. [**E**] Both inspectors' maximum likelihood estimates equal 0.10.
2. [**M**] Because the two MLEs coincide, B's estimator $2/N$ is unbiased for θ, just as A's $S/20$ is.
3. [**M**] With a uniform Beta(1,1) prior, A and B obtain the same posterior for θ.
4. [**P**] Testing $H_0 : \theta = 0.25$ against $\theta < 0.25$ at the 5% level, A and B necessarily reach the same decision, since their likelihoods are proportional.
5. [**M**] The Wald 95% intervals of A and B, built from the observed information at the MLE, are identical.

Answers.

1. **True.** Both likelihoods are proportional to $\theta^2(1-\theta)^{18}$ (L01-01 (1)–(3)). The score (5) vanishes at $\hat{\theta} = 2/20 = 0.10$ whichever constant, $\binom{20}{2}$ or $\binom{19}{1}$, multiplies the kernel. The MLE uses only the realised likelihood, so the sampling rule cannot change it.
2. **False.** The MLE is the same *function of the data*. Its *sampling distribution* is not. Under inverse sampling N is random, and $1/x$ is convex, so Jensen gives $E(2/N) > \theta$ (L01-01 (19)). At $\theta = 0.1$ the exact mean is 0.165. Haldane's $(s-1)/(N-1) = 1/(N-1)$ is the unbiased one. A's $S/20$ is unbiased by (10). The converse trap is "same likelihood ⇒ same bias".
3. **True.** The posterior is Beta(1 + 2, 1 + 18) = Beta(3,19), with mean 0.136, under either rule. The binomial coefficients cancel in the normalisation (L01-01 (21)). This is the likelihood principle at work: the stopping rule does not enter Bayesian inference.
4. **False.** "Extreme" is defined by each sample space. For A, $p = P(S \leq 2 \mid n = 20, \theta = 0.25) = 0.091$: **do not reject**. For B, $p = P(N \geq 20 \mid s = 2, \theta = 0.25) = P(\text{Bin}(19, 0.25) \leq 1) = 0.031$: **reject**. Identical vials, opposite verdicts. This is L01-01 §9 with new numbers.
5. **True.** $J(\hat{\theta}) = n/[\hat{\theta}(1-\hat{\theta})] = 20/(0.1 \times 0.9) = 222.2$ in both cases (L01-01 (8)), and the plug-in variances coincide by (18). The interval is also poor in both cases, $0.10 \pm 1.96 \times 0.067 = (-0.031,\ 0.231)$, which leaves $[0, 1]$. Being "the same" is not being "right".

Block B — the Poisson MLE

Setting (items 6–9). An analyst models the yearly number of offshore oil-spill incidents reported to the ANP as iid Poisson(λ) (illustrative).

6. **[E]** If the model is right, no unbiased estimator of λ has a smaller variance than $\bar{y}$.
7. **[M]** The sample variance s^2 is also unbiased for λ. Both it and $\bar{y}$ are consistent, so for large samples they are equally precise.
8. **[M]** For any data set, the Poisson likelihood has a maximiser in $(0, \infty)$.
9. **[P]** At $\lambda = 2$ and $n = 50$, Chebyshev's lower bound on $P(|\bar{y} - \lambda| < 0.25)$ is a close approximation to the exact probability.

Answers.

6. **True.** $V(\bar{y}) = \lambda/n$ equals the Cramér–Rao bound $1/I(\lambda) = \lambda/n$ (L01-02 (12), (17)–(18)), so $\bar{y}$ is efficient. Under the bound's regularity conditions no unbiased competitor can do better. (It is also the UMVUE, L01-02 §9.)
7. **False.** Both are unbiased and consistent, but $V(s^2) = \lambda/n + 2\lambda^2/(n-1)$ (L01-02 (19)). Its efficiency tends to $1/(1+2\lambda)$, which is 0.2 at $\lambda = 2$, so s^2 needs about **five times** the data. Consistency is a limit statement. It says nothing about precision at any n.
8. **False.** If every year has zero incidents ($s_n = 0$), then $\ell(\lambda) = \log c - n\lambda$ increases as $\lambda \downarrow 0$, and there is no maximum on the open space (L01-02 §4). This happens with probability $e^{-n\lambda}$, which is not negligible for small $n\lambda$.
9. **False.** Chebyshev gives $1 - 2/(50 \times 0.25^2) = 0.36$. The exact probability is 0.789, from $s_n \sim$ Poisson(100) (L01-02 §8). The bound proves consistency but is far too crude to quantify it.

Block C — the change point and the first prior

Setting (items 10–14). Yearly counts of track incidents on a metro network (illustrative) and the coal-mining series of L01-03.

10. **[M]** A dispersion index $s^2/\bar{y} = 1.6$ with dispersion-test $p < 0.001$ shows that the incident rate changed at some date.
11. **[P]** In the coal data, LR = 70.0 at the profile-maximising cut, so the correct p-value is $P(\chi^2_1 > 70)$.

12. [**E**] The single-rate model can never attain a higher maximised likelihood than the best two-regime model.
13. [**E**] With an Exp(a) prior and n iid Poisson counts summing to s_n, the posterior of λ is Gamma with shape $s_n + 1$ and rate $a + n$.
14. [**M**] With the Exp(1/2) prior, the posterior mean $(s_n + 1)/(n + 1/2)$ always exceeds the MLE s_n/n, because the prior adds one pseudo-event.

Answers.

10. **False.** The index ignores the order of the years. Shuffling the coal series leaves it at exactly 1.592, yet the shuffled best-cut LR has a 95% quantile near 14, against 70 in calendar order. Overdispersion rules out *one* Poisson rate. A change point is one explanation; year-to-year heterogeneity is another. Locating a break needs the time structure.
11. **False.** τ was **chosen to maximise** the statistic over 111 candidates. It is not identified under $\lambda_1 = \lambda_2$, and it is discrete, so Wilks does not apply (L01-03 §6). Under an iid Poisson null with no break, the maximised LR has a 95% quantile of 9.9, not 3.84. The single-rate model is still rejected, because 70 exceeds every simulated null draw.
12. **True.** One rate is the special case $\lambda_1 = \lambda_2$ of the two-regime model (3). Maximising over a larger set cannot give a lower maximum, so the dashed line in `L01-coal-profile-likelihood` lies below every cut.
13. **True.** Prior kernel $e^{-a\lambda}$ times likelihood kernel $\lambda^{s_n} e^{-n\lambda}$ gives $\lambda^{s_n} e^{-(a+n)\lambda}$, which is the Gamma($s_n + 1,\ a + n$) kernel (L01-04 (6)–(8)). Exp(a) is Gamma(1, a), so the update adds the count to the shape and the exposure to the rate.
14. **False.** The prior adds one event **and half a year**. By (13) the posterior mean is a weighted average of the prior mean $1/a = 2$ and $\bar{y}$, so it exceeds the MLE only when $\bar{y} < 2$. On the 1851–1891 data, $\bar{y} = 3.098 > 2$ and the posterior mean is 3.084, which is *below* the MLE. The pull is toward the prior centre, not always upward.

Block D — MSE and the bootstrap

Setting (items 15–19). A procurement auditor at the TCU models winning discounts in a sealed-bid auction as iid $U(0, \theta)$ and wants the upper bound θ (illustrative).

15. [**E**] An unbiased estimator of θ always has a smaller MSE than a biased one.
16. [**M**] The MLE of θ is found by setting the score $\partial \log L/\partial\theta$ to zero.
17. [**M**] Among all estimators of the form $c\, x_{(n)}$, the unbiased one, $c = (n + 1)/n$, has the smallest MSE.
18. [**M**] With $n = 2$ bids, $x_{(n)}$ has strictly smaller MSE than $2\bar{x}$.
19. [**P**] As n grows, the nonparametric bootstrap distribution of $x^*_{(n)}$ approaches the true sampling distribution of $x_{(n)}$.

Answers.

15. **False.** At $\theta = 3$, $n = 10$, the biased $x_{(n)}$ has MSE 0.136 and the unbiased $2\bar{x}$ has 0.300 (L01-05 (17)–(18)). MSE $= V + B^2$: a small bias can buy a large cut in variance.
16. **False.** $L(\theta) = \theta^{-n} \mathbb{1}(\theta \geq x_{(n)})$, so the score $-n/\theta$ never vanishes. The likelihood is decreasing wherever it is positive, so the maximum sits on the boundary $\theta = x_{(n)}$ (L01-05 (24)–(25); B Ex. 3.1.3).
17. **False.** MSE(c) is minimised at $c^* = (n + 2)/(n + 1)$, which is below $(n + 1)/n$. The optimum shrinks slightly and accepts a bias of $-\theta/(n+1)^2$. At $n = 10$: $\theta^2/121 < \theta^2/120$ (L01-05 (28)–(29)).
18. **False.** At $n = 2$ both MSEs equal $\theta^2/6$, because $(n - 1)(n - 2) = 0$ in (20). The strict ranking (22) needs $n \geq 3$. That is why the handwritten "$n \geq 2$" is wrong.

19. **False.** The resampled maximum equals the observed maximum with probability $1-(1-1/n)^n \to 0.632$ (L01-05 (32)). That point mass never disappears, while the true law of $x_{(n)}$ is continuous. The bootstrap is **inconsistent** here. The parametric bootstrap from $U(0,\hat{\theta})$ is not.

Block E — Gaussian linear regression

Setting (items 20–24). An IPEA analyst regresses municipal outcomes on several regressors by OLS with an intercept, as in the abstention study of L01-06 §8.

20. [**E**] Adding a regressor can never lower R^2.
21. [**M**] Adding a regressor raises adjusted R^2 only if its t-statistic is significant at 5%.
22. [**M**] Under Gaussian errors, the maximum likelihood estimator RSS$/n$ is unbiased for σ^2.
23. [**P**] For a fixed design and a correct linear model with p columns, the expected training MSE understates the expected test MSE at the same design by $2p\sigma^2/n$.
24. [**M**] Since the four size dummies are jointly significant ($F = 13.22$, $p < 0.0001$), adding them must raise the adjusted R^2 substantially.

Answers.

20. **True.** The smaller model is the larger one with the extra coefficient set to zero, so the minimised RSS cannot rise (L01-06 (21)). That is exactly why R^2 cannot select a model.
21. **False.** The exact condition is $|t| > 1$ (W p. 181; L01-06 §5), far below the 5% critical value. $\bar{R}^2$ is a lenient criterion: in the seeded best-subset simulation it keeps $d = 5$ where BIC picks the true $d = 3$.
22. **False.** $E(\text{RSS}) = \sigma^2(n-q)$ (L01-06 (18)), so $E(\text{RSS}/n) = \sigma^2(n-q)/n < \sigma^2$. The unbiased estimator divides by $n-q$. The MLE and the unbiased estimator are different here, while the $\hat{\beta}$'s are the same.
23. **True.** $E(\text{MSE}_{in}) = \sigma^2(1-p/n)$ and $E(\text{MSE}_{out}) = \sigma^2(1+p/n)$ (L01-06 (26)–(27)). The gap is $2p\sigma^2/n$: the training error is optimistic in proportion to the number of parameters, and C_p adds exactly that back.
24. **False.** $\bar{R}^2$ moves from 0.754 to 0.756. With $n = 5{,}565$, a very small share of variance is estimated precisely. Significance and explanatory share answer different questions. The group's $F > 1$ guarantees only that $\bar{R}^2$ rises, not by how much.

Block F — AR and VAR

Setting (items 25–28). A Banco Central analyst models a monthly inflation-gap series.

25. [**E**] An AR(1) with $\phi = 0.95$ is nonstationary because shocks to it are very persistent.
26. [**M**] For a Gaussian AR(1) with $|\phi| < 1$, the distribution of y_t given any starting value y_0 converges to $N(\mu, \sigma^2/(1-\phi^2))$.
27. [**P**] A VAR(1) whose coefficient matrix has an entry greater than 1 is necessarily explosive.
28. [**M**] OLS of y_t on $(1, y_{t-1})$ gives the exact Gaussian maximum likelihood estimates of an AR(1).

Answers.

25. **False.** Stationarity needs only $|\phi| < 1$ (L01-07 (9)). At $\phi = 0.95$ the equilibrium variance is $\sigma^2/(1-0.9025) = 10.26\sigma^2$, finite, and the half-life is about 13.5 periods. Persistent is not the same as a unit root. Only at $\phi = 1$ does the variance grow like $t\sigma^2$.
26. **True.** From (15), the conditional mean $\mu + \phi^t(y_0 - \mu) \to \mu$ and the conditional variance $\sigma^2(1-\phi^{2t})/(1-\phi^2) \to \sigma^2/(1-\phi^2)$. The start is forgotten geometrically (L01-07 (16)).
27. **False.** Stationarity depends on the eigenvalues of Φ, not on its entries. Tsay's Example 8.3 has $\Phi_{22} = 1.1$ and eigenvalues 0.8 and 0.5, so it is stationary with mean $(1,2)'$ (L01-07 §9.1).

28. **False.** OLS maximises the **conditional** likelihood given y_1 (L01-07 (13)). The exact likelihood also contains $p(y_1) = N(\mu, \sigma^2/(1-\phi^2))$, which depends on ϕ and imposes $|\phi| < 1$. The two estimates differ slightly and coincide asymptotically.

Block G — p-values and evidence

29. **[E]** Under H_0, the p-value of a test with a continuous statistic is uniformly distributed on $[0, 1]$.
30. **[P]** With $X \sim N(\theta, 1)$, $H_0 : \theta = 0$, $H_1 : \theta = 4$ and equal prior probabilities, an observation whose one-sided p-value is 0.03 is evidence in favour of H_1.

Answers.

29. **True.** $p = 1 - F_0(T)$, and $F_0(T) \sim U(0, 1)$ under H_0 by the probability integral transform, so $P_{H_0}(p \leq \alpha) = \alpha$ (L01-08 (1)). This is what makes the *minimum* of many p-values misleading: 32 independent uniforms fall below 0.05 at least once with probability 0.806.
30. **False.** $p = 0.03$ means $x = 1.881$. By (8), $BF_{10} = e^{4(1.881)-8} = 0.621 < 1$, so the data favour H_0, and $P(H_1 \mid x) = 0.383$. Any $x \in (1.645, 2)$ is significant at 5% yet closer to 0 than to 4. The p-value never looks at where H_1 is.

Part III — the distinctions that cost most points

Pair	The difference
Likelihood vs sampling distribution	the observed curve vs the average over outcomes that did not happen
$J(\hat{\theta})$ vs $I(\theta)$	realised curvature (design-free) vs expected curvature (design-dependent)
Unbiased vs minimum MSE	the centre alone vs centre and spread together
Unbiased vs consistent	a fixed-n statement vs a limit
Profile vs integrated likelihood	the peak height vs the area under the curve
Overdispersion vs change point	"not one rate" vs "two rates, in this order"
p-value vs Bayes factor vs posterior	a tail area under H_0 vs a ratio of heights vs a probability of the hypothesis
R^2 vs test error	can only rise with p vs rises once noise is fitted
Persistent vs unit root	$\|\phi\| < 1$ near 1 vs $\phi = 1$
Nonparametric vs parametric bootstrap	resample the data vs resample the fitted model

---

Links

Map: L01-00-map · Formulas: L01-09-formulas · Code: L01-11-python · Class 2 pitfalls: L02 §8 · Grading conventions: rules/00.

Class 1 · Computation in Python — every routine, with the assert that must pass

> Every computation the unit uses, written in Python (numpy, scipy, statsmodels; no R), and each checked with an `assert` against the number printed in the deck or the section note. The code below is the script `Notas/L01/L01_python_checks.py`, split into its sections. Run `python Notas/L01/L01_python_checks.py` (about 10 s). It prints one `ok` line per section and ends with `ALL L01 CHECKS PASSED`, which is the output of the run on 2026-10-19. Prev: L01-10-pitfalls · Map: L01-00-map · Formulas: L01-09-formulas

What is reused, not rewritten.

Already in the unit	What it computes	How this note uses it
`companions/check_companions.py`	the coal series loaded from `.agent/notes/coal-mining-data.md`; `cp_python()`, the Change-Point Dial's read-outs (LR, $w(\tau)$, rates, dispersion p-values); `fe_python()` for the Four Estimators	**imported** (`import check_companions as cc`). §3 takes `cc.Y` as its data and checks its own LR against `cc.cp_python(...)`
`figures/make_figures.py`	the 14 figures, each with its own assert: the negative-binomial means 0.256/0.461, $\Pr(y = 2 \mid \lambda = 2)$, the coal MLE lines, $\tau = 1891$ and LR 70.0, the posterior 3.084/0.273, the four MSEs at $n = 10$, the bootstrap jump 0.634, BIC choosing $d = 3$, the AR(1) bands, $p = 0.0359$	**cited**, not imported: importing it would redraw every figure. §5 and §6 reuse its seeds (2026 and 128), so the samples are the same ones the figures show

The section notes' asserts check **figures**. This script checks the **formulas and printed numbers**: exact distributions against closed forms, Monte Carlo against theory, and `numpy.linalg` against `statsmodels`. Where the two overlap (the coal LR, the MSEs at $n = 10$), both sides must agree.

Conventions. `near(x, printed, tol)` passes when `x` matches a printed value to half a unit in its last decimal (`tol` defaults to 5e-4). Monte Carlo checks are seeded and allow about four standard errors. Equation numbers "(k)" are those of the section note named in each heading.

---

0 · Setup

```
"""Class 1 (L01) reference implementation, with assert self-checks against the decks'
↪ numbers.

Run:  python Notas/L01/L01_python_checks.py      (stops at the first failed assert)

Sections §1-§9 are the code blocks of L01-11-python.md, in the same order. The coal
↪ series
```

```
and the Change-Point Dial's read-out function are imported from
 ↪  companions/check_companions.py
(reuse, not a copy). Equation numbers "(k)" refer to the section note named in each
 ↪  header.
Every Monte Carlo draw is seeded; MC checks use tolerances of about four standard
 ↪  errors.
"""
import pathlib
import sys
from fractions import Fraction
from math import comb

import numpy as np
import statsmodels.api as sm
from scipy import linalg, stats
from scipy.special import gammaln, xlogy
from statsmodels.tsa.api import VAR
from statsmodels.tsa.ar_model import AutoReg

HERE = pathlib.Path(__file__).resolve().parent
sys.path.insert(0, str(HERE / "companions"))
import check_companions as cc  # noqa: E402  coal series (from .agent/notes) +
 ↪  companion read-outs


sys.stdout.reconfigure(encoding="utf-8")


def near(x, printed, tol=5e-4):
    """True when x agrees with a printed value to within tol (default: half a unit in
     ↪  the 3rd decimal)."""
    return abs(float(x) - printed) <= tol
```

---

1 · Sampling rules → L01-01-likelihood-sampling-rules

The exact pmf of $\hat{\theta} = s/N$ is the negative binomial (13), summed far into its tail. Its mean and sd reproduce slide 9's table. The closed form (20) and Haldane's unbiased $(s-1)/(N-1)$ hold to machine precision. The p-values use `binom.cdf` for "few successes" and `nbinom.sf` for "many trials", and each also equals the hand fraction of §9 (11/1024, 1/512, 386/1024, 130/512).

```
# === §1 Sampling rules: binomial vs negative binomial (L01-01) ===
def negbin_theta(s, theta, m_max=5000):
    """Support s/m and exact pmf of theta_hat = s/N, N = trials to the s-th success:
     ↪  L01-01 (13)."""
    m = np.arange(s, m_max)
    return s / m, stats.nbinom.pmf(m - s, s, theta)   # scipy counts failures, m - s


def negbin_moments(s, theta):
```

```
    x, p = negbin_theta(s, theta)
    mean = (x * p).sum()
    return mean, np.sqrt((x**2 * p).sum() - mean**2)


# slide 9 table: exact E and sd of s/N at theta = theta_hat
for s, mean, sd in [(1, 0.256, 0.281), (4, 0.461, 0.178)]:
    m_, sd_ = negbin_moments(s, s / 10)
    assert near(m_, mean) and near(sd_, sd), (s, m_, sd_)
th = 0.1
assert abs(negbin_moments(1, th)[0] - (-th * np.log(th) / (1 - th))) < 1e-12   #
 ↪  closed form (20)
x, p = negbin_theta(4, 0.4)
assert abs(((3 / (4 / x - 1)) * p).sum() - 0.4) < 1e-12                         #
 ↪  Haldane: (s-1)/(N-1) unbiased

J = lambda s, n: n**3 / (s * (n - s))                                           #
 ↪  observed information (8)
assert near(J(1, 10), 111.1, 0.05) and abs(J(1, 10) - 10 / (0.1 * 0.9)) < 1e-9


def wald(th_hat, n, z=1.96):
    """Wald interval th_hat +- z sqrt(th_hat(1-th_hat)/n): same under both rules,
     ↪  (18)."""
    se = np.sqrt(th_hat * (1 - th_hat) / n)
    return th_hat - z * se, th_hat + z * se


assert np.allclose(wald(0.1, 10), (-0.086, 0.286), atol=5e-4)
assert np.allclose(wald(0.4, 10), (0.096, 0.704), atol=5e-4)

p_bin = lambda s, n: stats.binom.cdf(s, n, 0.5)          # P(S <= s | n), extreme =
 ↪  few successes
p_nb = lambda s, n: stats.nbinom.sf(n - s - 1, s, 0.5)   # P(N >= n | s), extreme =
 ↪  many trials
assert abs(p_bin(1, 10) - 11 / 1024) < 1e-15 and near(p_bin(1, 10), 0.0107, 5e-5)
assert abs(p_nb(1, 10) - 0.5**9) < 1e-15 and near(p_nb(1, 10), 0.0020, 5e-5)
assert abs(p_bin(4, 10) - 386 / 1024) < 1e-15 and near(p_bin(4, 10), 0.377)
assert abs(p_nb(4, 10) - 130 / 512) < 1e-15 and near(p_nb(4, 10), 0.254)

# Beta posterior is the same under all three rules: Beta(a+s, b+n-s), (21)
assert near(stats.beta(1 + 1, 1 + 9).mean(), 0.167) and near(stats.beta(1 + 4, 1 +
 ↪  6).mean(), 0.417)
# the site's Bernoulli code: s = 7 of n = 10, Beta(3,3) prior
for z, lo, hi in [(1.645, 0.462, 0.938), (1.960, 0.416, 0.984), (2.576, 0.327,
 ↪  1.073)]:
    assert np.allclose(wald(0.7, 10, z), (lo, hi), atol=5e-4)
post = stats.beta(3 + 7, 3 + 3)
assert near(stats.beta(3, 3).std(), 0.189) and near(post.mean(), 0.625) and
 ↪  near(post.std(), 0.117)
```

```
assert np.allclose(post.ppf([0.025, 0.975]), (0.384, 0.837), atol=5e-4)
assert np.ceil(1.96**2 * 0.25 / 0.05**2) == 385                                  #
 ↪  sample size (23)
print("§1 sampling rules: ok")
```

The two `p_*` lambdas are the whole of slide 10. Same (s, n), same likelihood, different tail, because each one sums over a different sample space.

---

2 · Poisson MLE, Cramér–Rao and the efficiency of s^2 → L01-02-poisson-mle-properties

The Fisher information and the fourth central moment are summed exactly over the pmf. The general $V(s^2)$ formula then reproduces (19) for every n. The Monte Carlo repeats slide 8's design with 20,000 replications instead of 1,000, so the tolerances can be tight. The ± 0.25 coverage is exact because $n\hat{\lambda} \sim \text{Poisson}(n\lambda)$.

```
# === §2 Poisson MLE, CRLB and the efficiency of s^2 (L01-02) ===
assert near(stats.poisson.pmf(2, 2), 0.271) and near(stats.poisson.cdf(3, 2), 0.857)
assert near(stats.poisson.cdf(3, 1), 0.981) and near(stats.poisson.pmf(3, 2),
 ↪  0.18045, 5e-6)

lam = 2.0
k = np.arange(0, 200)
pk = stats.poisson.pmf(k, lam)
assert abs(((k / lam - 1) ** 2 * pk).sum() - 1 / lam) < 1e-12        # Fisher info
 ↪  per obs, (15)-(17)
mu4 = ((k - lam) ** 4 * pk).sum()
assert abs(mu4 - (lam + 3 * lam**2)) < 1e-9                           # mu4 = lambda
 ↪  + 3 lambda^2


def var_s2(lam, n):
    """V(s^2) for iid Poisson, L01-02 (19)."""
    return lam / n + 2 * lam**2 / (n - 1)


def eff_s2(lam, n):
    """CRLB / V(s^2), L01-02 (20); tends to 1/(1+2 lambda), (21)."""
    return (lam / n) / var_s2(lam, n)


for n in (10, 50, 200):
    assert abs(var_s2(lam, n) - (mu4 / n - lam**2 * (n - 3) / (n * (n - 1)))) < 1e-12
assert [round(var_s2(2, n), 3) for n in (10, 50, 200)] == [1.089, 0.203, 0.050]
assert [round(eff_s2(2, n), 3) for n in (10, 50, 200)] == [0.184, 0.197, 0.199]
assert abs(1 / (1 + 2 * lam) - 0.2) < 1e-15
# the deck's MC efficiencies: CRLB over its MC variance of s^2 (slide 10)
assert [round(c / v, 3) for c, v in [(0.2, 1.245), (0.04, 0.187), (0.01, 0.051)]] ==
 ↪  [0.161, 0.214, 0.196]
```

```
# Monte Carlo (slides 8-10 design, more replications): ybar unbiased with V =
 ↪  lambda/n; s^2 per (19)
rng = np.random.default_rng(10)
N = 20000
for n in (10, 50, 200):
    y = rng.poisson(lam, size=(N, n))
    ybar, s2 = y.mean(axis=1), y.var(axis=1, ddof=1)
    assert abs(ybar.mean() - lam) < 4 * np.sqrt(lam / n / N)
    assert abs(ybar.var(ddof=1) / (lam / n) - 1) < 0.05
    assert abs(s2.mean() - lam) < 4 * s2.std() / np.sqrt(N)          # s^2 also
     ↪  unbiased
    assert abs(s2.var(ddof=1) / var_s2(lam, n) - 1) < 0.07

# slide 12: P(|lambda_hat - 2| < 0.25), exact via s_n ~ Poisson(2n) and by the CLT
 ↪  (14)
for n, exact, normal in [(10, 0.424, 0.424), (50, 0.789, 0.789), (200, 0.987,
 ↪  0.988)]:
    S = np.arange(0, 10 * n)
    pe = stats.poisson.pmf(S, lam * n)[np.abs(S / n - lam) < 0.25].sum()
    pn = 2 * stats.norm.cdf(0.25 / np.sqrt(lam / n)) - 1
    assert near(pe, exact) and near(pn, normal), (n, pe, pn)
print("§2 Poisson MLE: ok")
```

---

3 · Coal mining: dispersion, profile likelihood, LR → L01-03-coal-mining-change-point

The data come from `check_companions.py`, which reads the verified series in `.agent/notes/coal-mining-data.md`, so the notes, the companion and this script share one source. `seg_kernel` is $s\log(s/n) - s$, the maximised kernel of each segment in (6), with `xlogy` giving $0\log 0 = 0$.

```
# === §3 Coal mining: dispersion test, split, profile likelihood, LR (L01-03) ===
Y, YEARS = cc.Y, cc.YEARS                # Jarrett (1979), read from
 ↪  .agent/notes/coal-mining-data.md
assert len(Y) == 112 and Y.sum() == 191 and Y[:40].sum() == 125 and Y[:41].sum() ==
 ↪  127


def dispersion(seg):
    """Index s^2/ybar and upper-tail p-value of D = (n-1)s^2/ybar ~ chi2_{n-1},
     ↪  L01-03 (1)."""
    n, m, v = len(seg), seg.mean(), seg.var(ddof=1)
    D = (n - 1) * v / m
    return m, v, v / m, D, stats.chi2.sf(D, n - 1)


rows = {"1851-1962": Y, "1851-1890": Y[YEARS <= 1890], "1891-1962": Y[YEARS > 1890]}
printed = {"1851-1962": (1.705, 2.714, 1.592, 176.7), "1851-1890": (3.125, 2.574,
 ↪  0.824, 32.1),
```

```
           "1891-1962": (0.917, 1.063, 1.160, 82.4)}
pvals = {}
for key, seg in rows.items():
    m, v, idx, D, pv = dispersion(seg)
    m0, v0, i0, D0 = printed[key]
    assert near(m, m0) and near(v, v0) and near(idx, i0) and near(D, D0, 0.05), key
    lo, hi = m - 1.96 * np.sqrt(m / len(seg)), m + 1.96 * np.sqrt(m / len(seg))   #
 ↪  Wald CI, L01-02 (14)
    pvals[key] = pv
assert pvals["1851-1962"] < 0.001 and near(pvals["1851-1890"], 0.774) and
 ↪  near(pvals["1891-1962"], 0.168)


def seg_kernel(seg):
    """Maximised Poisson log-likelihood kernel s log(s/n) - s of one segment (0 log 0
     ↪  = 0)."""
    s = seg.sum()
    return xlogy(s, s / len(seg)) - s


def profile(y, years):
    """Profile log-likelihood l_p(tau) of L01-03 (6) for every admissible cut tau."""
    taus = years[:-1]
    return taus, np.array([seg_kernel(y[years <= t]) + seg_kernel(y[years > t]) for t
     ↪  in taus])


taus, lp = profile(Y, YEARS)
l0 = seg_kernel(Y)
assert taus[lp.argmax()] == 1891
assert near(lp.max(), -54.055) and near(l0, -89.049)
LR = 2 * (lp.max() - l0)                                                    # (8)
assert near(LR, 69.99, 5e-3) and round(LR, 1) == 70.0
s1, s2_ = Y[:41].sum(), Y[41:].sum()
lr_compact = 2 * (s1 * np.log((s1 / 41) / (191 / 112)) + s2_ * np.log((s2_ / 71) /
 ↪  (191 / 112)))
assert abs(lr_compact - LR) < 1e-9                                          # compact
 ↪  form (9)
LR1890 = 2 * (lp[taus == 1890][0] - l0)
assert near(LR1890, 69.47, 5e-3)
assert 7e-17 < stats.chi2.sf(LR1890, 1) < 9e-17 and 5e-16 < stats.chi2.sf(70.0, 2) <
 ↪  7e-16

win = (taus >= 1885) & (taus <= 1895)
w = np.exp(lp[win] - lp[win].max())
assert near(np.exp(np.round(lp[win] - lp.max(), 2)).sum(), 3.983)   # L01-03's sum,
 ↪  from the rounded gaps
assert near(w.sum(), 3.985)                                                 # unrounded;
 ↪  same weights to 3 dp
w /= w.sum()                                                                # (7)
```

```
slide18 = [0.008, 0.091, 0.105, 0.038, 0.154, 0.194, 0.251, 0.098, 0.039, 0.016,
 ↪  0.007]
assert np.allclose(w, slide18, atol=5e-4)
gaps = [-3.50, -1.02, -0.87, -1.89, -0.49, -0.26, 0.00, -0.94, -1.86, -2.74, -3.59]
assert np.allclose(lp[win] - lp.max(), gaps, atol=5e-3)
assert near(w[4:8].sum(), 0.696)                                          #
 ↪  1889-1892
# reuse: the Change-Point Dial's own read-out function gives the same LR, w and rates
d = cc.cp_python(1851, 1962, 1891, 5, 0.95)
assert (d["LR2"], d["w"], d["s1_lam"], d["s2_lam"], d["ratio"]) == (f"{LR:.2f}",
 ↪  "0.251", "3.098", "0.901", "3.436")

# why a mixture is overdispersed: law of total variance (2), and the sample's own
 ↪  split
pi, l1, l2 = 40 / 112, 125 / 40, 66 / 72
assert near(pi * l1 + (1 - pi) * l2 + pi * (1 - pi) * (l1 - l2) ** 2, 2.825)
a_, b_ = rows["1851-1890"], rows["1891-1962"]
within = 39 * a_.var(ddof=1) + 71 * b_.var(ddof=1)
between = 40 * (a_.mean() - Y.mean()) ** 2 + 72 * (b_.mean() - Y.mean()) ** 2
assert near(111 * Y.var(ddof=1), 301.3, 0.05) and near(within, 175.9, 0.05) and
 ↪  near(between, 125.4, 0.05)
assert near(np.exp(-191 / 112), 0.182) and (Y == 0).sum() == 33
assert near(stats.poisson.sf(3, 191 / 112), 0.094) and (Y >= 4).sum() == 20
lhat = 191 / 112
rel = lambda s, n, l: s * np.log(lhat / l) - n * (lhat - l)               # log
 ↪  L(lhat_full)/L(l_segment)
assert near(rel(125, 40, 125 / 40), -18.92, 5e-3) and near(rel(66, 72, 66 / 72),
 ↪  -15.81, 5e-3)

# G layer: integrated (not profiled) likelihood of tau with Exp(a) priors on both
 ↪  rates, (10)
a = 0.5


def log_int(seg):
    s, n = seg.sum(), len(seg)
    return np.log(a) + gammaln(s + 1) - (s + 1) * np.log(n + a)


li = np.array([log_int(Y[YEARS <= t]) + log_int(Y[YEARS > t]) for t in taus[win]])
wi = np.exp(li - li.max())
wi /= wi.sum()
assert np.allclose(wi, [0.008, 0.089, 0.104, 0.038, 0.153, 0.194, 0.251, 0.099,
 ↪  0.040, 0.017, 0.007], atol=5e-4)
print("§3 coal mining: ok")
```

One correction to the printed arithmetic. L01-03 §5 sums e^{d_τ} over the **rounded** gaps and gets 3.983. The unrounded gaps give **3.985**. Every $w(\tau)$ is the same to three decimals, so the table stands. Both values are asserted above.

---

4 · Exponential prior, Gamma posterior → L01-04-first-bayesian-poisson

The conjugate result (8) is checked two ways: by `scipy.stats.gamma` with **rate** $a+n$ (scipy takes `scale = 1/rate`), and by normalising prior × likelihood on a grid. That second route is the brute-force version of the Bayes theorem in (4).

```
# === §4 First Bayesian example: Exp(a) prior -> Gamma(s+1, a+n) posterior (L01-04)
 ↪  ===
y41 = Y[:41]                                          # 1851-1891, the site's
 ↪  class1-poisson.html
n, s = len(y41), int(y41.sum())
assert (n, s) == (41, 127) and near(y41.mean(), 3.097561, 5e-7)
a = 0.5
prior = stats.expon(scale=1 / a)
assert prior.mean() == 2 and prior.var() == 4                                  #
 ↪  (3)
assert np.allclose(prior.ppf([0.025, 0.975]), (0.051, 7.38), atol=5e-3)
post = stats.gamma(s + 1, scale=1 / (a + n))                                   #
 ↪  (8), rate a + n
mode = s / (a + n)                                                             #
 ↪  (11)
assert near(post.mean(), 3.084) and near(mode, 3.060) and near(post.std(), 0.273)
assert near(s / n, 3.098)
assert np.allclose(post.ppf([0.025, 0.975]), (2.573, 3.641), atol=5e-4)
se = np.sqrt((s / n) / n)
assert near(se, 0.275) and np.allclose((s / n - 1.96 * se, s / n + 1.96 * se),
 ↪  (2.559, 3.636), atol=5e-4)
wp, wd = a / (n + a), n / (n + a)                                              #
 ↪  shrinkage weights (13)
assert abs(wp * (1 / a) + wd * (s / n) - post.mean()) < 1e-12 and near(wp, 0.01205,
 ↪  5e-6)
# a grid posterior (prior x likelihood, normalised numerically) agrees with the
 ↪  conjugate result
g = np.linspace(1e-6, 8, 200001)
logk = s * np.log(g) - n * g - a * g
dens = np.exp(logk - logk.max())
dens /= np.trapezoid(dens, g)
assert near(np.trapezoid(g * dens, g), post.mean(), 1e-6)
assert near((191 + 1) / (112 + a), 1.707)                                      #
 ↪  all 112 years
print("§4 Gamma posterior: ok")
```

---

5 · Four estimators, Monte Carlo, the two bootstraps → L01-05-comparing-estimators-mse-bootstrap

The MSEs are exact `Fractions`. That makes the ranking (22) and the two ties at $n = 1, 2$ checks of exact equality, not of floating-point closeness. The bootstrap uses figure `L01-bootstrap-max-pointmass`'s seeded sample ($x_{(n)} = 2.969$). The nonparametric resampled maximum lands **exactly** on $x_{(n)}$ about 63% of the time. The parametric one never does, and its SD matches the plug-in formula.

```
# === §5 Four estimators of the U(0, theta) bound; Monte Carlo; bootstrap (L01-05) ===
def mse_exact(n, theta=1):
    """Exact MSEs (16)-(18) of xbar, x(n), 2 xbar, (n+1)/n x(n); Fractions when theta is."""
    return [theta**2 * Fraction(1 + 3 * n, 12 * n), theta**2 * Fraction(2, (n + 1) * (n + 2)),
            theta**2 * Fraction(1, 3 * n), theta**2 * Fraction(1, n * (n + 2))]


for n in range(1, 201):
    m1, m2, m3, m4 = mse_exact(n)
    if n == 1:
        assert m1 == m2 == m3 == m4
    elif n == 2:
        assert m4 < m2 == m3 < m1
    else:
        assert m4 < m2 < m3 < m1                                  # ranking (22), n >= 3
assert [round(float(m), 3) for m in mse_exact(10, 3)] == [2.325, 0.136, 0.300, 0.075]
sd_th = [3 / np.sqrt(120), 3 * np.sqrt(10 / (121 * 12)), 3 / np.sqrt(30), 3 / np.sqrt(120)]
assert [round(v, 3) for v in sd_th] == [0.274, 0.249, 0.548, 0.274]

# Bolfarine Table 1.1 (p. 12): MSE[x(n)] at n = 10, 20 and the ratio column
assert mse_exact(10)[1] == Fraction(1, 66) and mse_exact(20)[1] == Fraction(1, 231)
assert [mse_exact(n)[0] for n in (3, 5, 10, 20)] == [Fraction(5, 18), Fraction(4, 15), Fraction(31, 120), Fraction(61, 240)]
assert [round(float(mse_exact(n)[1] / mse_exact(n)[0]), 3) for n in (3, 5, 10, 20)] == [0.36, 0.179, 0.059, 0.017]

# G layer: best multiple c x(n), (27)-(29)
mse_c = lambda c, n: c**2 * Fraction(n, n + 2) - 2 * c * Fraction(n, n + 1) + 1     # in units of theta^2
c_star = Fraction(12, 11)
assert mse_c(c_star, 10) == Fraction(1, 121) and near(9 / 121, 0.0744, 5e-5)
assert mse_c(c_star, 10) < mse_c(Fraction(11, 10), 10) == Fraction(1, 120)

# Monte Carlo at theta = 3, n = 10 (slides 9-12), many more replications than the deck's 1000
rng = np.random.default_rng(5)
theta, n, N = 3.0, 10, 200000
xs = rng.uniform(0, theta, size=(N, n))
est = np.column_stack([xs.mean(1), xs.max(1), 2 * xs.mean(1), (n + 1) / n * xs.max(1)])
mse_mc = ((est - theta) ** 2).mean(0)                                                  # (26)
assert np.allclose(mse_mc / np.array([float(m) for m in mse_exact(n, 3)]), 1, atol=0.03)
assert np.allclose(est.std(0, ddof=1) / sd_th, 1, atol=0.02)
```

```
# nonparametric bootstrap: point mass at x(n), (32)
pm = lambda n: 1 - (1 - 1 / n) ** n
assert [round(pm(n), 3) for n in (10, 50, 100)] == [0.651, 0.636, 0.634]
assert near(1 - np.exp(-1), 0.632)
assert [round((o - pm(n)) / 0.015, 1) for n, o in [(10, 0.666), (50, 0.622), (100,
 ↪  0.632)]] == [1.0, -0.9, -0.1]

rng = np.random.default_rng(2026)          # the same seeded sample as figure
 ↪  L01-bootstrap-max-pointmass
m = 100
xb = np.sort(rng.uniform(0, 3.0, m))
th4 = (m + 1) / m * xb[-1]
assert near(xb[-1], 2.969) and near(th4, 2.999)
B = 20000
boot = rng.choice(xb, size=(B, m), replace=True)
share = (boot.max(1) == xb[-1]).mean()
assert abs(share - pm(m)) < 4 * np.sqrt(pm(m) * (1 - pm(m)) / B)               # ~63%
 ↪  point mass
par = rng.uniform(0, th4, size=(B, m)).max(1) * (m + 1) / m                      #
 ↪  parametric bootstrap of theta_4
assert abs(par.std(ddof=1) / (th4 / np.sqrt(m * (m + 2))) - 1) < 0.03            #
 ↪  matches the plug-in SD
assert (par == th4).mean() == 0                                                  # no
 ↪  point mass


def plug_in_sd(th, n):
    """SDs of the four estimators with theta replaced by theta_4 (slide 19's plug-in
     ↪  row)."""
    return [th / np.sqrt(12 * n), th * np.sqrt(n / ((n + 1) ** 2 * (n + 2))), 2 * th
     ↪  / np.sqrt(12 * n), th / np.sqrt(n * (n + 2))]


assert np.allclose(plug_in_sd(2.568, 10), [0.234, 0.213, 0.469, 0.234], atol=5e-4)
assert np.allclose(plug_in_sd(2.926, 50), [0.119, 0.056, 0.239, 0.057], atol=1e-3)
 ↪  # 0.11945: on the rounding edge
assert np.allclose(plug_in_sd(3.029, 100), [0.087, 0.030, 0.175, 0.030], atol=5e-4)
assert np.allclose([3 / np.sqrt(600), 3 * np.sqrt(50 / (51**2 * 52)), 3 /
 ↪  np.sqrt(150), 3 / np.sqrt(2600)],
                   [0.122, 0.058, 0.245, 0.059], atol=5e-4)

# comparing-estimators.html: triangular weights, n = 21; median under t4; Beta(3,9)
 ↪  median
wts = np.array([i if i <= 11 else 22 - i for i in range(1, 22)], float)
assert wts.sum() == 121 and (wts**2).sum() == 891
assert near((wts**2).sum() / 121**2, 0.0609, 5e-5) and near((1 / 21) / (891 / 14641),
 ↪  0.78, 5e-3)
f0 = stats.t(4).pdf(0)
assert near(f0, 0.375) and near(1 / (4 * f0**2), 1.78, 5e-3) and stats.t(4).var() ==
 ↪  2
```

```
assert near(stats.beta(3, 9).median(), 0.236)
# bootstrap-example.html: exact t and chi-square intervals from the page's summaries
tq = stats.t.ppf(0.975, 49)
assert near(tq, 2.00958, 5e-6)
assert np.allclose((0.10006 - tq * 0.9129307 / np.sqrt(50), 0.10006 + tq * 0.9129307
 ↪  / np.sqrt(50)),
                   (-0.15939, 0.35951), atol=5e-5)
assert np.allclose((50 * 0.8334424 / stats.chi2.ppf(0.975, 49), 50 * 0.8334424 /
 ↪  stats.chi2.ppf(0.025, 49)),
                   (0.5934305, 1.3206221), atol=5e-6)
print("§5 four estimators and the bootstrap: ok")
```

Rounding note. At $n = 50$, $\hat{\theta}_4 = 2.926$ gives a plug-in SD of 0.11945 for $\hat{\theta}_1$, which L01-05 §11 prints as 0.119. The value sits 0.00005 from the rounding edge, so the assert allows 10^{-3} for that row. Every other entry is checked to half a unit.

---

6 · OLS, projection, R^2, criteria, HC1 → L01-06-gaussian-linear-regression

Three routes to $\hat{\beta}$ must agree: the normal equations (`numpy.linalg.solve`), `numpy.linalg.lstsq`, and `statsmodels.OLS`. Gaussian log-likelihood (15) at $(\hat{\beta}, \mathrm{RSS}/n)$ equals statsmodels' `llf`, which is "OLS = MLE" as a number. The optimism result (26)–(27) is checked at a fixed design, $p = 10$, $n = 50$: training MSE $\approx 0.8\sigma^2$, test MSE $\approx 1.2\sigma^2$. The best-subset block regenerates figure `L01-r2-vs-criteria`'s data (seed 128) and checks the numbers quoted in L01-06 §6.

```
# === §6 Gaussian linear regression: OLS, projection, R^2, criteria, HC1 (L01-06)
 ↪  ===
def ols(X, y):
    """beta_hat from the normal equations (6)-(7), residuals, RSS, unbiased sigma^2
     ↪  (19)."""
    b = np.linalg.solve(X.T @ X, X.T @ y)
    e = y - X @ b
    return b, e, e @ e, e @ e / (len(y) - X.shape[1])


rng = np.random.default_rng(6)
n, q = 200, 4
X = np.column_stack([np.ones(n), rng.standard_normal((n, q - 1))])
y = X @ np.array([1.0, 2.0, -1.0, 0.5]) + 1.5 * rng.standard_normal(n)
b, e, rss, s2 = ols(X, y)
fit = sm.OLS(y, X).fit()
assert np.allclose(b, np.linalg.lstsq(X, y, rcond=None)[0]) and np.allclose(b,
 ↪  fit.params)

P = X @ np.linalg.solve(X.T @ X, X.T)                                   # hat
 ↪  matrix (9)
assert np.allclose(P, P.T) and np.allclose(P @ P, P) and np.isclose(np.trace(P), q)
assert np.allclose(X.T @ e, 0) and abs(e.sum()) < 1e-9                   # (10)
tss = ((y - y.mean()) ** 2).sum()
r2 = 1 - rss / tss                                                      # (20)
```

```
adj = 1 - (1 - r2) * (n - 1) / (n - q)                                  # (23), q
 ↪  = k + 1
assert np.isclose(r2, fit.rsquared) and np.isclose(adj, fit.rsquared_adj) and
 ↪  np.isclose(s2, fit.scale)
llf = -n / 2 * np.log(2 * np.pi * rss / n) - n / 2                      # (15) at
 ↪  the MLE (16)-(17)
assert np.isclose(llf, fit.llf)                                         # OLS =
 ↪  Gaussian MLE

# HC1 by formula (29) against statsmodels
XtXi = np.linalg.inv(X.T @ X)
yh = X @ np.array([1.0, 2.0, -1.0, 0.5]) + rng.standard_normal(n) * (0.5 +
 ↪  np.abs(X[:, 1]))
bh, eh, _, _ = ols(X, yh)
V_hc1 = n / (n - q) * XtXi @ (X.T * eh**2) @ X @ XtXi
assert np.allclose(np.sqrt(np.diag(V_hc1)), sm.OLS(yh, X).fit(cov_type="HC1").bse)

# E(RSS) = sigma^2 (n - q) (18); in- and out-of-sample MSE at a fixed design,
 ↪  (26)-(27)
rng = np.random.default_rng(26)
n, p_, R = 50, 10, 20000
Xf = rng.standard_normal((n, p_))
Pf = Xf @ np.linalg.solve(Xf.T @ Xf, Xf.T)
eps, eps_new = rng.standard_normal((R, n)), rng.standard_normal((R, n))
mse_in = (((eps - eps @ Pf) ** 2).sum(1) / n).mean()
mse_out = (((eps_new - eps @ Pf) ** 2).sum(1) / n).mean()
assert abs(mse_in - (1 - p_ / n)) < 0.01 and abs(mse_out - (1 + p_ / n)) < 0.01


def criteria(rss, d, n, sig2):
    """Deck slide 16 / L01-06 (24): Cp, AIC, BIC for a model with d predictors."""
    return (rss + 2 * d * sig2) / n, (rss + 2 * d * sig2) / (n * sig2), (rss +
     ↪  np.log(n) * d * sig2) / n


# best subset on the same simulated data as figure L01-r2-vs-criteria (seed 128)
from itertools import combinations  # noqa: E402

rng = np.random.default_rng(128)
n, Pc = 128, 7
Xs = rng.standard_normal((n, Pc))
ys = 1.0 + Xs[:, :3] @ np.array([1.0, 0.7, 0.5]) + rng.standard_normal(n)
rss_of = lambda cols: ols(np.column_stack([np.ones(n), Xs[:, list(cols)]]), ys)[2]
sig2 = rss_of(range(Pc)) / (n - Pc - 1)
best = np.array([min(rss_of(c) for c in combinations(range(Pc), d)) for d in range(1,
 ↪  Pc + 1)])
dd = np.arange(1, Pc + 1)
tss = ((ys - ys.mean()) ** 2).sum()
r2s = 1 - best / tss
adjs = 1 - (1 - r2s) * (n - 1) / (n - dd - 1)
```

```
cp, aic, bic = criteria(best, dd, n, sig2)
assert np.all(np.diff(r2s) > 0) and near(r2s[0], 0.337) and near(r2s[-1], 0.644)
 ↪  # (21)
assert dd[adjs.argmax()] == 5 and near(adjs.max(), 0.627)
assert dd[cp.argmin()] == 4 and near(cp[3], 1.1356, 5e-5) and near(cp[2], 1.1361,
 ↪  5e-5)
assert np.array_equal(np.argsort(cp), np.argsort(aic))
 ↪  # AIC = Cp / sigma^2
assert dd[bic.argmin()] == 3 and near(bic.min(), 1.211)

# printed numbers of L01-06
assert near(1 - (1 - 0.30) * 50 / 40, 0.125) and near(1 - (1 - 0.10) * 50 / 40,
 ↪  -0.125)   # W p. 181
assert near(128 / 122, 1.049) and near(np.log(128), 4.85, 5e-3) and near(np.log(128)
 ↪  / 2, 2.4, 0.05)
assert [1 + sum(comb(13, j) for j in range(1, kk + 1)) for kk in range(1, 6)] == [14,
 ↪  92, 378, 1093, 2380]
assert near(47.13 / 4.48, 10.5, 0.05)
assert near(-2.200 + 0.717 * 21.27 + 0.225 * 4.45 + 1.013 * 6.59, 20.728)
 ↪  # plane through the means
assert near(0.684 / 0.009, 76.0, 0.05) and np.allclose((0.684 - 1.96 * 0.009, 0.684 +
 ↪  1.96 * 0.009), (0.666, 0.702), atol=5e-4)
for c_, s_, t_, lo_, hi_ in [(0.069, 0.067, 1.03, -0.062, 0.200), (0.439, 0.112,
 ↪  3.92, 0.219, 0.659),
                             (0.960, 0.152, 6.32, 0.662, 1.258), (1.140, 0.317, 3.60,
 ↪  0.519, 1.761)]:
    assert near(c_ / s_, t_, 5e-3) and near(c_ - 1.96 * s_, lo_) and near(c_ + 1.96
     ↪  * s_, hi_)
F = lambda d_r2: (d_r2 / 4) / ((1 - 0.757) / 5531)
 ↪  # W (4.41)
assert near(F(0.002), 11.4, 0.05) and near(F(0.001), 5.7, 0.05) and near(F(0.003),
 ↪  17.1, 0.05)
assert near(stats.f.ppf(0.95, 4, 5531), 2.37, 5e-3)
assert near(1 - 0.245 * 5564 / 5535, 0.754) and near(1 - 0.243 * 5564 / 5531, 0.756)
print("§6 Gaussian linear regression: ok")
```

> **Use `lstsq` (or `solve`), never `inv(X.T @ X) @ X.T @ y`, in real work.** The explicit inverse appears above only inside the HC1 sandwich (29), where the formula itself needs it.

---

7 · AR(1), unit root, VAR(1) → L01-07-ar-var-review

OLS on the lagged series equals `AutoReg`'s estimates, which is Tsay's conditional least squares in (13). The stationary mean and variance come back from a long simulation. Tsay's Example 8.3 is solved by the vec formula (25) and checked independently by `scipy.linalg.solve_discrete_lyapunov`, which solves (24) directly. A long simulation of the same VAR then shows equation-by-equation OLS reproducing `statsmodels`' `VAR.fit`.

```
# === §7 AR(1), unit root, AR(p) and VAR(1) (L01-07) ===
def ar1_moments(phi0, phi, sig2):
```

```
    """Stationary mean (6) and variance (9) of an AR(1); requires |phi| < 1."""
    assert abs(phi) < 1, "no equilibrium distribution at |phi| >= 1"
    return phi0 / (1 - phi), sig2 / (1 - phi**2)


assert near(ar1_moments(0, 0.9, 1)[1], 5.263) and near(np.sqrt(5.263), 2.294)
assert near(ar1_moments(0, 0.5, 1)[1], 1.333) and near(2 / np.sqrt(1 - 0.9**2),
 ↪  4.588)
half = lambda phi: np.log(0.5) / np.log(abs(phi))
 ↪  # (17)
assert near(half(0.9), 6.58, 5e-3) and abs(half(0.5) - 1) < 1e-12
assert near(0.9**10 * 8, 2.79, 5e-3) and near((1 - 0.9**20) / 0.19, 4.62, 5e-3)
 ↪  # (15) at t = 10
assert near(0.9**5, 0.59049, 5e-6) and near(np.sqrt(1000 / 1100), 0.9535, 5e-5)
 ↪  # W (11.4), p. 355
assert near(0.0091 / (1 - 0.116 + 0.019 + 0.104), 0.00904, 5e-6)
 ↪  # Tsay AR(3) mean (21)
assert near(stats.chi2.sf(38.241, 30), 0.1437, 5e-5) and stats.chi2.sf(182.12, 30) <
 ↪  1e-22   # Ljung-Box (3)

# simulate a stationary AR(1); OLS on the lag = AutoReg's conditional MLE (5), (13)
rng = np.random.default_rng(7)
phi0, phi, T = 0.5, 0.9, 100000
r = np.empty(T)
r[0] = phi0 / (1 - phi)
shocks = rng.standard_normal(T)
for t in range(1, T):
    r[t] = phi0 + phi * r[t - 1] + shocks[t]
Xa = np.column_stack([np.ones(T - 1), r[:-1]])
ba = ols(Xa, r[1:])[0]
assert np.allclose(ba, AutoReg(r, lags=1).fit().params)
mu_, g0 = ar1_moments(phi0, phi, 1.0)
assert abs(r.mean() - mu_) < 0.15 and abs(r.var() / g0 - 1) < 0.06 and abs(ba[1] -
 ↪  phi) < 0.01
assert abs(np.corrcoef(r[5:], r[:-5])[0, 1] - phi**5) < 0.02                    #
 ↪  ACF (11)

# unit root: V(y_t) = t sigma^2 across many random walks (18)
rw = rng.standard_normal((20000, 100)).cumsum(1)
assert abs(rw[:, -1].var() / 100 - 1) < 0.04 and abs(rw[:, 9].var() / 10 - 1) < 0.04

# VAR(1): Tsay Example 8.3 (p. 400)
phi0v = np.array([0.2, 0.4])
Phi = np.array([[0.2, 0.3], [-0.6, 1.1]])
Sig = np.array([[2.0, 1.0], [1.0, 1.0]])
assert np.allclose(sorted(np.linalg.eigvals(Phi).real), [0.5, 0.8])
 ↪  # stationary
muv = np.linalg.solve(np.eye(2) - Phi, phi0v)
 ↪  # (23)
assert np.allclose(muv, [1.0, 2.0])
```

```
G0 = np.linalg.solve(np.eye(4) - np.kron(Phi, Phi),
 ↪  Sig.flatten(order="F")).reshape(2, 2, order="F")   # (25)
assert np.allclose(G0, np.array([[25, 20], [20, 40]]) / 9)
assert np.allclose(G0, linalg.solve_discrete_lyapunov(Phi, Sig))
 ↪  # G0 = Phi G0 Phi' + Sigma (24)
assert np.allclose(Phi @ G0, np.array([[11, 16], [7, 32]]) / 9)
 ↪  # Gamma_1 (26)
assert near(G0[0, 1] / np.sqrt(G0[0, 0] * G0[1, 1]), 0.632)

# simulate it; equation-by-equation OLS = statsmodels VAR; sample moments near
 ↪  (23)-(25)
T = 100000
L = np.linalg.cholesky(Sig)
rv = np.empty((T, 2))
rv[0] = muv
for t in range(1, T):
    rv[t] = phi0v + Phi @ rv[t - 1] + L @ rng.standard_normal(2)
Xv = np.column_stack([np.ones(T - 1), rv[:-1]])
Bv = np.linalg.lstsq(Xv, rv[1:], rcond=None)[0]
 ↪  # (3 x 2): const, lag1, lag2
vfit = VAR(rv).fit(1)
assert np.allclose(Bv, vfit.params)
assert np.allclose(Bv[1:].T, Phi, atol=0.02) and np.allclose(rv.mean(0), muv,
 ↪  atol=0.2)
assert np.allclose(np.cov(rv.T), G0, rtol=0.08)
print("§7 AR / VAR: ok")
```

> `ar1_moments` **refuses** $|\phi| \geq 1$ **on purpose.** At a unit root there is no equilibrium distribution to return. The random-walk block checks that the variance grows like t instead.

---

8 · p-hacking and the Bayes factor → L01-08-p-hacking-evidence

Under H_0 every p-value is $U(0,1)$, as in (1). So 32 independent null tests are 32 uniform draws, and the share whose minimum falls below 0.05 is the family-wise rate (3). The run prints `family-wise rate: independent 0.806, Bonferroni 0.048, correlated (rho = 0.5) 0.405`, against $1 - 0.95^{32} = 0.806$. **Correlated tests**, here 32 one-sided z-tests with pairwise correlation 0.5 standing in for "the same 30 students", give 0.405. That lies between 0.05 and 0.806, as L01-08 §3 argues.

```
# === §8 p-hacking: family-wise error, Bonferroni, p-value vs Bayes factor (L01-08)
 ↪  ===
fwer = lambda m, alpha=0.05: 1 - (1 - alpha) ** m
 ↪  # (3), independent tests
assert near(fwer(32), 0.806) and round(fwer(32), 2) == 0.81
assert [round(fwer(m), 3) for m in (1, 5, 10, 32)] == [0.050, 0.226, 0.401, 0.806]
assert 3 * 3 * 3 * 2 * 3 == 162 and near(fwer(162), 0.9998, 5e-5)
assert 0.05 / 32 == 0.0015625

rng = np.random.default_rng(32)
```

```
pv = rng.uniform(size=(200000, 32))
 ↪  # null p-values are U(0,1), (1)
assert abs((pv.min(1) < 0.05).mean() - fwer(32)) < 0.005
assert (pv.min(1) < 0.05 / 32).mean() <= 0.05 + 0.002
 ↪  # Bonferroni (4)
# the same 32 tests on correlated statistics (shared data): the rate falls between
 ↪  0.05 and 0.806
zc = np.sqrt(0.5) * rng.standard_normal((200000, 1)) + np.sqrt(0.5) *
 ↪  rng.standard_normal((200000, 32))
fw_corr = (zc.max(1) > stats.norm.isf(0.05)).mean()
assert 0.05 < fw_corr < fwer(32)
print(f"   family-wise rate: independent {(pv.min(1) < 0.05).mean():.3f}, Bonferroni
 ↪  {(pv.min(1) < 0.05 / 32).mean():.3f}, "
      f"correlated (rho = 0.5) {fw_corr:.3f}")

x0 = 1.8
p_val = stats.norm.sf(x0)
 ↪  # (5)
assert near(p_val, 0.0359, 5e-5) and near(stats.norm.cdf(x0 - 4), 0.0139, 5e-5)     #
 ↪  (6)
BF01 = stats.norm.pdf(x0) / stats.norm.pdf(x0 - 4)
 ↪  # (7)
assert near(BF01, 2.2255, 5e-5) and abs(np.log(1 / BF01) - (4 * x0 - 8)) < 1e-12
 ↪  # (8)
assert near(1 / BF01, 0.449)
post_h1 = lambda bf01, prior_odds_10=1.0: 1 / (1 + bf01 / prior_odds_10)
 ↪  # (11)
assert near(post_h1(BF01), 0.310)
 ↪  # (10)
odds_needed = 9 * BF01
 ↪  # (12)-(13)
assert near(odds_needed, 20.03, 5e-3) and near(post_h1(BF01, odds_needed), 0.9,
 ↪  1e-12)
assert near(20 / 21, 0.952) and near(1 / (9 * 2.23), 1 / 20.07, 1e-5)
assert near(stats.norm.isf(0.05), 1.645)
 ↪  # the band (1.645, 2)
print("§8 p-hacking and evidence: ok")
```

---

9 · The numbers behind the pitfalls and the self-test → L01-10-pitfalls

Every new number in L01-10-pitfalls is computed here. That covers the illustrative inspectors of Block A, the reference distribution of a **searched** LR, the profile-vs-integrated toy and the unbiased-but-inconsistent $2x_1$. `max_lr` does all 111 cuts at once with cumulative sums. The iid-Poisson null and the permutation null each take 4,000 draws, and the run prints their 95% quantiles.

```
# === §9 Numbers behind the pitfalls and the self-test (L01-10) ===
def max_lr(y):
    """LR of the best single cut against one rate, (8), vectorised over all cuts by
     ↪  cumulative sums."""
```

```
    n, s = len(y), y.sum()
    s1, n1 = np.cumsum(y)[:-1], np.arange(1, n)
    lp_all = xlogy(s1, s1 / n1) + xlogy(s - s1, (s - s1) / (n - n1)) - s
    return 2 * (lp_all.max() - (xlogy(s, s / n) - s))


assert abs(max_lr(Y) - LR) < 1e-9
rng = np.random.default_rng(1962)
null_lr = np.array([max_lr(rng.poisson(191 / 112, 112)) for _ in range(4000)])   #
 ↪  iid Poisson: no break
perm_lr = np.array([max_lr(rng.permutation(Y)) for _ in range(4000)])           #
 ↪  same counts, order destroyed
q_null, q_perm = np.quantile(null_lr, 0.95), np.quantile(perm_lr, 0.95)
assert stats.chi2.isf(0.05, 1) < q_null < q_perm < 20 and null_lr.max() < 70 and
 ↪  perm_lr.max() < 70
assert abs(dispersion(rng.permutation(Y))[2] - dispersion(Y)[2]) < 1e-12           #
 ↪  s^2/ybar ignores the order
print(f"   max-LR 95% quantiles: iid Poisson null {q_null:.1f}, permuted coal
 ↪  {q_perm:.1f} (chi2_1: 3.84; observed 70.0)")

# profiling vs integrating on a toy series: one high first year, then ones (uniform
 ↪  prior on the cut)
toy = np.array([6, 1, 1, 1, 1, 1, 1, 1])
cuts = np.arange(1, len(toy))
lp_toy = np.array([seg_kernel(toy[:c]) + seg_kernel(toy[c:]) for c in cuts])
li_toy = np.array([log_int(toy[:c]) + log_int(toy[c:]) for c in cuts])            # a
 ↪  = 1/2 as in §3
wp_toy, wi_toy = np.exp(lp_toy - lp_toy.max()), np.exp(li_toy - li_toy.max())
wp_toy, wi_toy = wp_toy / wp_toy.sum(), wi_toy / wi_toy.sum()
assert near(wp_toy[0], 0.783) and near(wi_toy[0], 0.653)

# expected information depends on the design, at the same theta: binomial n = 10 vs
 ↪  neg. binomial s = 1
assert 10 / (0.5 * 0.5) == 40 and 1 / (0.5**2 * 0.5) == 8                         #
 ↪  LO1-01 (11), (14)
assert near(np.sqrt(1 / 40), 0.158) and near(np.sqrt(1 / 8), 0.354)
# unbiased but not consistent: 2 x_1 for U(0, theta); MSE = theta^2/3 for every n
xs1 = np.random.default_rng(11).uniform(0, 3, size=(200000, 50))
assert abs(((2 * xs1[:, 0] - 3) ** 2).mean() / 3 - 1) < 0.02
 ↪  # 9/3 = 3 at theta = 3

# self-test: two inspectors, 2 defective vials in 20 (fixed n) vs 20 vials to the 2nd
 ↪  defective (fixed s)
assert near(J(2, 20), 222.2, 0.05) and near(stats.beta(1 + 2, 1 + 18).mean(), 0.136)
assert near(negbin_moments(2, 0.1)[0], 0.165)
 ↪  # E(2/N) > 0.1
assert near(stats.binom.cdf(2, 20, 0.25), 0.091) and near(stats.nbinom.sf(20 - 2 - 1,
 ↪  2, 0.25), 0.031)
x_03 = stats.norm.isf(0.03)
 ↪  # one-sided p = 0.03
```

```
assert near(x_03, 1.881) and near(np.exp(4 * x_03 - 8), 0.621) and near(1 / (1 +
 ↪  np.exp(-(4 * x_03 - 8))), 0.383)
assert near(1 / (1 - 0.95**2), 10.256) and near(half(0.95), 13.5, 0.05)
 ↪  # AR(1), phi = 0.95
assert np.allclose(wald(0.1, 20), (-0.031, 0.231), atol=5e-4)
 ↪  # leaves [0, 1]
assert near(1 - 2 / (50 * 0.25**2), 0.36, 1e-12)
 ↪  # Chebyshev, item 9
print("§9 pitfall and self-test numbers: ok")

print("\nALL L01 CHECKS PASSED")
```

Run output for this section: `max-LR 95% quantiles: iid Poisson null 9.9, permuted coal 14.2 (chi2_1: 3.84; observed 70.0)`. A searched LR has its own reference distribution, and 70 is far outside it.

Links

- Unit: L01-00-map · L01-09-formulas (every number asserted here is listed there with its source) · L01-10-pitfalls (each Part I counterexample is computed above).
- Class 2 code: L02 §9 (cross-validation by hand, LOOCV shortcut, IRLS logit, LDA/QDA, Poisson GLM).
- Companion check: `companions/check_companions.py` (run it with `python Notas/L01/companions/check_co` it needs `node`).
- Rules: rules/08 (Monte Carlo) · rules/04 · rules/05.